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Universal Aspects of Quantum-Critical Dynamics In and Out of Equilibrium

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Universal Aspects of Quantum-Critical Dynamics In and Out of Equilibrium

A DISSERTATION PRESENTED
BY
JULIA ALLISON STEINBERG
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Universal Aspects of Quantum-Critical Dynamics In and Out of Equilibrium

ABSTRACT

Quantum phase transitions occur strictly at zero temperature and are accessed by varying a physical parameter such as a magnetic field or chemical potential. While these transitions themselves are experimentally inaccessible, the existence of a zero temperature quantum critical point has remarkable effects on the dynamical properties of interacting quantum many-body systems at finite temperatures. In this thesis, we study dynamics in four strongly interacting many-body systems near quantum critical points in the context of NMR relaxation, quantum quenches, many-body quantum chaos, and charge diffusion.

In chapter 1, we begin by revisiting the issue of NMR relaxation in the quantum disordered phase of Ising spin chains. We find that the imaginary part of the spin susceptibility at low frequencies is dominated by rare quantum processes which leads to an NMR relaxation rate that scales as $1/T_1 \sim \exp(-\frac{2\Delta}{T})$ near the quantum critical point rather than the previously conjectured $1/T_1 \sim \exp(-\frac{\Delta}{T})$. These results resolve certain discrepancies between the energy scales measured with different experimental probes in the quantum disordered paramagnetic phase of the Ising chain system CoNb_2O_6 .

In chapter 2, we study the non-equilibrium dynamics of Sachdev-Ye-Kitaev (SYK) models with q body interactions after a quantum quench, considering a quench protocol which drives the system from an initial state with long-lived quasiparticle excitations to a final state lacking any long-lived quasiparticle excitations. For $q = 4$, we find that the timescale for the system to locally thermalize $\tau_{eq} \sim \frac{\hbar}{k_B T}$ and for $q \rightarrow \infty$ we find that the system thermalizes instantaneously.

In chapter 3, we study the real time dynamics of quantum electrodynamics in $2+1$ dimensions with N_f flavors of fermions. We focus on thermalization and the onset of many-body quantum chaos and compute the quantum Lyapunov exponent in a $\frac{1}{N_f}$ expansion. We briefly comment on chaos, locality, and gauge invariance.

Finally, in Chapter 4 we consider whether the butterfly velocity - a speed at which quantum information propagates - may provide a fundamental bound on diffusion constants in dirty incoherent metals by computing the charge diffusion constant and the butterfly velocity in charge-neutral holographic matter with long wavelength “hydrodynamic” disorder. In this limit, we find that the butterfly velocity does not set a sharp lower bound for the charge diffusion constant.

Contents

o	INTRODUCTION	I
o.1	Emergent Phenomena and Universality	1
o.2	Quantum Phase Transitions	3
o.3	The Transverse Field Ising Chain: Free Fermions in Disguise	8
o.4	The Sachdev-Ye-Kitaev Model: A System Without Quasiparticles	14
o.5	QED ₃ : An Emergent Description of Metallic Quantum Critical States	20
o.6	Holographic Quantum Matter: Insights from Black Holes	25
o.7	Organization of Thesis	33
I	NMR RELAXATION RATES IN ISING SPIN CHAINS	35
I.1	Introduction	35
I.2	Exact Spectrum of the Quantum Ising chain	37
I.3	Form Factors	39
I.4	NMR relaxation rate	40
I.5	Quantum criticality	47
I.6	Experiments	49
I.7	Conclusions	52
2	NON-EQUILIBRIUM DYNAMICS IN THE SACHDEV-YE-KITAEV MODEL	54
2.1	Introduction	54
2.2	Preliminaries	58
2.3	Kadanoff-Baym equations: Numerical Study	67
2.4	Kadanoff-Baym equations: Large q limit	73
2.5	Conclusions	84
3	THERMALIZATION AND CHAOS IN A CONFORMAL GAUGE THEORY	87
3.1	Introduction	87
3.2	Preliminaries	93
3.3	Diagrammatic expansion for the squared fermion anti-commutator	100
3.4	Bethe Salpeter equation and the Lyapunov exponent	105
3.5	Numerics	110

3.6	Discussion	110
4	BOUNDING TRANSPORT IN DISORDERED HOLOGRAPHIC MATTER	114
4.1	Introduction	114
4.2	Striped Black Holes	120
4.3	Charge Diffusion	123
4.4	Butterfly Velocity	128
4.5	Scaling Geometries	135
4.6	Conclusions	140
	APPENDIX A APPENDIX FOR CHAPTER 1	142
A.1	Evaluation of $e^{-3\Delta/T}$ contribution	142
	APPENDIX B APPENDIX FOR CHAPTER 2	146
B.1	Spectral function and Green's functions of random hopping model	146
B.2	Spectral Function in the Conformal Limit	147
B.3	Details on the Numerical Solution of Kadanoff-Baym equation	148
B.4	Numerical quenches: simpler cases	150
B.5	Conjugacy property of $g(t_1, t_2)$	153
B.6	Initial state in the large q limit	155
	APPENDIX C APPENDIX FOR CHAPTER 3	159
C.1	Real time Polarization bubble	159
C.2	Fermion damping rate	169
C.3	Full expressions for the scalar kernel	171
C.4	Feynman rules for ladder diagrams	174
C.5	Details of numerical computations	181
	APPENDIX D APPENDIX FOR CHAPTER 4	183
D.1	Diffusion in an Inhomogeneous Fluid	183
D.2	Higher Order Corrections to the Butterfly Velocity	189
	REFERENCES	206

Citations to Previously Published Work

Much of the material in this thesis has been previously published in print. Details for particular chapters are given below.

Chapter 1

- Julia Steinberg, N. P. Armitage, Fabian H. L. Essler, and Subir Sachdev. “*NMR relaxation in Ising spin chains*”, Phys. Rev. B **99**, 035156 (2019), arXiv:1810.11466.

Chapter 2

- Andreas Eberlein, Valentin Kasper, Subir Sachdev, and Julia Steinberg. “*Quantum quench of the Sachdev-Ye-Kitaev model*”, Phys. Rev. B **96**, 205123 (2017), arXiv:1706.07803.

Chapter 3

- Julia Steinberg and Brian Swingle. “*Thermalization and chaos in QED_3* ”, Phys. Rev. D **99**, 076007 (2019), arXiv: 1901.04984.

Chapter 4

- Andrew Lucas and Julia Steinberg. “*Charge diffusion and the butterfly effect in striped holographic matter*”, JHEP **1610**, 143 (2016), arXiv:1608.03286.

Other work by this author not included in this thesis

- Shubhayu Chatterjee, Yang Qi, Subir Sachdev, Julia Steinberg. “*Superconductivity from a confinement transition out of a fractionalized Fermi liquid with $\mathbb{Z}_2$ topological and Ising-nematic orders*”, Phys. Rev. B **94**, 024502 (2016), arXiv:1603.03041.
- Julia Steinberg, Madhu Advani, and Haim Sompolsky. “*A new role for sparse expansion in neural networks*”, in preparation.
- Julia Steinberg and Haim Sompolsky. “*Associative memory of structured knowledge*”, in preparation.

Listing of figures

1	Finite temperature phase diagram of the transverse field Ising model.	14
2	Diagrammatic depiction of the Schwinger-Dyson equation for the $q = 4$ SYK model to leading order in the large N limit	16
1.1	Crossover phase diagram of the Ising chain in a transverse field, b . There is a quantum critical point at $T = 0$ and $b = b_c$ (for the Hamiltonian in Eq. (1.1), $b_c = 1$) between a ferromagnetic ($b < b_c$) which breaks the Ising symmetry, and a paramagnetic phase. This chapter focuses on the “quantum disordered” regime above the paramagnetic phase for $b > b_c$. The other regimes were described in Ref. ¹¹⁹	37
1.2	$c_1(T)$ for $b = 1.5$ and several temperatures. The thermodynamic limit result (blue dots) is seen to be in good agreement with the extrapolation of numerical results for $L = 192$ and $0.1 \leq \eta \leq 0.115$ (yellow dots).	47
1.3	$c_2(T)$ for $b = 1.5$ and several temperatures. The results of extrapolating numerical results for $0.1 \leq \eta \leq 0.115$ and $L = 144$ are in good agreement with those for $L = 192$	48
1.4	Phase diagram and energy scales of the Ising chain system CoNb_2O_6 as a function of transverse field. Energy scales from different experimental probes as well as the transition temperature of the 3D incommensurate Neel order are given. The transition is to a state with ferromagnetic chains that are ordered antiferromagnetically in the b . On the paramagnetic side of the transition, one can see an excellent agreement between various spectroscopic probes (THz absorption ²¹⁰ and inelastic neutron scattering ¹⁷⁰ and heat capacity ¹³¹). In contrast one can see clearly that the gap extracted from the temperature dependence of the NMR spin relaxation time is approximately twice as large ¹¹⁹	50
2.1	Schematic of the closed time contour $\mathcal{C}$ adapted from ¹¹⁴	59
2.2	The ratio of the Keldysh and spectral function is shown before and after a quench from a J_2+J_4 model to a purely J_4 model. Fits to this data allow determination of the effective temperature as a function of time after the quench, $\beta_{\text{eff}}(\mathcal{T})$ using Eq. (2.52).	71
2.3	Fits to the values of the effective temperature $\beta_{\text{eff}}(\mathcal{T}) = 1/T_{\text{eff}}(\mathcal{T})$ from results as in Fig. 2.2 to Eq. (2.53) to allow determination of the thermalization rate Γ for each quench.	71

2.4	The values of the thermalization rate Γ obtained from Fig. 2.3 are shown as a function of the final temperature, T_f for each quench. One of our main numerical results is the proportionality of Γ to T_f at small $T_f \ll J$. At low temperatures the relaxation towards equilibrium is expected to be controlled by quantum and classical fluctuations and hence we expect an equilibration rate proportional to T_f . At higher temperatures Γ is of order J_4 and since at high temperatures the relaxation is essentially dominated by thermal fluctuations the relaxation rate is dominated by the microscopic interaction energy-scale of the fermions.	72
2.5	The regions in the t_1 - t_2 plane.	76
3.1	A represents the time ordered correlation function on the two fold contour and B represents the OTOC. The red represent A^μ and blue represents A^ν . These are two of the terms in the squared commutator for the photon field. There are two more times, one which is time ordered and one which is out of time order.	90
3.2	The fermion polarization bubble is the $O(1)$ contribution to the gauge field propagator and dominates over the bare term at low energies.	95
3.3	The imaginary and real parts of the full polarization component $\Pi^{xx}(k^0, \mathbf{k})$ are plotted against the high temperature limits in A and B respectively.	98
3.4	The leading contribution to the fermion self energy is $O(\frac{1}{N_f})$ so the correction to the mass can be neglected.	99
3.5	A plot of the fermion decay rate $\Gamma_{\mathbf{k}}$ in units of temperature which vanishes as $\mathbf{k} \rightarrow 0$	101
3.6	A diagrammatic picture of the square of the fermion anticommutator. The lower rails and retarded propagators and upper rails are advanced propagators and the rails are Wightman functions.	108
3.7	We get four types of terms from expanding the squared current-current commutator to first order. The first nontrivial diagrams can be divided into two groups and are shown above. The black dots are current insertions. Each group contains a diagram corresponding to each direction of particle flow. Only the diagrams with uncrossed rungs will contribute to chaos.	111
3.8	Terms contributing to the current current commutator at higher order.	112
4.1	(a) The global Kruskal geometry. The smooth black lines on the left/right side of the diagram denote the “AdS” boundaries of the left/right field theories. The rough red lines at the top and bottom denote the singularities inside of a black hole and a white hole respectively. The dashed purple lines denote event horizons at $U = 0$ and $V = 0$. Blue arrows denote the direction as well as the standard “bulk radial” coordinate r . Eddington-Finkelstein coordinates only cover the region $V \geq 0$ in this diagram. The properties of this black hole vary smoothly in the x direction, which runs out of the page. (b) The effect of the “geometric shockwave” is a shift $b(x)$ between the horizons as one crosses $U = 0$	130

B.1	The spectral function of the random hopping model A long before ($\mathcal{T} = -14.7$) and after ($\mathcal{T} = 14.7$) a parameter quench from $J_{2,i} = 1$ to $J_{2,f} = 0.5$ and 2.0 . The width depends only on the value of J_f , becoming wider if $J_f > J_i$ and narrower if $J_f < J_i$. . .	150
B.2	The Keldysh component of the Green's function of the random hopping model A long before ($\mathcal{T} = -14.7$) and after ($\mathcal{T} = 14.7$) a parameter quench from $J_{2,i} = 1$ to $J_{2,f} = 0.5$ and 2.0	151
B.3	The spectral function for Majorana fermions is shown long after a quench starting with free Majorana fermions and switching on a $q = 4$ SYK interaction term. Here $J_{4,f} = 1$	152
B.4	The propagation of the conjugacy property in the t_1 - t_2 plane.	153
C.1	The decay rate is related to the imaginary part of the fermion self energy via the optical theorem.	170

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*Poets say science takes away from the beauty of the stars -
mere globs of gas atoms. I too can see the stars on a desert
night, and feel them. But do I see less or more?*

Richard P. Feynman

0

Introduction

0.1 EMERGENT PHENOMENA AND UNIVERSALITY

The overarching goal of condensed matter physics is to understand how interactions of microscopic degrees of freedom on the lattice level can lead to macroscopic phenomena which has a simplified description. To achieve this understanding, it is not enough to know how the individual microscopic degrees of freedom interact with each other. At each level of complexity, new interactions

and symmetries emerge which lead to more and richer phenomena⁶. In BCS superconductors, the charge carriers are no longer single electrons but Cooper pairs which are formed via attractive interactions mediated by the lattice vibrations. In CoNb_2O_6 structures, the low energy excitations can be described in terms of *spinless* free fermions which are formed from nonlocal collections of locally interacting spin degrees of freedom¹¹⁹. In spin liquids, fractionalization of the electron leads to chargeless spinon excitations which interact with each other via an emergent gauge field. Finally, in the “strange metal” phase of cuprate superconductors, the emergent low energy excitations have no single particle description at all and behave like a highly entangled quantum fluid.

The study of critical phenomena has led to the development of many tools which allow us to systematically identify the relevant collective degrees of freedom from the microscopic physics. As a system approaches criticality, fluctuations persist throughout the entire system leading to a loss of microscopic scales. Through a coarse graining procedure, the low energy description of a system can often be described via a continuum field theory and the symmetries of this field theory strongly constrain scaling forms of correlations, susceptibilities, and other quantities. Many theories with vastly different microscopic degrees of freedom are described by the same continuum theory leading to identical scaling behavior, known as “universality”. While the concept of universality class was first introduced to explain the scaling of static quantities such as spin susceptibility, it extends to the description of dynamic quantities and responses of systems near criticality^{97,111}.

In summary, the study of condensed matter shows us that more is different but more is also the same¹¹². This dichotomy is especially pronounced in the study of quantum critical phenomena where strong quantum fluctuations provide new phenomena that may also be conceptually similar to the dissipative dynamics of classical black holes. This thesis will focus on universal dynamics of systems in the vicinity of a quantum critical point. In this regime, characteristic timescales emerge from the interplay of quantum and thermal fluctuations and their dependence on microscopic properties of the system is constrained to fixed scaling forms.

In the rest of this chapter, we will provide a general overview of quantum phase transitions and relevant background on the four examples of quantum critical matter that will be addressed in this thesis. The first two examples are lattice models start while the latter two are start directly from a universal continuum description. We will end by summarizing the organization of this thesis.

0.2 QUANTUM PHASE TRANSITIONS

A general definition of a phase transitions is a change in the state of a thermodynamic system from one phase to another accompanied by a non-analyticity in the thermodynamic free energy. In first order phase transitions, the non-analyticity is manifested in a latent heat and in second order phase transitions it is manifested in a diverging susceptibility. At finite temperatures, phase transitions occurring in equilibrium are most often driven by thermal fluctuations reflected in a balancing of energetic and entropic contributions to the free energy.

By contrast, quantum phase transition are transitions between different quantum states of matter at zero temperature and are driven by varying a physical parameter such as a magnetic field. In the infinite lattice limit, quantum phase transitions are characterized by a non-analyticity in the ground state energy of the system in which an excited state becomes the ground state.

We will focus on second order quantum phase transitions in which low energy fluctuations are characterized by an energy scale denoted as Δ . To illustrate generic scaling behaviors, we consider a Hamiltonian of the form $H = H_0 + gH_1$ which can be driven through a quantum phase transition by tuning the dimensionless coupling g to a critical value g_c . In most cases of interest, we find that the energy scale Δ vanishes as

$$\Delta \sim J|g - g_c|^{z\nu} \quad (1)$$

where J is a microscopic coupling scale and the quantity $z\nu$ is a critical exponent and z is called the dynamical critical exponent. Like classical phase transitions, quantum phase transitions also have a

diverging characteristic length scale ξ which is given by

$$\xi^{-1} \sim \Lambda |g - g_c|^\nu \quad (2)$$

where Λ is an inverse length scale defined by the inverse lattice spacing i.e. $\Lambda \sim \frac{1}{a}$. The divergence of ξ can be related to the disappearance of the energy scale Δ as

$$\Delta \sim \xi^{-z} \quad (3)$$

Strictly speaking, quantum phase transitions occur *only* at zero temperature. In fact, for many of the most interesting quantum critical systems, such as the transverse Ising chain, there is no phase transition at finite temperature meaning there is no long range order and the correlation length remains finite at values of the coupling. Furthermore, if there is a finite temperature phase transition, the scaling behavior of the transition can be obtained by a purely classical description at nonzero temperatures. Naively, one may think that unique signatures of quantum phase transitions are experimentally inaccessible or only accessible at very low temperatures. Throughout this thesis, we will see that this is *not* the case. In fact, the most interesting consequences of quantum critical points are reflect in the dynamics of quantum many body systems at finite temperature, in which the competition between thermal and quantum fluctuations produces profound phenomena completely absent in classical systems.

While much of the progress studying of quantum phase transitions is fairly recent, the study of classical phase transitions has a rich and celebrated history within the field of statistical physics^{218,220,219}.

It turns out that many of the tools used to study critical phenomena in classical systems can be applied (with limitations) to the study of quantum criticality. We can formalize this correspondence by introducing a quantum/classical mapping between the imaginary time Feynman path integral

and the partition function. This results in an “equivalence” of the partition function of a classical system in $D = d + 1$ spatial dimensions to the partition function of a quantum system with Hamiltonian H_Q in d spatial dimensions

$$\mathcal{Z} \sim \text{Tr} \left[e^{-\frac{H_Q}{k_B T}} \right] \quad (4)$$

This quantum/classical mapping works as follows: if we consider evolution of the quantum state in imaginary time, we see that the quantum partition function is equivalent to imaginary time evolution over a period of length L_τ .

$$L_\tau = \frac{\hbar}{k_B T} \quad (5)$$

The periodicity in time turns the sum over histories in the path integral into the sum over states in the partition function. In the classical model, τ is viewed as another dimension on equal footing with spatial dimensions (neglecting anisotropy in the couplings). So a quantum model at temperature T in d spatial dimensions is equivalent to a classical model in $d + 1$ spatial dimensions. The additional spatial dimensional in the classical model corresponds to the τ dimension in the quantum model. This dimension is distinct from the other spatial dimensions and is periodic with finite length L_τ .

While the quantum-classical mapping proves useful in studying thermodynamic properties of quantum phase transitions, it is inadequate for studying the dynamical properties of quantum critical phenomena^{177,205}. Unlike many-body classical systems, quantum systems are inherently dynamical. The state of a closed quantum system undergoes unitary time-evolution generated by its Hamiltonian and as a consequence, quantum fluctuations and conservation laws are both encoded in the emergent dynamics of quantum many-body systems. By contrast, the nature of equilibrium dynamics in classical systems is not fully specified by the Hamiltonian. Instead, one must specify transition rates between states and conservation of laws must be put in by hand through restricting allowed transitions.

At finite temperatures, systems near quantum criticality are characterized by a new intrinsic timescale that emerges from their microscopic dynamics. This timescale τ_ϕ is called the phase coherence time and is a measure of the timescale the wavefunction of a many-body quantum systems retains memory of its phase. In most cases, τ_ϕ is closely related to the local thermal equilibration time τ_{eq} , which is the characteristic time in which thermal equilibrium is established in the absence of globally conserved quantities. τ_ϕ can also be interpreted as a lifetime of *local* “quasiparticle excitations”, where a quasiparticle is defined as a collective single particle excitation with the same quantum numbers as the bare particles composing the system.

The manner in which $\tau_\phi \rightarrow \infty$ as $T \rightarrow 0$ characterizes the dynamical properties of different regimes in the vicinity of a quantum critical point. In general, there are two relevant energy scales determining the behavior of τ_ϕ , the microscopic energy scale Δ and $k_B T$. In the two regimes where $|\Delta| > k_B T$, we find that $\tau_\phi \gg \hbar/k_B T$ implying the low energy dynamics become effectively semi-classical: quantum interference effects occur on a longer timescale than thermal fluctuations. This allows us to describe the dynamics in terms of long-lived quasiparticle excitations.

For $\Delta \ll T$, known as the “quantum critical” regime, the microscopic scale Δ disappears from the dynamics and τ_ϕ is given by

$$\tau_\phi \geq \mathcal{C} \frac{\hbar}{k_B T} \quad (6)$$

where $\mathcal{C}$ is an $O(1)$ number independent of any microscopic couplings. We see from Eqn. 6 that $\tau_\phi \sim L_\tau$. This casts major doubt on the validity of applying the quantum classical mapping to study dynamics in real time. Since quantum fluctuations occur on the same timescale as the period of imaginary time evolution, it is essentially impossible to construct longtime correlations for the quantum system from the corresponding classical problem. In other words, analytic continuation from imaginary to real time does not commute with perturbative expansion in a control parameter with errors being increasingly amplified for longer time scales. Furthermore, describing the dynam-

ics in terms of single particle excitations is simply incorrect as these excitations decay on the same timescale as thermal equilibration.

Given the breakdown of the quantum/classical mapping in the quantum critical regime, one may wonder if there are other ways to study dynamics of quantum critical systems. The answer is yes, but only in specific settings can we obtain potentially reliable results. For example, if a system is integrable we can obtain exact results for correlators in imaginary time and analytically continue them to real time without any issues. In other cases, one may introduce a system with N flavors of a field making it possible to do a perturbative expansion about a free theory in powers of $\frac{1}{N}$. However, the physicality of the large N limit is unclear. A more general strategy is to compute dynamic observables directly in real time using the Keldysh formalism. However, this has its own complications and is generally limited to perturbative expansions about a free theory. In the past decade, several techniques have been developed within the framework of the AdS/CFT correspondence which allow direct computations of real-time observables in the gravity dual. While this direction may be promising, it raises as many questions as it answers.

While the dissipative dynamics of quantum critical systems make them difficult to study using conventional techniques, they lead to very rich physical phenomena, some of which will be discussed in the rest of this thesis. We will now introduce four distinct description of quantum systems near criticality. The first is the transverse field Ising chain (TFIC) a one dimensional lattice model of Ising spins with ferromagnetic exchange interactions in a transverse magnetic field. The second is the Sachdev-Ye-Kitaev model, a $0 + 1$ dimensional model of Majorana fermions with random all-to-all interactions. The third is quantum electrodynamics in $2 + 1$ dimensions with N_f flavors of fermions. Finally, we provide an overview of how insights from AdS/CFT correspondence can be applied to study universal properties of strongly interacting quantum matter.

0.3 THE TRANSVERSE FIELD ISING CHAIN: FREE FERMIONS IN DISGUISE

The transverse field Ising chain is a one dimensional chain of ferromagnetically interacting Ising spins with an on-site transverse field. The Hamiltonian can be written in terms of spin operators as

$$H_I = -J \sum_{i=1}^L [\sigma_i^z \sigma_{i+1}^z + b \sigma_i^x] \quad (7)$$

where σ_i^x 's and σ_i^z 's are represented by the Pauli matrices and obey the usual commutation relations. $J > 0$ is a ferromagnetic exchange constant setting the microscopic energy scale and $b > 0$ is a dimensionless coupling used to tune the system across the quantum phase transition. Physically, the first term in H_I represents a ferromagnetic interactions between neighboring spins on a one dimensional chain and the second term represents an onsite magnetic field perpendicular to the direction of the exchange interaction. CoNb_2O_6 provides an almost ideal material realization of such a system.

When $b = 0$, H_I is diagonal in the eigenbasis of σ_i^z and has two ground states:

$$|\uparrow\rangle = \prod_i |\uparrow\rangle_i, \text{ or } |\downarrow\rangle = \prod_i |\downarrow\rangle_i \quad (8)$$

which are related by global $\mathbb{Z}_2$ symmetry, i.e. flipping all of the spins. This is the ferromagnetic ground state. When $b > 0$, the Hamiltonian contains terms proportional to σ_i^x which does not commute with σ_i^z . As a consequence, the introduction of a finite transverse field introduces quantum mechanical tunneling events which flip the orientations of spins in the lattice and persist at zero temperature in the absence of any thermal fluctuations. In the large field limit, i.e. $b \gg 1$, to leading order in $\frac{1}{b}$, the ground state of H_I is

$$|0\rangle = \prod_i |\rightarrow\rangle_i \quad (9)$$

This state has unbroken $\mathbb{Z}_2$ symmetry and is called the quantum paramagnetic state.

At zero temperature, the TFIC exhibits a phase transition from an ordered state with $\mathbb{Z}_2$ symmetry to a quantum paramagnetic with unbroken $\mathbb{Z}_2$ symmetry. This transition is in the same universality class as the $d = 2$ classical Ising model which can be shown directly using the quantum classical mapping introduced in Section 0.2. The continuum theory of the TFIC near the quantum critical point is a theory of free fermions with the following Lagrangian

$$\mathcal{L}_I = \Psi^\dagger \frac{\partial \Psi}{\partial \tau} + \frac{c}{2} \left(\Psi^\dagger \frac{\partial \Psi^\dagger}{\partial x} - \Psi \frac{\partial \Psi}{\partial x} \right) + \Delta \Psi^\dagger \Psi \quad (10)$$

Here Ψ and $\Psi^\dagger$ are complex Grassman fields in imaginary time and the coupling constants Δ and c are given by

$$\Delta = 2J(1 - g), \quad c = 2Ja \quad (11)$$

At finite temperature, the TFIC does not have a ferromagnetic phase. Any long range order will be destroyed by thermal fluctuations as in the classical Ising chain. However, despite the absence of a phase transition, we can identify three regimes of the phase diagram (shown in Fig. 1) from the nature of the dynamics as characterized by the phase decoherence time τ_ϕ . These regimes are called “renormalized classical”, the “quantum disordered”, and “quantum critical” regime and are separated by smooth crossovers instead of sharp phase boundaries. In each regime, τ_ϕ can be extracted from time dependence of the two-point correlation function of the order parameter σ_ℓ^z defined as

$$C(x_\ell, t) = \langle \sigma_\ell^z(x_\ell, t) \sigma_0^z(0, 0) \rangle = \frac{1}{\mathcal{Z}} \text{Tr} \left[e^{-\beta H_I} e^{iH_I t} \sigma_\ell^z e^{-iH_I t} \sigma_0^z \right] \quad (12)$$

In the continuum limit near the critical point, the two-point correlator admits the following scaling form

$$C(x, t) = Z T^{\frac{1}{4}} \Phi_I \left(\frac{T x}{c}, T t, \frac{\Delta}{T} \right) \quad (13)$$

We can identify the universal dynamical behavior of $C(x, t)$ for each of the three regimes by considering the nature of the dominant low energy excitations.

In the “renormalized classical” or quantum ordered regime $T \ll \Delta$ and the low energy excitations above the ground state are domain walls between ordered regions^{181,177}. For finite b these domain walls are mobile excitations and can be viewed as fermionic particles whose mean separation is much larger than their de Broglie wavelength, which is found by equating their kinetic energy with the thermal energy using classical equipartition. The energy of each fermionic particle in the low momentum limit is $\varepsilon_k = \Delta + \frac{c^2 k^2}{2\Delta}$ and their density ρ can be obtained via classical Boltzmann statistics

We can compute the two point correlator $C(x, t)$ in a semiclassical limit of the first-quantized Feynman path integral. The stationary phase approximation of this integral simply selects the forward and backward classical trajectories. In the dilute limit, we only consider collisions between two particles. Due to momentum conservation before and after collisions, the classical trajectories remain straight lines. Each collision must be treated quantum mechanically and is characterized by a two particle S with matrix elements given by $S_{k_1 k_2} = -1$ for all momenta k_1, k_2 . Consequently the net phase factor for each collision is just $|S_{k_1 k_2}| = 1$. This implies that the classical trajectories are just independently distributed straight lines with uniform density ρ and slope $v_k = \frac{d\varepsilon_k}{dk}$, where v_k ’s are the velocities of the particles. After averaging over trajectories $C(x, t)$ can be written in the form

$$C(x, t) = \langle \sigma^z \rangle_{T=0}^2 R(x, t) \quad (14)$$

where $\langle \sigma^z \rangle_{T=0}$ is the zero temperature magnetization and $R(x, t)$ is a relaxation function which is defined as

$$R(x, t) \equiv \exp \left(- \int \frac{dk}{2\pi} e^{-\varepsilon_k/T} |x - v_k t| \right) \quad (15)$$

The equal-time and equal-space forms of $R(x, t)$ can be written as

$$R(x, 0) = e^{-|x|/\xi_c} \quad (16)$$

$$R(0, t) = e^{-|t|/\tau_\phi} \quad (17)$$

where $\xi_c = \frac{1}{2\rho}$ is the correlation length and τ_ϕ is the phase decoherence time

$$\tau_\phi = \frac{\pi\hbar}{2k_B T} e^{\frac{\Delta}{k_B T}} \quad (18)$$

From Eqn. 15, $R(x, t)$ satisfies the following scaling form

$$R(x, t) = \Phi_R\left(\frac{x}{\xi_c}, \frac{t}{\tau_\phi}\right) \quad (19)$$

where we can obtain the following closed form expression for Φ_R

$$\ln \Phi_R(\bar{x}, \bar{t}) = -\bar{x} \text{erf}\left(\frac{\bar{x}}{\bar{t}\sqrt{\pi}}\right) - \bar{t} e^{-\frac{\bar{x}^2}{\pi\bar{t}^2}} \quad (20)$$

The Fourier transform of $C(x, t)$ is the dynamic structure factor $S(k, \omega)$. For $T = 0$, $S(k, \omega)$ contains a delta function at $\omega = 0$ and at finite temperatures, this delta function broadens on a frequency scale of order τ_ϕ^{-1} .

In the quantum disordered or quantum paramagnet regime, $T \ll |\Delta|$ where $\Delta < 0$. The low energy excitations are spin flips opposing the direction of the transverse field. These excitations are created and annihilated by the operator σ^z . This operator can be represented in terms of operators $\psi(x, t)$ which create and annihilate single particle excitations above the ground state

$$\sigma^z(x, t) = (2cZ|\Delta|^{\frac{1}{4}})^{\frac{1}{2}} (\psi(x, t) + \psi^\dagger(x, t) + \dots) \quad (21)$$

where the ellipses represent multiparticle creation or annihilation terms. We can compute the two point correlator $C(x, t)$ in a semiclassical limit as we did in the quantum ordered regime if we ignore these terms. As we will show in Chapter 1, the semiclassical approach produces incorrect results for the NMR relaxation rate as the dominant contributions to the spin susceptibility actually comes from rare fermion decay processes at low frequencies.

The semiclassical calculation of the phase decoherence time on the quantum disordered side is quite similar in spirit to the calculation on the ordered side. However, there are a few key differences. There is a special trajectory where particles are created at $(0, 0)$ and annihilated at (x, t) which each contributed a zero temperature propagator $\langle \sigma^z(x, t) \sigma^z(0, 0) \rangle$. This trajectory occurs only forward in time. If a particles collides with thermally excited particle before it is annihilated, it accumulates a factor of the S matrix element $S_{k_1 k_2}$ at each collision. All other collisions from other trajectories occur both forward and backwards and give factors of one. Averaging over trajectories, we find $C(x, t)$ can be written in the form

$$C(x, t) = \langle \sigma^z(x, t) \sigma^z(0, 0) \rangle_{T=0} R(x, t) \quad (22)$$

where $\langle \sigma^z(x, t) \sigma^z(0, 0) \rangle_{T=0}$ is the zero temperature correlator and $R(x, t)$ is the same relaxation function in Eqn. 15 and the only change is in the sign of Δ . This implies that the phase coherence time is given by

$$\tau_\phi = \frac{\pi \hbar}{2 k_B T} e^{\frac{|\Delta|}{k_B T}} \quad (23)$$

Similarly, at finite temperature, we see a broadening of $S(k, \omega)$ on frequency scales $\omega \sim \tau_\phi^{-1}$.

The final regime in Fig. 1 is the quantum critical regime in which $\Delta \ll T$. In this regime, there is no effective classical model describing the dynamics. The de Broglie wavelength of the particles is in the same order as their spacing. To analyze the time dependence of $C(x, t)$, we can start from the continuum theory in Eqn. 10. For $\Delta = 0$, this is a conformal field theory of free fermions. In con-

formal field theories, two-point functions for primary operators are highly constrained by conformal symmetry. Since σ^ω is a primary operator, the zero temperature Euclidean correlator $C(x, \tau)$ must take the form

$$C(x, \tau) \sim \frac{1}{(\tau^2 + x^2/c^2)^{2\Delta_{\sigma^\omega}}} \quad (24)$$

where Δ_{σ^ω} is the scaling dimension of the operator σ^ω . By evaluating $C(x, 0)$, it can be shown that

$$\Delta_{\sigma^\omega} = \frac{1}{8} \quad (25)$$

We can obtain $C(x, \tau)$ at finite temperature from Eqn. 24 via a conformal mapping

$$C(x, \tau) \sim T^{\frac{1}{4}} \frac{1}{[\sin(\pi T(\tau - ix/c)) \sin(\pi T + ix/c))]^{\frac{1}{8}}} \quad (26)$$

We obtain the dynamic susceptibility $\chi(k, \omega)$ by Fourier transforming Eqn. 26 and analytically continuing from Matsubara frequencies by sending $\omega_n \rightarrow \omega + i\varepsilon$. At zero temperature $\chi(k, \omega)$ has a branch cut. At finite temperature we can understand the low frequency structure of $\chi(k, \omega)$, by expanding $\chi^{-1}(k, \omega)$ in powers of k and ω . This has the form¹⁷⁷

$$\chi(k, \omega) = \frac{\chi(0)}{1 - i(\omega/\omega_1) + k^2 \bar{\xi}^2 - (\omega/\omega_2)^2} \quad (27)$$

where $\omega_1 \sim T$, $\omega_2 \sim T$ and $\bar{\xi} \sim c/T$, and $\chi(0) = \chi(k=0, \omega=0) \sim T^{-\frac{7}{4}}$. Here ω_1 characterizes the frequency of damping. More generally, we can define the relaxation rate Γ_R as

$$\Gamma_R^{-1} \equiv i\chi(0) \frac{\partial \chi^{-1}(0, \omega)}{\partial \omega} \Big|_{\omega=0} \quad (28)$$

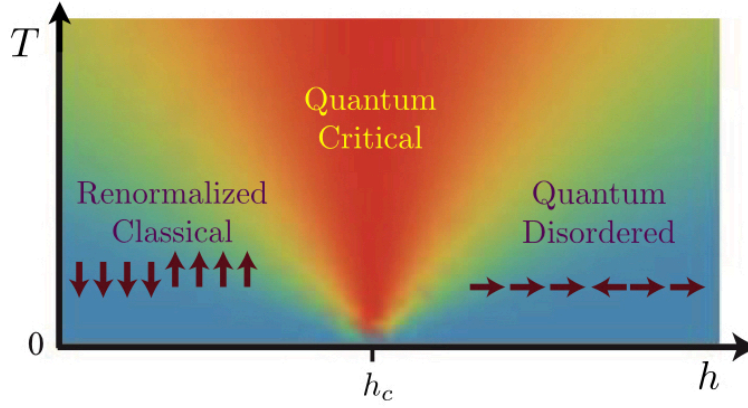


Figure 1: Finite temperature phase diagram of the transverse field Ising model.

Using the low frequency form of $\chi(k, \omega)$ in Eqn. 27 we see that $\Gamma_R = \omega_1$. We identify Γ_R as a phase coherence time and conclude

$$\tau_\phi \sim \frac{\hbar}{k_B T} \quad (29)$$

which is independent of the microscopic scale Δ and of the same order as thermal fluctuations.

0.4 THE SACHDEV-YE-KITAEV MODEL: A SYSTEM WITHOUT QUASIPARTICLES

The most general form of the Sachdev-Ye-Kitaev model is a Hamiltonian of N Majorana fermions with q body interactions in $0+1$ dimensions. This is a variant of a model first introduced by Sachdev and Ye in ¹⁸⁰ and later discussed by Kitaev in the context of holography ¹²¹. This model exhibits novel scaling behavior indicating a non-Fermi liquid state.

The Hamiltonian (using conventions from Ref. ¹⁴⁵) can be written as

$$H = (i)^{\frac{q}{2}} \sum_{1 \leq i_1 < i_2 < \dots < i_q \leq N} j_{i_1 i_2 \dots i_q} \psi_{i_1} \psi_{i_2} \dots \psi_{i_q} \quad (30)$$

where ψ_i are Majorana fermions on sites $i = 1 \dots N$ obeying

$$\{\psi_i, \psi_j\} = \delta_{ij} \quad (31)$$

and $j_{i_1 i_2 \dots i_q}$ are independent Gaussian random variables with zero mean and variance

$$\langle j_{i_1 \dots i_q}^2 \rangle = \frac{j^2(q-1)!}{N^{q-1}} = \frac{2^{q-1}}{q} \frac{\mathcal{J}^2(q-1)!}{N^{q-1}} \quad (32)$$

In the large N limit (for $q > 2$) the SYK model is exactly solvable model of quantum matter without quasi-particles.

We can define the Euclidean propagator as

$$G(\tau_1, \tau_2) \equiv \langle T_\tau \psi(\tau_1) \psi(\tau_2) \rangle \quad (33)$$

For a free Majorana fermions the propagator is

$$G_0(\tau_1, \tau_2) = G(\tau_1 - \tau_2) = \frac{1}{2} \text{sgn}(\tau_1 - \tau_2) \quad (34)$$

$$G_0(\omega) = \int_0^\beta d\tau e^{i\omega\tau} G_0(\tau) = -\frac{1}{i\omega} \quad (35)$$

where at finite temperature ω are fermionic Matsubara frequencies, i.e. $\omega = \frac{2(n+\frac{1}{2})}{\beta}$. Using the free propagator, we can obtain the full propagator $G(\tau_1, \tau_2)$ by summing diagrams in a perturbative expansion in powers of $\frac{1}{N}$. To leading order, $\frac{1}{N}$ only the “melon” diagrams survive. We obtain the following Schwinger-Dyson equation

$$G_0^{-1}(\tau_1, \tau_2) - G^{-1}(\tau_1, \tau_2) = \Sigma(\tau_1, \tau_2) \quad (36)$$

$$\Sigma(\tau_1, \tau_2) = j^2 [G(\tau_1, \tau_2)]^{q-1} \quad (37)$$

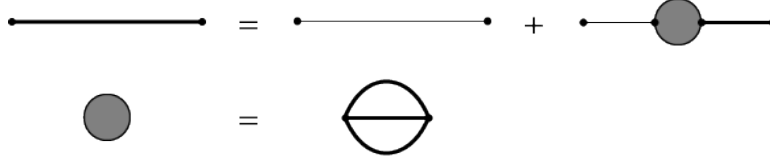


Figure 2: Diagrammatic depiction of the Schwinger-Dyson equation for the $q = 4$ SYK model to leading order in the large N limit

which is shown diagrammatically in Fig. 2. For time independent couplings $j_{i_1 i_2 \dots i_q}$, $G(\tau_1, \tau_2) = G(\tau_1 - \tau_2)$ and we can write Eqn. 36 in frequency space as

$$G(\omega)^{-1} = -i\omega - \Sigma(\omega) \quad (38)$$

For $q = 2$, these equations can be solved exactly as

$$G(\omega) = \frac{2}{i\omega + i\text{sgn}(\omega)\sqrt{J^2 + \omega^2}} \quad (39)$$

One can see from Eqn. 39 that the spectral density $A(\omega) = -2 \text{Im } G(\omega + i\varepsilon)$ produces a semicircular density of states as discussed Wigner in the context of nuclear physics²¹⁷.

At low energies the SYK model has an emergent time-reparameterization symmetry. One can see this by neglecting the first term on the right hand side of Eqn. 36, giving us the following two Schwinger-Dyson equations

$$\int d\tau' G(\tau_1, \tau') \Sigma(\tau', \tau_2) = -\delta(\tau_1 - \tau_2), \quad \Sigma(\tau_1, \tau_2) = J^2 [G(\tau_1, \tau_2)]^{q-1} \quad (40)$$

We see that the equations in 40 are invariant under reparameterizations in time $\tau \rightarrow f(\tau)$ provided G

and Σ transform as follows

$$G(\tau_1, \tau_2) = [f'(\tau_1)f'(\tau_2)]^\Delta G(f(\tau_1), f(\tau_2)) \quad (41)$$

$$\Sigma(\tau_1, \tau_2) = [f'(\tau_1)f'(\tau_2)]^{\Delta(q-1)} \Sigma(f(\tau_1), f(\tau_2)) \quad (42)$$

where $\Delta = \frac{1}{q}$ is the scaling dimension of the fermions. We refer to the solution of Eqns. 40 as the “conformal” Green’s function, which takes the form

$$G_c(\tau) = \frac{b}{|\tau|^{2\Delta}} \text{sgn}(\tau) \int^\tau b^q \pi = \left(\frac{1}{2} - \Delta \right) \tan \pi \Delta \quad (43)$$

For finite temperatures, the Green’s function for $\omega \ll T$ can be obtained from Eqn. 43 from applying the reparameterization

$$f(\tau) = \tan \frac{\tau\pi}{\beta} \quad (44)$$

The power law form of Eqn. 43 implies a spectral density $A(\omega) \sim \omega^{2\Delta-1}$, which is singular for $q \geq 4$. This mysterious low energy behavior of the spectral function and local spin susceptibility is what first sparked interest in the original SY model, which corresponds to $q = 4$ and replacing the Majorana fermions with complex fermions^{180,65}. In these studies, it was shown that the SY model has an *extensive* zero temperature entropy, which can be evaluated using the low energy conformal Green’s function. The extensive ground state entropy implies that the level spacing at low energies is $\partial E \sim e^{-Ns_0}$ for large N where s_0 is the zero temperature entropy density. A tenet of Fermi liquid theory is the Landau energy functional

$$E = E_0 + \sum_\alpha \delta n_\alpha \varepsilon_\alpha + \sum_{\alpha\beta} F_{\alpha\beta} \delta n_\alpha n_\beta + \dots \quad (45)$$

which expresses the change in the total energy of the system after a weak perturbation as an expan-

sion in terms of small changes in quasiparticle occupation numbers δn_α and ε_α is just the energy of an isolated quasiparticle^{46,155}. Eqn. 66 implies that for a generic system with a quasiparticle description, the energy is a sum of terms polynomial in N . This is impossible for the SYK model: the level spacing implies that there are exponentially many low energy states. Thus, we conclude that there is no quasiparticle description for the $q \geq 4$ SYK model.

The key to understanding the many exotic properties of the SYK model lies in its emergent reparameterization invariance and its *breaking*. While the full Green's function is invariant under arbitrary reparameterizations $f(\tau)$, we note that the conformal solution is only invariant under $f(\tau) \in SL(2, R)$ i.e.

$$f(\tau) = \frac{a\tau + b}{c\tau + d}, \quad ad - bc = 1 \quad (46)$$

We can view the reparameterization invariance of the low energy equations of motion in Eqn. 41 and Eqn. 42 as an emergent symmetry of the theory in the infrared which is spontaneously broken by the conformal solution G_c . However, if one naively uses G_c to compute higher order correlators such as the four point function, one finds unphysical divergences. To account for the full low energy properties of the SYK model, one must consider both implicit and explicit breaking of the reparameterization symmetry. This is easiest to see by starting directly from the fermionic path integral and obtaining an effective action. We can do this by integrating over disorder introducing the bi-local field $G(\tau_1, \tau_2)$ with a Lagrange multiplier Σ that imposes the equality

$$G(\tau_1, \tau_2) = \frac{1}{N} \sum_i \psi_i(\tau_1) \psi_i(\tau_2) \quad (47)$$

After integrating out the fermions, we obtain the following nonlocal effective action

$$\frac{S}{N} = -\frac{1}{2} \log \det(\partial_\tau) + \frac{1}{2} \int d\tau_1 d\tau_2 \left[\Sigma(\tau_1, \tau_2) G(\tau_1, \tau_2) - \frac{J^2}{q} G(\tau_1, \tau_2)^q \right] \quad (48)$$

This saddle point equations for this action reproduce the equations of motion in Eqns. 36 and 37.

At low energies, if we insert the conformal solutions $G = G_c$ and $\Sigma = \Sigma_c$ and drop the first term in Eqn. 48, we find that the action is zero. If we include the leading non-conformal correction to the action and evaluate fluctuations about the saddle point, we find that finite reparameterizations $f(\tau)$ obtain the action

$$S = -N \frac{\alpha_S}{\mathcal{J}} \int d\tau \{f, \tau\} \quad (49)$$

where $\{f, \tau\}$ is the Schwarzian derivative defined as

$$\{f, \tau\} \equiv \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2 \quad (50)$$

Eqn. 49 is known as the Schwarzian action which describes “nearly conformal quantum mechanics” (NCFT₁) and in very near extremal black holes which develops a “nearly NAdS₂” (NAdS₂) horizon. The Schwarzian effective action controls the dominant fluctuations of the Majorana fermions in the SYK model at low temperatures and is responsible for many of its remarkable properties including its specific heat, maximal chaos exponent, and local thermalization behavior out of equilibrium, which we will explore in Chapter 2. Many of these exotic properties persist in extended SYK models with spatial structure^{82,193,49,158}. Recently, analysis of a model of a doped random quantum Heisenberg magnet originally introduced in¹⁵⁶ in the large M limit identified a possible deconfined critical point or phase at finite doping with SYK criticality in spin correlations down to zero temperature^{110, 203} applied this limit to the Hubbard model and argued that the deconfined critical

point provides an understanding of the electron and spin spectral functions at the metal-insulator transition found in the numerical study³⁹. Most notably, this study found the presence of linear resistivity down to the lowest numerically accessible temperature. The presence of SYK criticality suggests that the reparameterization soft mode offers a promising and generic mechanism for linear resistivity in strange metals⁸⁴.

0.5 QED₃: AN EMERGENT DESCRIPTION OF METALLIC QUANTUM CRITICAL STATES

We now introduce a 2 + 1 dimensional gauge theory with a $U(1)$ gauge field coupled to N_f flavors of massless fermions (QED₃). The Lagrangian is given by

$$\mathcal{L} = -\frac{1}{4}F^2 + \sum_{a=1}^{N_f} i\bar{\psi}_a \gamma^\mu D_\mu \psi_a \quad (51)$$

where $\gamma^\mu = (\sigma^z, i\sigma^x, i\sigma^y)$ are the gamma matrices, $D_\mu = \partial_\mu - ieA_\mu$ is the covariant derivative, and ψ_a 's are two component spinors. This Lagrangian in Eqn. 51 is superrenormalizable. The coupling $e^2 N_f$ is dimensionful and provides the only scale. When N_f is taken to infinity, fluctuations of the gauge field are completely suppressed and we have a theory of free fermions. As a result QED₃ can be treated perturbatively in a $\frac{1}{N_f}$ expansion. At high temperatures, 3+1 dimensional quantum chromodynamics (QCD₃) can be dimensionally reduced to a 2+1 dimensional gauge theory (QCD₃). In the large N_f limit of QCD₃, non-Abelian interactions are suppressed by $\frac{1}{N_f}$ leading to some resemblance between QED₃ at large N_f to four dimensional gauge theories at high temperatures¹¹. In fact, QED₃ was the first gauge theory beyond two dimensions that permitted a systematic analysis of dynamical chiral-symmetry breaking and has been a useful theoretical tool used to gain insight into dynamical symmetry breaking in four-dimensional theories^{12,160,9}.

Due to the scaling of the gauge coupling, any mass scale that is generated dynamically will be a pure number times e^2 . In the $\frac{1}{N_f}$ expansion, this mass will be suppressed by a power of $\frac{1}{N_f}$. It has

been shown that dynamical mass generation occurs only if the number of fermion flavors is below some critical value N_c . In ¹⁰ it was shown in calculations to leading order in $\frac{1}{N_f}$ that $N_c \geq \frac{32}{\pi^2}$. However, there is no general consensus on the value of N_c due to the breakdown of the perturbative expansion ¹⁰³.

Recently, QED₃ has been studied in the context of several condensed matter systems including $U(1)$ spin liquids, systems near deconfined quantum critical points, transitions between quantum hall states, and surface states of three dimensional topological insulators ^{163,189,188,108,73}. In these systems, the fermions are not fundamental particles but emergent low energy excitations coupled to an emergent gauge field associated with fractionalization of the order parameters. QED₃ displays many novel metallic phases in the presence of disorder which have been studied perturbatively in the large N_f limit at zero temperature ^{206,67}. It is reasonable to expect that these phases also occur in condensed matter systems in which the universal low energy physics is described by QED₃ and the $\frac{1}{N_f}$ expansion gives us a controlled setting to study them.

To compute quantities perturbatively for large N_f , it is useful to implement the following rescaling of the gauge field

$$A_\mu \rightarrow \frac{1}{e\sqrt{N_f}} A_\mu$$

It is often convenient to work in Euclidean spacetime to compute thermodynamic quantities at finite temperature. By convention, we use only lower indices. The Euclidean action is then given by

$$S_E = \int d^2x d\tau \left[\frac{1}{4e^2 N_f} F^2 - \sum_{a=1}^{N_f} \bar{\psi}_a \tilde{\gamma}_\mu D_\mu \psi_a \right] \quad (52)$$

where $D_\mu = \partial_\mu - \frac{iA_\mu}{\sqrt{N_f}}$ and the Euclidean Dirac matrices are $\tilde{\gamma}_\mu = (\sigma^z, \sigma^x, \sigma^y)$ with $\tilde{\gamma}^\mu = \tilde{\gamma}_\mu$. The free

fermion propagator is

$$G_E(\not{P}) = \frac{-i\tilde{\gamma}_\mu P_\mu}{P^2} \quad (53)$$

where $P = (\omega, \mathbf{p})$ is a Euclidean four momentum. To obtain the bare Euclidean photon propagator, we must add a gauge fixing term S_{gf} to the action in Eqn. 52. The standard local gauge fixing term is

$$S_{gf} = \int d\tau d^2x \frac{1}{2e^2 N_f} \frac{1}{\xi} (\partial_\mu A_\mu)^2 \quad (54)$$

which gives us the following bare Euclidean photon propagator

$$\mathcal{D}_{\mu\nu}^0 = \frac{Ne^2}{P^2} \left(\delta_{\mu\nu} - (1 - \xi) \frac{P_\mu P_\nu}{P^2} \right) \quad (55)$$

We can also choose to add a non-local gauge fixing term⁴³

$$S_{gf} = \frac{1}{32(\xi - 1)} \int \frac{d^3P}{(2\pi)^3} \frac{P_\mu P_\nu}{|P|} A^\mu(P) A^\nu(-P) \quad (56)$$

Resulting in the propagator

$$\mathcal{D}_{\mu\nu}^0 = \frac{Ne^2}{P^2} \left(\delta_{\mu\nu} - \frac{P_\mu P_\nu}{P^2} \right) + \frac{16(\xi - 1)}{|P|} \frac{P_\mu P_\nu}{P^2} \quad (57)$$

Since the kinetic term for the gauge field is suppressed by $\frac{1}{N_f}$ in the action in Eqn. 52, one must include the resummation of fermion polarization bubble diagrams in the propagator. The fermion polarization bubble $\Pi_{\mu\nu}(P)$ can be obtained using the free fermion propagator in Eqn. 53, includ-

ing factors of $\frac{\tilde{\gamma}_\mu}{\sqrt{N_f}}$ for each interaction vertex. At zero temperature $\Pi_{\mu\nu}(P)$ evaluates to

$$\Pi_{\mu\nu}(P) = \frac{|P|}{16} \left(\delta_{\mu\nu} - \frac{P_\mu P_\nu}{P^2} \right) \quad (58)$$

As we see, $\Pi_{\mu\nu}(P)$ is $O(1)$ in $\frac{1}{N_f}$ and hence a nonperturbative object. The full resummed photon propagator can be written as

$$D_{\mu\nu}(P) = \left((\mathcal{D}_{\mu\nu}^0(P))^{-1} + \Pi_{\mu\nu}(P) \right)^{-1} \quad (59)$$

For the local gauge fixing term in Eqn. 54, the propagator becomes

$$D_{\mu\nu}(P) = - \left(\delta_{\mu\nu} - \frac{P_\mu P_\nu}{P^2} \right) \frac{1}{\frac{P^2}{e^2 N_f^2} - \frac{|P|}{16}} + \frac{\xi e^2 N_f P_\mu P_\nu}{P^2} \quad (60)$$

For perturbative calculations at zero temperature, it is often convenient to choose the non-local gauge fixing condition so one can take the limit $e \rightarrow 0$ in the beginning of the calculation without choosing a specific value of ξ to specify a gauge. In the limit $N_f \rightarrow \infty$ with the nonlocal gauge fixing condition in 56, $D_{\mu\nu}(P)$ becomes

$$D_{\mu\nu}(P) = \frac{16}{|P|} \left(\delta_{\mu\nu} - \xi \frac{P_\mu P_\nu}{P^2} \right) + O\left(\frac{P^2}{e^2}\right) \quad (61)$$

The full propagator in Eqn. 60 is $O(1)$ in $\frac{1}{N_f}$. Since e is a relevant coupling, the term of $O\left(\frac{P^2}{e^2}\right)$ in Eqn. 60 are suppressed in the infrared.

At finite temperature, $\tau \in [0, \beta]$ and the frequency ω is replaced by the discrete Matsubara frequency $p_n = \frac{2n\pi}{\beta}$. The introduction of a thermal medium breaks Lorentz invariance, and the photon polarization bubble no longer takes the simple form in Eqn. 58. We note that current conserva-

tion requires that $\Pi_{\mu\nu}$ be transverse, i.e.

$$P_\mu \Pi_{\mu\nu} = 0 \quad (62)$$

which allows us to write it in the following form

$$\Pi_{\mu\nu}(p_n, \mathbf{p}) = \frac{\mathbf{p}^2}{p^2} \Pi_{00}(p_n, \mathbf{p}) P_{\mu\nu}^L + \Pi_{yy}(p_n, \mathbf{p}) P_{\mu\nu}^T \quad (63)$$

where $P_{\mu\nu}^L$ and $P_{\mu\nu}^T$ are given by

$$P_{\mu\nu}^T \equiv \delta_{\mu i} \delta_{\nu j} \left(\delta_{ij} - \frac{P_i P_j}{\mathbf{p}^2} \right) \quad (64)$$

$$P_{\mu\nu}^L \equiv \left(\delta_{\mu\nu} - \frac{P_\mu P_\nu}{p^2} \right) - P_{\mu\nu}^T \quad (65)$$

and the full propagator is determined by the two components Π_{00} and Π_{yy} . The full propagator in Landau gauge ($\xi = 0$) can then be written in the form

$$D_{\mu\nu}(p_n, \mathbf{p}) = -\frac{1}{\frac{p^2}{e^2 N_f} - \frac{\mathbf{p}^2}{p^2} \Pi_{00}(p_n, \mathbf{p})} P_{\mu\nu}^L - \frac{1}{\frac{p^2}{e^2 N_f} - \Pi_{yy}(p_n, \mathbf{p})} P_{\mu\nu}^T \quad (66)$$

where $\Pi_{00}(p_n, \mathbf{p})$ and $\Pi_{yy}(p_n, \mathbf{p})$ must be computed numerically. At finite temperature, QED₃ exhibits many typical thermodynamic and dynamic properties of a quantum critical theory. We will study some of these dynamic properties in Chapter 3. If we also consider theories which also contain N_b bosons that also interact with the $U(1)$ gauge field, we can obtain for different values of the ratio $\frac{N_b}{N_f}$ ¹¹⁵.

0.6 HOLOGRAPHIC QUANTUM MATTER: INSIGHTS FROM BLACK HOLES

The AdS/CFT correspondence as it was originally proposed, is a duality between a theory of quantum gravity in anti-de-Sitter (AdS) space and a class of strongly coupled quantum field theories living on the boundary¹⁴³. For quantum field theories containing rank N matrix fields, the gravity dual behaves classically in the limit $N \rightarrow \infty$. The radial dimension of behaves like an energy scale and movement from the boundary into the interior of the bulk can be interpreted as a renormalization group flow between ultraviolet and infrared fixed points.

In many ways, the AdS/CFT correspondence appears to be quite far removed from condensed matter physics. On the microscopic lattice level condensed matter systems have many broken symmetries which differentiate them from the large N supersymmetric $SU(N)$ gauge theories originally studied in the correspondence⁷. However, while the microscopic nature of these gauge theories is obviously very different from the microscopic description of a metal, there are several compelling reasons to expect that the emergent macroscopic phenomena may be qualitatively similar¹³⁴. In practice, the AdS/CFT correspondence allows us to study strongly coupled quantum matter in regimes where perturbative methods are known to fail. So while there are many valid reasons to be sceptical, there are equally many reasons to take a further look into this new direction, also known as AdS/CMT.

Perhaps the most promising application of AdS/CMT is the study of the universal properties of the anomalous “strange metal” phase of high temperature superconductors, occurring at temperatures above T_c near optimal doping in the cuprates, pnictides and other related compounds. It is thought that the presence of the strange metal phase is a manifestation of a zero temperature quantum critical point underneath the superconducting dome at optimal doping¹⁷⁹. The transport properties of the strange metal strongly deviate from the predictions of Fermi liquid theory and defy any long-lived quasiparticle description as seen in spectral probes, implying that local equilibration

occurs on timescale τ_ϕ that saturates the bound proposed in Eqn. 6¹⁷⁷. Remarkably, fast relaxation is a property shared by strange metals and black holes,^{98,211} whose thermodynamic and dissipative behavior appears even at the classical level^{17,18}. The holographic approach has shed light on several more surprising properties shared by strongly coupled quantum systems and classical black hole horizons and led to the development of several useful tools to study transport in condensed matter systems^{87,230,92}. AdS/CMT has also led to a resurgence in the study of hydrodynamic transport beyond the setting of the correspondence^{137,141,140}.

To apply the AdS/CFT correspondence to the study of condensed matter, one must know how to relate observables in the boundary quantum field theory to the bulk gravitational theory. In the large N limit when the gravitation theory becomes classical, there exists a set of operators $\mathcal{O}_i$ in the quantum field theory which exhibit no fluctuations to leading order in large N . These are called single trace operators and there is a one to one mapping identifying each $\mathcal{O}_i$ to a dynamical field ϕ_i in the bulk gravitational theory. This mapping is often called the AdS/CFT “dictionary”. The most important entries for our purposes are summarized in Table 1. The dynamics of the bulk fields ϕ_i are governed by the action of the gravitational bulk theory. One can identify this action by either a “top-down” or “bottom-up” approach²³⁰. In a top-down approach, one starts with the full string theory description of the bulk and performs a consistent truncation of this theory to lead to a particular AdS action with essentially no free parameters. In a bottom up approach, one instead constructs a gravitational action using a combination of symmetry principles and consistency requirements of general relativity. This construction is similar in spirit to Landau-Ginzburg theory in that the structure of the interactions are constrained by emergent symmetries, but the coupling constants themselves depend on the unspecified microscopic origin of the theory. It has been explicitly shown in most cases of interest that the actions obtained in the bottom-up construction match the truncated action obtained from the top down construction, so it is unnecessary to start directly from the complete string theory to study universal phenomena and scaling properties. One can sim-

AdS/CFT Dictionary	
Boundary: field theory	Bulk: gravity
Partition function	Partition function
Order parameter/operator (scalar or fermion)	Bulk field (scalar or Dirac)
Source of operator	Boundary value of field
Conformal dimension of operator	Mass of field
Spin/charge of operator	Spin/charge of bulk field
Global spacetime symmetry	Local isometry
Global internal symmetry	Local gauge symmetry
Energy momentum tensor $T^{\mu\nu}$	Metric field $g_{\mu\nu}$
Global internal symmetry current	Gauge field
Finite temperature	Hawking temperature or radius of Euclidean time circle
Charge density/chemical potential	Boundary values of the electrostatic potential $\mathcal{A}_t$
Free energy	On-shell value of the action
Entropy	Area of black-hole horizon

Table 1: The AdS/CFT dictionary adapted from²³⁰.

ply construct the bulk action by identifying each relevant quantity in the field theory and translate it into a bulk action using the AdS/CFT dictionary. Once we have obtain a bulk action, the GPKW rule, proposed independently by Gubser, Klebanov, and Polyakov in ⁸³ and Witten in ²²² provides a means of obtaining Green's functions in the quantum field theory by conjecturing an equivalence between the partitions functions of the quantum field theory and the effective gravitational theory evaluated at the boundary of spacetime. Formally, it identifies the partition function of a quantum field theory in the presence of source b with the partition function of a bulk gravitational theory, where the asymptotically anti-de Sitter boundary value of the fields φ are equated with the sources b i.e.

$$\left\langle e^{i \sum_i \int d^D x b(x) \mathcal{O}(x)} \right\rangle = \int \mathcal{D}\varphi e^{i S_{bulk}[\varphi(x, r=0) \rightarrow b(x)]} \quad (67)$$

where we have chosen the boundary to be located at $r = 0$. We can use the GKPW rule to calculate correlation functions in the quantum field theory by evaluating the on-shell bulk action with the boundary values of the field equal to the source in the quantum field theory. We start with the bulk action of the metric for pure AdS with radius L

$$S[g] = \frac{1}{2\kappa^2} \int d^{d+2}x \sqrt{-g} \left(R + \frac{d(d+1)}{L^2} \right) \quad (68)$$

where $\kappa = \sqrt{8\pi G_N}$ is the surface gravity. Placing the boundary at $r = 0$, the simplest solution to the Einstein equation is

$$ds^2 = L^2 \left(\frac{-dt^2 + dr^2 + d\mathbf{x}^2}{r^2} \right) \quad (69)$$

Ignoring any back-reaction of the metric, we consider the following bulk action of a massive free

scalar field φ

$$S[\varphi] = - \int d^{d+2}x \sqrt{g} \left(\frac{1}{2} (\nabla \varphi)^2 + \frac{m^2}{2} \varphi^2 \right) \quad (70)$$

The equation of motion for φ in the bulk is

$$\nabla^\mu \nabla_\mu \varphi = m^2 \varphi \quad (71)$$

where ∇_μ is the covariant derivative in AdS_{d+2} . The bulk action can be divided into a “bulk” and “boundary” contribution. The bulk contribution will vanish on the equations of motion, but the boundary contribution will give a nonzero contribution. To evaluate it, we can write the solution for φ in a series expansion for $r \rightarrow 0$

$$\varphi(r \rightarrow 0) = \frac{\varphi^{(0)}}{L^{\frac{d}{2}}} r^{d+1-\Delta} + \dots + \frac{\varphi^{(1)}}{L^{\frac{d}{2}}} r^\Delta + \dots \quad (72)$$

$$\Delta(\Delta - d - 1) = mL^2 \quad (73)$$

We can identify $\varphi^{(0)}$, the coefficient of the leading term of the expansion in Eqn. 72 as the boundary value of the field, i.e. the “source” h in the field theory. Since h couples linearly to the field theory operator $\mathcal{O}$, we find from rescaling symmetry that Δ is the scaling dimension of $\mathcal{O}$. After suitable regularization of UV divergences, we find the expectation value of $\mathcal{O}$ in the presence of source h is given by

$$\langle \mathcal{O} \rangle_h = i \frac{\delta}{\delta h} S_{bulk} [\varphi(x, r=0) \rightarrow h(x)] = \frac{2\Delta - d - 1}{2L} \varphi^{(1)} \quad (74)$$

where we can identify $\varphi^{(1)}$, the coefficient of the subleading term of the expansion in Eqn. 72, as the “response”. The retarded Green’s function can be written as a ratio of the response and source and

takes the form

$$G^R(\omega, \mathbf{k}) = \frac{\langle \mathcal{O} \rangle_b}{b} = \frac{2\Delta - d - 1}{2L} \frac{\varphi^{(1)}}{\varphi^{(0)}} \sim (k^2 - \omega^2)^{\Delta - (d+1)/2} \quad (75)$$

Eqn. 75 takes the form we expect for conformal field theory. Higher order correlation functions can be computed using a technique called “Witten” diagrams²²². Finally, we note that some quantities such as *DC* conductivities, can be computed in an effective low energy description of the system that does not refer to the boundary degrees of freedom at all. In a black hole, the low energy behavior is determined by the near-horizon dynamics. In many situations, for a uniform boundary source (corresponding to b), we find that the resulting bulk current (corresponding to $\langle \mathcal{O} \rangle_b$) is radially conserved. In a black hole geometry, this current can be evaluated at the horizon rather than the boundary at $r = 0$ allowing it to be expressed completely in terms of “horizon data”^{102,53}. In Chapter 4, we will use this prescription to calculate charge diffusion coefficients in with holographic matter containing long-wavelength disorder.

The simplest model that can describe quantum matter for generic temperature and chemical potential is Einstein-Maxwell theory describing coupling of the metric g to a $U(1)$ gauge field A

$$S = \int d^{d+2}x \sqrt{-g} \left[\frac{1}{2\kappa^2} \left(R + \frac{d(d+1)}{L^2} \right) - \frac{1}{4e^2} F^2 \right] \quad (76)$$

The solution to the equations of motion is called the Reissner-Nördstrom black brane. A black brane is analogous to a black hole in flat space, but with a metric is asymptotically AdS far away

from the horizon. The metric and gauge potential A_t of the black brane are given by

$$ds^2 = \frac{L^2}{r^2} \left(-f(r)dt^2 + \frac{dr^2}{f(r)} + d\mathbf{x}^2 \right) \quad (77)$$

$$A_t = \mu \left(1 - \left(\frac{r}{r_+} \right)^{d-1} \right) \quad (78)$$

$$f(r) = 1 + Q^2 r^{2d} - Mr^{d+1} \quad (79)$$

$f(r)$ is called an emblackening factor with horizon located at $f(r_+) = 0$. The total mass M and total electric charge Q are given by

$$Q = \frac{\mu}{r_+^{d-1}\gamma} \quad (80)$$

$$M = r_+^{-d-1} + Q^2 r_+^{d-1} \quad (81)$$

where γ is defined as

$$\gamma = \sqrt{\frac{d}{d-1}} \frac{eL}{\kappa} \quad (82)$$

Note that $A_t(0) = \mu$ as expected from the AdS/CFT dictionary. A_t vanishes at the horizon r_+ , which is required for non-singular behavior. The Hawking temperature and entropy density of the black brane are

$$T = \frac{|f'(r_+)|}{4\pi} = \frac{1}{4\pi r_+} \left(d+1 - \frac{(d-1)\mu^2 r_+^2}{\gamma^2} \right) \quad (83)$$

$$s = \frac{2\pi L^d}{\kappa^2 r_+^d} \quad (84)$$

At low temperatures we find that $s \sim \mu^d$, meaning that the entropy is extensive at $T = 0$ in violation of the third law of thermodynamics. Defining r_* as the zero temperature horizon location, at low

temperatures, we can perturb r_+ about r_* as follows

$$r_+ = r_* \left(1 - \frac{2\pi T}{\mu^2 r_*} \right) \equiv r_* (1 - \delta) \quad (85)$$

Defining the new coordinate ζ as

$$\zeta = \frac{1}{r_* (1 - \delta(d-1)/d) - r} \frac{r_*^2}{d(d+1)} \quad (86)$$

we find that the near-horizon metric takes the form

$$ds^2 \approx \frac{L_2}{\zeta^2} \left(-f_2(\zeta) dt^2 + \frac{d\zeta^2}{f_2(\zeta)} + \frac{L^2}{r_*^2} d\mathbf{x}^2 \right) \quad (87)$$

$$f_2(\zeta) = 1 - \frac{\zeta^2}{\zeta_+^2} \quad (88)$$

$$\zeta_+ \equiv \frac{1}{2\pi T} \quad (89)$$

At low temperatures, this metric is $\mathbb{R}^d$ times an AdS_2 -Schwarzschild black brane. At low temperatures and energies i.e. $\omega, T \ll \mu$, the physics of the boundary quantum theory is controlled by the near horizon geometry of the black brane. At $T = 0$, the near horizon geometry is simply $\text{AdS}_2 \times \mathbb{R}^d$, which is invariant under a new emergent symmetry: reparameterizations in time. This implies that the dynamical critical exponent $z \rightarrow \infty$, a property called local criticality. Such behavior was first observed in marginal Fermi liquids²⁰⁹. Nearly AdS_2 geometries were also found to match the low energy effective action of reparameterization modes in SYK models with Majorana or complex fermions^{121,146,106,49,174}. In the complex SYK model, the ground state entropy per charge exactly matches the Bekenstein-Hawking entropy of the Reissner-Nordstrom black brane¹⁷⁸. With some slight modifications of the action in Eqn. 76 one can access a full zoo of holographic phases with parallels to Fermi liquids, strange metals, and superconductors^{135,47,89}. These states share much in

common with their solid state counterparts, the extent of which is still an active area of research.

0.7 ORGANIZATION OF THESIS

In this thesis, we study dynamics in four strongly interacting many-body systems near quantum critical points in the context of NMR relaxation, quantum quenches, many-body quantum chaos, and charge diffusion

The rest of this thesis explores dynamical aspects of quantum criticality in the four distinct quantum many-body states discussed in the previous sections. In Chapter 1 we revisit the issue of NMR relaxation in the quantum paramagnet phase of an Ising spin chain and resolve a discrepancy in the determination of the TFIC spin gap using data obtained from different experimental probes. We examine the low frequency spin susceptibility of the paramagnetic phase of the quantum Ising chain in transverse field at temperatures well below the energy gap. We find that the imaginary part is dominated by rare quantum processes in which the number of quasiparticles changes by an odd number. We obtain exact results for the NMR relaxation rate in the low temperature limit for the integrable model with nearest-neighbor Ising interactions, and derive exact universal scaling results applicable to generic Ising chains near the quantum critical point. We find that the NMR relaxation rate near the quantum critical point is $1/T_1 \sim \exp(-\frac{2\Delta}{T})$ rather than the previously conjectured $1/T_1 \sim \exp(-\frac{\Delta}{T})$. These results resolve certain discrepancies between the energy scales measured with different experimental probes in the quantum disordered paramagnetic phase of the Ising chain system CoNb_2O_6 .

In Chapter 2 we study the non-equilibrium dynamics of the Sachdev-Ye-Kitaev (SYK) model after a quantum quench. Our quench protocol drives the system from an initial state with a quasiparticle description to a final state lacking any long-lived quasiparticle excitations. The Kadanoff-Baym equations show that the final state is thermal, and their numerical analysis is consistent with

a thermalization rate proportional to the absolute temperature of the final state. We also obtain an exact analytic solution of the quench dynamics in the large q limit of a model with q fermion interactions: in this limit, the thermalization of the fermion Green's function is instantaneous.

In Chapter 3, we study the real time dynamics of quantum electrodynamics in $2+1$ dimensions with N_f flavors of fermions to leading order in a $1/N_f$ expansion. We focus on thermalization and the onset of many-body quantum chaos which can be diagnosed from the growth of initially anti-commuting fermion field operators. We compute such anti-commutators in this gauge theory to leading order in $1/N_f$. We find that the anti-commutator grows exponentially in time and compute the quantum Lyapunov exponent. We briefly comment on chaos, locality, and gauge invariance.

Finally, in Chapter 4 we consider whether the butterfly velocity - a speed at which quantum information propagates - may provide a fundamental bound on diffusion constants in dirty incoherent metals. We analytically compute the charge diffusion constant and the butterfly velocity in charge-neutral holographic matter with long wavelength “hydrodynamic” disorder in a single spatial direction. In this limit, we find that the butterfly velocity does not set a sharp lower bound for the charge diffusion constant.

1

NMR Relaxation Rates in Ising Spin Chains

1.1 INTRODUCTION

The transverse field Ising chain is an ideal setting to study the dynamics of quantum criticality¹⁷⁷ as many observable properties can be computed either exactly, or reliably in a semiclassical ap-

proach^{181,61,101}. In recent years, there have been several experimental realizations of the transverse Ising chain that make theoretical predictions testable^{45,20}. For instance, it has been found that CoNb_2O_6 is for many purposes an almost ideal realization of the one-dimensional ferromagnetic Ising chain. Experiments have studied properties of the transverse Ising chain across the different regimes of the phase diagram as a function of transverse field and temperature^{45,35,119,152,131,170}. In this chapter, we revisit the issue of the nuclear magnetic resonance (NMR) relaxation in an Ising spin chain in its gapped state without ferromagnetic order (the ‘quantum disordered’ regime of Fig. 1.1). A recent NMR experiment on CoNb_2O_6 ¹¹⁹ studied the NMR relaxation rate, $1/T_1$, in all three regimes near the quantum critical point of the phase diagram in Fig. 1.1. Experimental results agreed quantitatively with theoretical predictions in the ‘renormalized classical’ and ‘quantum critical’ regimes. While there were no firm theoretical predictions in the quantum disordered regime, it was conjectured¹¹⁹ that the low temperature (T) behavior was $1/T_1 \sim \exp(-\Delta/T)$, where Δ is the energy gap to single spin flips. However, other experiments probing the large field transverse paramagnetic regime show discrepancies with the energy scales probed by NMR. The NMR experiments¹¹⁹ measured an activation energy that was approximately two times larger than the gap inferred from heat capacity¹³¹, neutron scattering¹⁷⁰, and THz/infrared experiments²¹⁰.

Here we will show that near the critical point the behavior of the NMR relaxation rate is in fact $1/T_1 \sim \exp(-2\Delta/T)$, and compute the precise prefactor for the integrable nearest-neighbor Hamiltonian. We find that the result is compatible with the universal relativistic quantum field theory, and obtain the universal behavior of $1/T_1$ at $T \ll \Delta$ for a generic Ising Hamiltonian. We begin by recalling some exact results on the nearest-neighbor Ising chain in Section 2.4.4. In particular, the lattice form factors computed in Ref.¹⁰¹ will be crucial ingredients in our results. The computation of the NMR relaxation rate of the nearest-neighbor Ising chain appears in Section 1.4. Section 1.5 describes the universal behavior of the NMR relaxation rate across the quantum critical point. The experimental situation is discussed in Section 1.6.

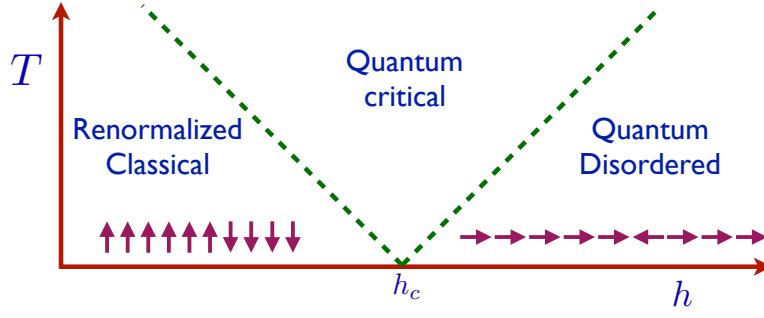


Figure 1.1: Crossover phase diagram of the Ising chain in a transverse field, h . There is a quantum critical point at $T = 0$ and $h = h_c$ (for the Hamiltonian in Eq. (1.1), $h_c = 1$) between a ferromagnetic ($h < h_c$) which breaks the Ising symmetry, and a paramagnetic phase. This chapter focuses on the “quantum disordered” regime above the paramagnetic phase for $h > h_c$. The other regimes were described in Ref. ¹¹⁹.

1.2 EXACT SPECTRUM OF THE QUANTUM ISING CHAIN

We work with the integrable Ising chain Hamiltonian

$$H = -J \sum_{\ell=1}^L [\sigma_{\ell}^z \sigma_{\ell+1}^z + h \sigma_{\ell}^x] \quad (1.1)$$

where σ_{ℓ}^{α} with $\alpha = x, y, z$ are Pauli matrices acting on the 2-state spins on site ℓ . For $h < 1$, this model has a ferromagnetic ground state with $\langle \sigma_{\ell}^z \rangle = N_0 \neq 0$. We are interested in the low T behavior in the paramagnetic state for $h > 1$, where $\langle \sigma_{\ell}^z \rangle = 0$ at $T = 0$.

The spectrum of H can be computed exactly by a Jordan-Wigner transformation which maps it onto a theory of spinless fermions. Defining $\sigma_{\ell}^{\pm} = \frac{1}{2}(-\sigma_{\ell}^z \pm i\sigma_{\ell}^y)$, we can define spinless fermion creation and annihilation operators by

$$c_{\ell} = \left(\prod_{\ell' < \ell} \sigma_{\ell'}^x \right) \sigma_{\ell}^+ \quad (1.2)$$

$$c_{\ell}^{\dagger} = c_i \left(\prod_{\ell' < \ell} \sigma_{\ell'}^x \right) \sigma_{\ell}^- \quad (1.3)$$

The spin operators σ_ℓ^x and σ_ℓ^z can be written in terms of the fermion operators as

$$\sigma_\ell^x = 1 - 2c_\ell^\dagger c_\ell \quad (1.4)$$

$$\sigma_\ell^z = - \prod_{\ell' < \ell} (1 - 2c_{\ell'}^\dagger c_{\ell'}) (c_\ell + c_\ell^\dagger) \quad (1.5)$$

where $\{c_\ell^\dagger, c_{\ell'}^\dagger\} = \{c_\ell, c_{\ell'}\} = 0$ and $\{c_\ell, c_{\ell'}^\dagger\}$. This allows us to write the Hamiltonian in terms of the fermion operators as follows

$$H = -J \sum_{\ell}^L ((c_\ell^\dagger - c_\ell)(c_{\ell+1} + c_{\ell+1}^\dagger) - 2bc_\ell^\dagger c_\ell + b) \quad (1.6)$$

In general we must treat the cases of even and odd fermions separately as they will impose different boundary conditions. More details are included in ³⁷. Here, we work in the infinite lattice limit $L \rightarrow \infty$ so we can neglect boundary conditions.

Setting the lattice spacing equal to one, we can write the fermions in Fourier space as

$$c_p = \frac{1}{\sqrt{L}} \sum_{\ell} c_\ell e^{ip\ell} \quad (1.7)$$

The Hamiltonian becomes

$$H_I = J \sum_p (2(g - \cos p)c_p^\dagger c_p + i \sin p (c_{-p}^\dagger c_p^\dagger + c_{-p} c_p - g)) \quad (1.8)$$

Since this Hamiltonian is quadratic in the fermion operators, we can diagonalize it via the following Bogoliubov transformation on $c_p, c_{-p}^\dagger$

$$\gamma_p = u_p c_p - i v_p c_{-p}^\dagger \quad (1.9)$$

where $u_p^2 + v_p^2 = 1$, $u_{-p} = u_p$, $v_{-p} = -v_p$ and the γ_p 's satisfy the same fermion anticommutation relations as the c_p 's. We write $u_p = \cos \frac{\theta_p}{2}$ and $v_p = \sin \frac{\theta_p}{2}$ to ensure the above conditions hold.

Demanding conservation of the γ fermions i.e. all $\gamma\gamma$ and $\gamma^\dagger\gamma^\dagger$ terms vanish, we find

$$\tan \theta_p = \frac{\sin p}{g - \cos p} \quad (1.10)$$

The final form of the Hamiltonian is

$$H_I = \sum_p \varepsilon_p (\gamma_p^\dagger \gamma_p - \frac{1}{2}) \quad (1.11)$$

with single particle dispersion

$$\varepsilon_p = 2J\sqrt{1 + b^2 - 2b\cos(p)} \quad (1.12)$$

as a function of crystal momentum $-\pi < p < \pi$. This dispersion implies an energy gap

$$\Delta = 2J|b - 1|. \quad (1.13)$$

The complete set of excited states are described by n fermion states $|p_1, p_2, \dots, p_n\rangle$, where all fermion momenta must be unequal.

1.3 FORM FACTORS

Remarkably, all matrix elements of the ferromagnetic order parameter, σ_ℓ^z , between all many-body states have been computed exactly by Iorgov *et al.*¹⁰¹. For our purposes, we need the matrix ele-

ments in the limit $L \rightarrow \infty$, which can be written as

$$\begin{aligned} \langle q_1, \dots, q_{2n} | \sigma_\ell^z | p_1, \dots, p_m \rangle &= \frac{(4f^2 b)^{(m-2n)^2/4}}{L^{n+m/2}} |1 - b^2|^{1/8} i^{\lfloor n+m/2 \rfloor} e^{-i\ell [\sum_{j=1}^{2n} - \sum_{l=1}^m p_l]} \\ &\times \prod_{j=1}^{2n} \frac{1}{\sqrt{\varepsilon_{q_j}}} \prod_{l=1}^m \frac{1}{\sqrt{\varepsilon_{p_l}}} \prod_{j < j'=1}^{2n} \frac{2 \sin(q_j - q_{j'})}{\varepsilon_{q_j} + \varepsilon_{q_{j'}}} \prod_{l < l'=1}^m \frac{2 \sin(p_l - p_{l'})}{\varepsilon_{p_l} + \varepsilon_{p_{l'}}} \prod_{j=1}^{2n} \prod_{l=1}^m \frac{\varepsilon_{q_j} + \varepsilon_{p_l}}{2 \sin(q_j - p_l)} \end{aligned} \quad (1.14)$$

where m is even (odd) for $b < 1$ ($b > 1$). We will be able to compute the NMR relaxation rate in the $T \rightarrow 0$ limit for $b > 1$ by a direct application of Eq. (1.14) in the Lehmann spectral representation.

1.4 NMR RELAXATION RATE

The NMR relaxation rate is determined by the low frequency behavior of local spin susceptibility.

We define the imaginary time (τ) susceptibility by

$$\chi(\tau) = \int_0^{1/T} d\tau \langle \sigma_0^z(\tau) \sigma_0^z(0) \rangle. \quad (1.15)$$

After a Fourier transform and analytic continuation to real frequencies (ω), we obtain the NMR relaxation rate from ¹¹⁹

$$\frac{1}{T_1} = \lim_{\omega \rightarrow 0} \frac{2T}{\omega} |a_{bf}|^2 \text{Im} \chi(\omega), \quad (1.16)$$

where a_{bf} is the hyperfine coupling between the nuclei and the Ising spins.

1.4.1 LOW-TEMPERATURE EXPANSION

We are interested in the retarded two-point order parameter autocorrelator

$$\chi(\omega) = \int_0^{1/T} d\tau e^{i\bar{\omega}\tau} \frac{1}{\mathcal{Z}} \text{Tr} \left[e^{-H/T} \sigma_j^z(\tau) \sigma_j^z \right] \Big|_{\bar{\omega} \rightarrow \eta - i\omega}, \quad (1.17)$$

where $\mathcal{Z}$ is the partition function. The idea of Ref.⁶¹ is to develop a linked cluster expansion for this quantity. The starting point is the Lehmann representation

$$\chi(\omega) = \frac{1}{\mathcal{Z}} \sum_{n,m=0}^{\infty} C_{n,m}(\omega), \quad (1.18)$$

where

$$C_{n,m}(\omega) = \frac{1}{n!} \sum_{\{k_1, \dots, k_n\}} \frac{1}{m!} \sum_{\{p_1, \dots, p_m\}} |\langle k_1 \dots k_n | \sigma_0^z | p_1 \dots p_m \rangle|^2 \frac{e^{-E(\{p_i\})/T} - e^{-E(\{k_j\})/T}}{\omega + i\eta - E(\{p_i\}) + E(\{k_j\})} \quad (1.19)$$

Here η is a positive infinitesimal. The expansion of the partition function reads

$$\mathcal{Z} = 1 + \sum_{p \in \mathbb{R}} e^{-\varepsilon_p/T} + \sum_{p_1 < p_2 \in \text{NS}} e^{-[\varepsilon_{p_1} + \varepsilon_{p_2}]/T} + \dots \equiv \sum_{n=0}^{\infty} \mathcal{Z}_n. \quad (1.20)$$

By construction the contribution $\mathcal{Z}_n$ scales with system size as L^n . Here the subscripts refer to Ramond and Neveu-Schwartz boundary conditions, which will not matter in the infinite L limit we take. The individual terms $C_{n,m}(\omega)$ in the expansion (1.18) diverge with the system size L because the matrix elements (1.14) become singular when $k_r \rightarrow p_s$. Ref.⁶¹ re-casts the expansion in terms of *linked clusters*, which are finite in the thermodynamic limit. The linked clusters relevant for a low-temperature expansion of $\chi(\omega)$ are

$$\begin{aligned} \mathbf{C}_{2n+1,0}(\omega) &= C_{2n+1,0}(\omega), & \mathbf{C}_{0,2n+1}(\omega) &= C_{0,2n+1}(\omega), \\ \mathbf{C}_{1,2n}(\omega) &= C_{1,2n}(\omega) - \mathcal{Z}_1 C_{0,2n-1}(\omega), \\ \mathbf{C}_{2,2n+1}(\omega) &= C_{2,2n+1}(\omega) - \mathcal{Z}_1 C_{1,2n}(\omega) - (\mathcal{Z}_2 - \mathcal{Z}_1^2) C_{0,2n-1}(\omega), \\ \mathbf{C}_{3,2n}(\omega) &= C_{3,2n}(\omega) - \mathcal{Z}_1 C_{2,2n-1}(\omega) - (\mathcal{Z}_2 - \mathcal{Z}_1)^2 C_{1,2n-2}(\omega). \end{aligned} \quad (1.21)$$

In terms of the linked clusters we have

$$\text{Im } \chi(\omega) = \sum_{n,m=0}^{\infty} \text{Im } \mathbf{C}_{n,m}(\omega) . \quad (1.22)$$

The leading terms at low temperatures and $\omega \approx 0$ are

$$\begin{aligned} \mathcal{C}_1(\omega) &= \mathbf{C}_{1,2}(\omega) + \mathbf{C}_{2,1}(\omega) , \\ \mathcal{C}_2(\omega) &= \mathbf{C}_{2,3}(\omega) + \mathbf{C}_{3,2}(\omega) , \\ \mathcal{C}_3(\omega) &= \mathbf{C}_{1,4}(\omega) + \mathbf{C}_{4,1}(\omega) + \mathbf{C}_{3,4}(\omega) + \mathbf{C}_{4,3}(\omega) . \end{aligned} \quad (1.23)$$

As $\varepsilon_k > \Delta = 2J|b-1|$ the formal temperature dependence of these terms at $T \ll \Delta$ is

$$\mathcal{C}_n(\omega) = \mathcal{O}(e^{-(n+1)\Delta/T}) . \quad (1.24)$$

As we are interested in the NMR relaxation rate we focus on the quantities

$$c_n(T) = \lim_{\omega \rightarrow 0} \frac{\text{Im}(\mathcal{C}_n(\omega))}{\omega} . \quad (1.25)$$

LEADING TERM

We begin with some qualitative considerations on the physical processes which lead to the dominant contributions to Eq. (1.16) as $T \rightarrow 0$ for $b > 1$. A thermal excitation with energy E_i will appear with a probability $e^{-E_i/T}$ as an initial state in the relaxation process. We should focus on the states with the lowest possible E_i . Because of the $\omega \rightarrow 0$ limit, the final states will also have an energy $E_f = E_i$ and are reached by the action of the σ_0^z operator. We notice from Eq. (1.14) that for $b > 1$ the matrix element is non-zero only between states with distinct parities in the number of fermions. Therefore

the initial and final states must differ by an odd number of fermions which places strong constraints on the ranges of allowed values of E_i and E_f .

We first consider the process $1 \rightarrow 2$, from an initial state with one fermion to a final state with 2 fermions. This process (and its inverse) will dominate as $T \rightarrow 0$. The single fermion excitations are in the energy range $(\varepsilon_{\min}, \varepsilon_{\max}) \equiv 2J(b-1, b+1)$. In the most optimal conditions, both fermions in the 2 particle state will have energy close to $\varepsilon_{\min}$. So for a $1 \rightarrow 2$ particle process to be allowed, we need $\varepsilon_{\max} > 2\varepsilon_{\min}$ or $b < 3$. This process will have probability $\exp(-2\varepsilon_{\min}/T)$.

For $b > 3$ we need to consider processes with larger numbers of fermions to obtain the leading contribution. In general, the $n \rightarrow m$ process is allowed for $b < (m+n)(m-n)$, where $m > n$ and $m-n$ is odd. Such a process occurs with probability $\exp(-m\varepsilon_{\min}/T)$. Thus for $3 < b < 5$ the most probable process is $2 \rightarrow 3$ with probability $\exp(-3\varepsilon_{\min}/T)$. There are also processes at smaller b , such as $1 \rightarrow 4$ for $b < 5/3$ with probability $\exp(-4\varepsilon_{\min}/T)$.

We now consider the $1 \rightarrow 2$ process which has a prefactor of $\exp(-2\Delta/T)$. Importantly for $\omega \approx 0$ and in the limit $\eta \rightarrow 0$ we have

$$\lim_{\eta \rightarrow 0} [\text{Im} C_1(\omega)] = \lim_{\eta \rightarrow 0} \text{Im} [C_{1,2}(\omega) + C_{2,1}(\omega)] \quad (1.26)$$

i.e. the “disconnected” contributions $\mathcal{Z}_1 C_{01}(\omega)$ and $\mathcal{Z}_1 C_{10}(\omega)$ vanish in the limit $\eta \rightarrow 0$. This is related to the fact that the kinematic poles in the form factors do not contribute to the momentum sums by virtue of the energy-conservation delta function. Using the explicit expression for the form factors (1.14) and turning momentum sums into integrals in the $L \rightarrow \infty$ limit we have

$$\begin{aligned} \text{Im}(C_1(\omega)) &= \frac{J\pi}{4} [b^2(b^2-1)]^{\frac{1}{4}} \int_{-\pi}^{\pi} \frac{dp_1 dp_2 dq}{(2\pi)^3} [\delta(\omega + \varepsilon_{p_1} + \varepsilon_{p_2} - \varepsilon_q) - \delta(\omega - \varepsilon_{p_1} - \varepsilon_{p_2} + \varepsilon_q)] \\ &\times \frac{(\varepsilon_{p_1} + \varepsilon_q)^2 (\varepsilon_{p_2} + \varepsilon_q)^2 \sin^2\left(\frac{p_1 - p_2}{2}\right)}{\varepsilon_{p_1} \varepsilon_{p_2} \varepsilon_q (\varepsilon_{p_1} + \varepsilon_{p_2})^2 \sin^2\left(\frac{p_1 - q}{2}\right) \sin^2\left(\frac{p_2 - q}{2}\right)} \left[e^{-(\varepsilon_{p_1} + \varepsilon_{p_2})/T} - e^{-\varepsilon_q/T} \right] \end{aligned} \quad (1.27)$$

Carrying out one of the integrals using the energy conservation delta-function we obtain

$$c_1(T) = \frac{J}{2T} [b^2(b^2 - 1)]^{\frac{1}{4}} \int_{-\pi}^{\pi} \frac{dp_1 dp_2}{(2\pi)^2} \Theta_H(2J(b+1) - \varepsilon_{p_1} - \varepsilon_{p_2}) \frac{e^{-(\varepsilon_{p_1} + \varepsilon_{p_2})/T}}{|\varepsilon'_{p_0}|} \\ \times \frac{(2\varepsilon_{p_1} + \varepsilon_{p_2})^2 (\varepsilon_{p_1} + 2\varepsilon_{p_2})^2 \sin^2\left(\frac{p_1 - p_2}{2}\right)}{\varepsilon_{p_1} \varepsilon_{p_2} (\varepsilon_{p_1} + \varepsilon_{p_2})^3 \sin^2\left(\frac{p_1 - p_0}{2}\right) \sin^2\left(\frac{p_2 - p_0}{2}\right)}, \quad (1.28)$$

where

$$p_0 = \arccos \left[\frac{1 + b^2 - \left(\frac{\varepsilon_{p_1} + \varepsilon_{p_2}}{2J} \right)^2}{2b} \right]. \quad (1.29)$$

In the low-temperature limit $T \ll 2J(b-1)$ and $b < 3$ we can carry out the remaining two integrals as follows. The integration will be dominated by small $p_{1,2} \sim \sqrt{T}$. For these small p , we can expand the dispersion as

$$\varepsilon_p = \Delta + \frac{p^2}{2m} + \dots \quad (1.30)$$

where $\Delta = 2J(b-1)$ and $m = (b-1)/(2J)$. Expanding the rest of the integrand around $p_1 = p_2 = 0$ we obtain

$$c_1(T) = \frac{J}{T} e^{-2\Delta/T} \frac{81[b^2(b^2 - 1)]^{\frac{1}{4}}}{64|\varepsilon'_{q_0}| \Delta \sin^4(q_0/2)} \int_{-\pi}^{\pi} \frac{dp_1 dp_2}{(2\pi)^2} e^{-\frac{p_1^2 + p_2^2}{2mT}} (p_1 - p_2)^2, \\ = \frac{81m^2 T J [b^2(b^2 - 1)]^{\frac{1}{4}}}{64\pi \Delta \sin^4(q_0/2) |\varepsilon'_{q_0}|} e^{-2\Delta/T}, \quad (1.31)$$

where we have defined

$$q_0 = \arccos \left[\frac{1 + b^2 - 4(b-1)^2}{2b} \right]. \quad (1.32)$$

Using Eq. (1.16), we obtain our main result

$$\frac{1}{T_1} = |a_{bf}|^2 \left[\frac{81f m^2 [b^2(b^2 - 1)]^{1/4}}{32\pi\Delta |v| \sin^4(q_0/2)} \right] T^2 e^{-2\Delta/T} \quad , \quad 1 < b < 3 \quad , \quad T \ll \Delta. \quad (1.33)$$

Note that Eq. (1.33) does not require Δ to be much smaller than J .

SUBLEADING TERM

We now turn to the term of order $e^{-3\Delta/T}$. For $1 < b < 3$ this will be smaller than the $e^{-2\Delta/T}$ term computed in Sec. 1.4.1, while for $3 < b < 5$ this turns out to be the largest non-zero term.

We will evaluate

$$c_2(T) = \lim_{\omega \rightarrow 0} \frac{\text{Im}(\mathcal{C}_2(\omega))}{\omega}. \quad (1.34)$$

This can be cast in the form

$$c_2(T) = \lim_{\eta \rightarrow 0} [c_2(T, \eta) - \mathcal{Z}_1 c_1(T, \eta) - (\mathcal{Z}_2 - \mathcal{Z}_1^2) c_0(T, \eta)], \quad (1.35)$$

where we have defined

$$c_n(T, \eta) = \lim_{\omega \rightarrow 0} \text{Im} \left[\frac{C_{n,n+1}(\omega) + C_{n+1,n}(\omega)}{\omega} \right], \quad n = 0, 1, 2, \quad (1.36)$$

and $\mathcal{Z}_n$ are the contributions of n -particle states to the partition function

$$\mathcal{Z}_1 = \sum_{p \in \mathbb{R}} e^{-\varepsilon_p/T}, \quad \mathcal{Z}_2 = \sum_{p_1 < p_2 \in \text{NS}} e^{-[\varepsilon_{p_1} + \varepsilon_{p_2}]/T}. \quad (1.37)$$

The explicit expressions for $c_n(T, \eta)$ are

$$\begin{aligned}
c_0(T, \eta) &= \sum_q |\langle q | \sigma_0^z | 0 \rangle|^2 \left[1 - e^{-\varepsilon_q/T} \right] \frac{4\eta\varepsilon_q}{[\varepsilon_q^2 + \eta^2]^2}, \\
c_1(T, \eta) &= \sum_{p_1 < p_2} \sum_q |\langle p_1, p_2 | \sigma_0^z | q \rangle|^2 \left[e^{-\varepsilon_q/T} - e^{-(\varepsilon_{p_1} + \varepsilon_{p_2})/T} \right] \frac{4\eta(\varepsilon_{p_1} + \varepsilon_{p_2} - \varepsilon_q)}{[(\varepsilon_{p_1} + \varepsilon_{p_2} - \varepsilon_q)^2 + \eta^2]^2}, \\
c_2(T, \eta) &= \sum_{p_1 < p_2 < p_3} \sum_{q_1 < q_2} |\langle p_1, p_2, p_3 | \sigma_0^z | q_1, q_2 \rangle|^2 \left[e^{-(\varepsilon_{q_1} + \varepsilon_{q_2})/T} - e^{-(\varepsilon_{p_1} + \varepsilon_{p_2} + \varepsilon_{p_3})/T} \right] \\
&\quad \times \frac{4\eta(\varepsilon_{p_1} + \varepsilon_{p_2} + \varepsilon_{p_3} - \varepsilon_{q_1} - \varepsilon_{q_2})}{[(\varepsilon_{p_1} + \varepsilon_{p_2} + \varepsilon_{p_3} - \varepsilon_{q_1} - \varepsilon_{q_2})^2 + \eta^2]^2}, \tag{1.38}
\end{aligned}$$

where the form factors are given in Eqn (1.14). Note that $c_0(T, \eta) \rightarrow 0$ as $\eta \rightarrow 0$ because it is not possible to satisfy the energy conservation delta function. Also in this limit, $c_1(T, \eta) \rightarrow c_1(T)$ computed in Eq. (1.28). On the other hand $c_1(T)$ vanishes for $3 < b < 5$ which implies that in this range of magnetic fields no “disconnected” contributions arise in Eq. (1.35) and we simply have $c_2(T) = \lim_{\eta \rightarrow 0} c_2(T, \eta)$. It is then straightforward to turn sums into integrals and we examine some properties of the resulting expression for $c_2(T)$ in Appendix A.1. By contrast, for $1 < b < 3$ $c_2(T, \eta)$ diverges with system size and the disconnected contributions in (1.35) need to be taken into account in order to obtain a finite expression. In principle it is possible to obtain a multiple contour integral representation for $c_2(T)$, but here we confine ourselves to a numerical evaluation of the momentum sums. We proceed as follows:

1. We evaluate $c_n(T, \eta)$ for several values of η and system sizes up to $L = 256$. We require η to be sufficiently large so that the finite-size corrections are negligible for our largest system sizes. We find that $\eta \approx 0.1$ is an appropriate order of magnitude.
2. We numerically extrapolate our results to $\eta = 0$ using a third order polynomial.

A useful check on this procedure is obtained by carrying it out for $c_1(T)$ and comparing it to the numerically evaluated expression (1.28), which is the result in the thermodynamic limit at $\eta \rightarrow 0$.

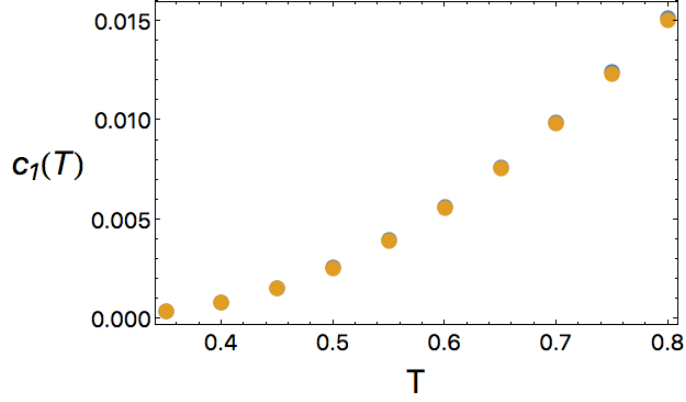


Figure 1.2: $c_1(T)$ for $h = 1.5$ and several temperatures. The thermodynamic limit result (blue dots) is seen to be in good agreement with the extrapolation of numerical results for $L = 192$ and $0.1 \leq \eta \leq 0.115$ (yellow dots).

Results for $c_2(T)$ for $h = 1.5$ are shown in Fig. 1.3, where we compare extrapolations for system sizes $L = 144$ and $L = 192$.

1.5 QUANTUM CRITICALITY

In this section we consider the approach to the quantum critical point at $h = 1$ in Fig. 1.1. It is useful to first review the analysis on the ferromagnetic side, $h < 1$, which was presented in Ref. ¹¹⁹. Then as $\Delta \ll J$, the $1/T_1$ rate obeys the universal scaling form

$$\frac{1}{T_1} = |a_{bf}|^2 \frac{Z}{T^{3/4}} \Phi_1(\Delta/T) \quad , \quad h < 1 \quad , \quad T, \Delta \ll J, \quad (1.39)$$

where Z is a non-universal constant, while Φ_1 is a universal function describing the crossovers between the quantum critical and renormalized classical regions. For the nearest-neighbor Ising model in Eq. (1.1) we take $Z = J^{-1/4}$ for our normalization of Φ_1 . While other microscopic models will have different values of Z , the function $\Phi_1(\Delta/T)$ is independent of the specific microscopic Hamil-

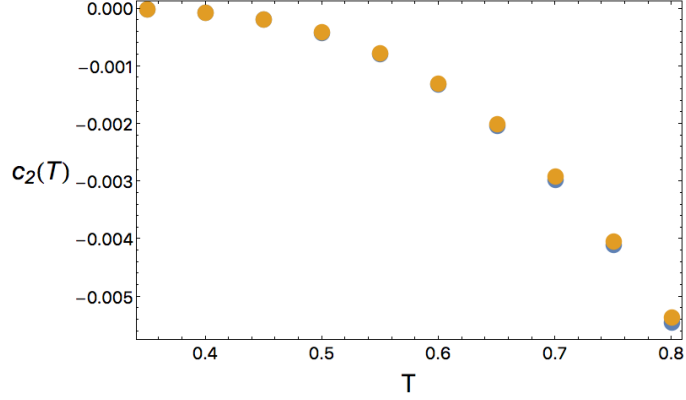


Figure 1.3: $c_2(T)$ for $b = 1.5$ and several temperatures. The results of extrapolating numerical results for $0.1 \leq \eta \leq 0.115$ and $L = 144$ are in good agreement with those for $L = 192$.

tonian. The limiting forms for Φ_1 in the two regimes are known exactly:

$$\Phi_1(\Delta/T) = \begin{cases} 2.1396\dots & , \quad \Delta \ll T \ll J \\ \pi(\Delta/T)^{1/4} e^{\Delta/T} & , \quad T \ll \Delta \ll J \end{cases} . \quad (1.40)$$

Furthermore, these theoretical predictions were found to be in good agreement with experimental observations¹¹⁹.

Now let us examine the paramagnetic phase $b > 1$. As in Eq. (1.39), the scaling form is

$$\frac{1}{T_1} = |a_{bf}|^2 \frac{Z}{T^{3/4}} \Phi_2(\Delta/T) \quad , \quad b > 1 \quad , \quad T, \Delta \ll J, \quad (1.41)$$

is expected to describe the crossovers in the NMR relaxation between the quantum disordered and quantum critical regimes. Matching with Eq. (1.39) in the quantum critical regime we have

$$\Phi_2(\Delta/T) = 2.1396\dots \quad , \quad \Delta \ll T \ll J. \quad (1.42)$$

For the form of Φ_2 in the quantum disordered regime, we examine only the leading term in Eq. (1.33) in the limit $T \ll \Delta \ll J$. In this limit, the fermion dispersion in Eq. (1.12) takes a relativistic form

$$\varepsilon_p = \sqrt{\Delta^2 + c^2 p^2} \quad (1.43)$$

with $c = 2J$. We evaluate the other parameters introduced above Eq. (1.33) for this dispersion and find

$$m = \Delta/c^2, \quad p_0 = \sqrt{3}\Delta/c, \quad v = \sqrt{3}c/2. \quad (1.44)$$

Finally, inserting in Eq. (1.33) we obtain

$$\frac{1}{T_1} = |a_{bf}|^2 \frac{3\sqrt{3}T^2}{2\pi J^{1/4} \Delta^{11/4}} e^{-2\Delta/T}, \quad T \ll \Delta \ll J, \quad b > 1 \quad (1.45)$$

This result is compatible with the scaling form in Eq. (1.41), and we have

$$\Phi_2(\Delta/T) = \frac{3\sqrt{3}}{2\pi} \left(\frac{T}{\Delta} \right)^{11/4} e^{-2\Delta/T}, \quad T \ll \Delta \ll J. \quad (1.46)$$

Note that the result in Eq. (1.46) applies to a generic ferromagnetic quantum Ising chain near its transverse field quantum critical point, and not just the nearest-neighbor model.

1.6 EXPERIMENTS

As mentioned above CoNb_2O_6 has been discovered to be an almost ideal realization of a 1D ferromagnetic Ising chain⁴⁵. It is quasi-1D material characterized by zig-zag chains of Co^{+2} ions with effective spin 1/2 moments. The spins lie in the ac plane at an angle of $\pm 31^\circ$ to the c -axis^{95,123} with the chains extending along the c -direction. A dominant ferromagnetic exchange between nearest-neighbor Co^{+2} ions along the c axis cause strong 1D ferromagnetic correlations to develop below

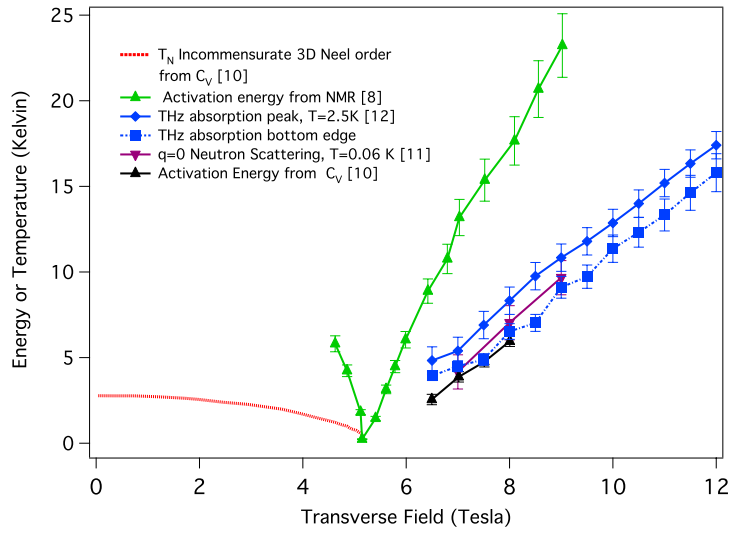


Figure 1.4: Phase diagram and energy scales of the Ising chain system CoNb_2O_6 as a function of transverse field. Energy scales from different experimental probes as well as the transition temperature of the 3D incommensurate Neel order are given. The transition is to a state with ferromagnetic chains that are ordered antiferromagnetically in the b . On the paramagnetic side of the transition, one can see an excellent agreement between various spectroscopic probes (THz absorption²¹⁰ and inelastic neutron scattering¹⁷⁰ and heat capacity¹³¹). In contrast one can see clearly that the gap extracted from the temperature dependence of the NMR spin relaxation time is approximately twice as large¹¹⁹.

~ 2.5 K⁸⁶. At zero transverse field, weak AF inter-chain exchange interactions stabilize an incommensurate spin-density waves at 2.95 K along the b -direction with a temperature-dependent ordering wave vector Q , and then a commensurate spin-density wave at 1.97 K. But at temperatures above these scales at zero field (and at temperatures much lower near the 1D quantum critical point (QCP)), the system can be described as a 1D Ising system. The effective 1D phase transition to a quantum disordered phase has been inferred to be at the relatively modest critical transverse field of 5.2 T in the b direction. Note that the 3D phase is believed to extend out slightly past the effective 1D QCP, a feature necessary for the observation of “kink” bound states in the spectrum near the critical point⁴⁵.

A number of measurements have been made of the various energy scales on both sides of the transition of the 1D QCP in CoNb_2O_6 . We will concentrate on the paramagnetic regime. As shown in Fig. 1.4 neutron scattering¹⁷⁰ and THz absorption²¹⁰ experiments have given evidence for a $q = 0$ (or symmetry equivalent) mode which increases in energy roughly linearly with field from the critical point. This may be identified straightforwardly with the zone center excitation described by Eq. 1.12. Heat capacity experiments have also been performed¹³¹ and data fit to the nearest-neighbor Ising model. As seen in Fig. 1.4, the extracted gap scale from heat capacity is in excellent agreement with the spectroscopic probes. In contrast to these experiments, the scale of the lowest energy excitation extracted from the temperature dependence of the $1/T_1$ in NMR is greater by approximately a factor of two than the other probes. As explained above, this data was fit to an activated functional form which was effectively $1/T_1 \sim \exp(-\Delta_{\text{NMR}}/T)$. However, we have shown here that near the critical point the expectation is in fact $1/T_1 \sim \exp(-2\Delta/T)$. This means that the activation energy scale from $1/T_1$ will be double that extracted from the other probes. This is precisely as observed experimentally. Also note that the differences between energy scales are far bigger than anything that could be explained by inter-chain couplings or 1D vs. 3D regimes of behavior. The coupling in the transverse b direction has been found to be smaller than $1/60$ of J ³⁵. At low temper-

ature on the paramagnetic side of the transition, this gives a minimum of the dispersion at finite q_b , but except very near the critical point this band width in directions perpendicular to the chain is a very small fraction of Δ ³⁵. The third direction has frustrated antiferromagnetic couplings and has even smaller effect on the dispersions (although it is presumably responsible for stabilizing different magnetically ordered states at low transverse field¹²⁸). The scenario put forward in the current work comes with a distinct prediction. At fields three times the critical field $1/T_1$ should crossover to a form that goes as $\exp(-3\Delta/T)$ e.g. a much faster dependence. This is at fields greater than 15.6 T in CoNb_2O_6 and should be easily testable.

1.7 CONCLUSIONS

The quantum Ising chain has been an essential model to understand the low frequency, non-zero temperature dynamics of a strongly interacting system^{181,61,37}. The integrability of the model allows for exact solutions, and yet many local observables exhibit generic dissipative dynamics at long times. Here we have examined the NMR relaxation rates in the quantum disordered region. It is given by the imaginary part of the local spin susceptibility, at frequencies far below the quasiparticle gap, Δ . Therefore it is not directly amenable to a quasi-classical computation involving collisions of a dilute gas of quasiparticles¹⁸¹. Instead, we showed here that it is dominated by rare processes in which one quasiparticle has sufficient energy to decay into two quasiparticles (and vice versa) near the nucleus. Consequently we found that the NMR relaxation is suppressed by a thermal Boltzmann factor of $\exp(-2\Delta/T)$ for not too large a transverse field, $1 < b < 3$ (the suppression is stronger for larger b). We also computed the precise prefactor of this exponential for the nearest-neighbor Ising chain, and its universal form near the quantum critical point. Finally we compared our results to the experimental probes. The scenario put forth here is in excellent agreement with the experimental results. Spectroscopic and thermodynamic probes show agreement as to the size of

the gap, whereas $1/T_1$ from NMR shows an activation energy, which is approximately twice as large.

2

Non-equilibrium Dynamics in the Sachdev-Ye-Kitaev Model

2.1 INTRODUCTION

One of the great remaining challenges in condensed matter physics is to understand how universal phenomena can emerge in many body systems out of equilibrium. A further challenge is to extend

this understanding to the quantum regime where quantum fluctuations provide yet another source of novelty. Identifying universal non-equilibrium behaviors in strongly interacting quantum systems is important for understanding a large variety of phenomena in particle cosmology²⁰⁷, heavy ion collisions at large hadron colliders¹⁴⁸, and pump and probe experiments⁶⁸ in condensed matter systems.

An interesting way to probe the dynamics of an isolated quantum system is to do a quantum quench. A quantum quench is a protocol in which one starts the system in a pure or thermal state evolving unitarily under one Hamiltonian and then starts evolving the system unitarily under a different Hamiltonian over some timescale which is short compared to the energy gap to the first excited state³⁶. A long time after the quench, we expect the system to end in a steady state. The nature of this state will depend on both interactions and conserved quantities and can be analyzed by looking at the time dependence of macroscopic observables and response functions.

In the long time limit, if time averages of macroscopic observables coincide with expectation values in the microcanonical ensemble, the system has thermalized. The equivalence of time averages and microcanonical expectation values provides us with a notion of quantum “ergodicity” that is in some ways analogous to its classical counterpart¹⁶¹. However, for many reasons, the interpretation of quantum thermalization is much more subtle and appears to contradict the constraints of unitary time evolution. This can be resolved as follows: while unitary time evolution preserves full information about the initial state of the system, the spreading of entanglement delocalizes this information so that it becomes inaccessible at long times. The final state of the system can be completely described by a small number of parameters, one for each extensive conserved quantity^{154,66}. In other words, each subsystem looks like it is in thermal equilibrium with a reservoir characterized by temperature T , chemical potential μ , etc. with the reservoir being the rest of the isolated system. Likewise, we expect response functions to be related by KMS conditions and fluctuation dissipation relations as expected in thermal equilibrium.

In weakly coupled quantum many body systems, thermalization is typically controlled by locally conserved quantities. If a system contains few locally conserved quantities, it will quickly thermalize. By contrast, if there is an extensive number of locally conserved quantities, a system may exhibit non-ergodic behavior and fail to thermalize retaining memory of its initial conditions. However, recent theoretical and experimental studies of strongly correlated quantum systems suggest the emergence of new behaviors out of equilibrium which go beyond previous notions of ergodicity. For instance, the emergence of a time-scale leading to a quasistationary (or ‘prethermal’) state^{118,19}, which has fundamentally different properties from true thermal equilibrium. Other examples, which go beyond this naive categorization, are one dimensional systems close to integrability⁷², systems with slow modes preventing the thermalization of fast modes¹⁸⁵ or interacting disordered systems²¹³.

The non-equilibrium dynamics of strongly interacting quantum many-particle systems are usually studied within the Schwinger-Keldysh formalism, which can describe evolution from a generic initial state to a final state which reaches thermal equilibrium at long times¹¹⁴. The thermodynamic parameters of the final state (*e.g.* temperature) are determined by the values of the conjugate conserved quantities (*e.g.* energy). The Kadanoff-Baym equations obtained from this formalism describe the manner and rate by which this final thermal state is reached.

The Kadanoff-Baym equations are usually too difficult to solve in their full generality. Frequently, a quasiparticle structure has been imposed on the spectral functions, so that the Kadanoff-Baym equations reduce to a quantum Boltzmann equation for the quasiparticle distribution functions. Clearly such an approach cannot be employed for final states of Hamiltonians which describe critical quantum matter *without* quasiparticle excitations. The most common approach is then to employ an expansion away from a regime where quasiparticles exist, using a small parameter such as the deviation of dimensionality from the critical dimension, or the inverse of the number of field components: the analysis is then still carried out using quasiparticle distribution functions^{48,172}.

In this chapter, we will examine the non-equilibrium dynamics towards a final state without

quasiparticles. We will *not* employ a quasiparticle decomposition, and instead provide solutions of the full Kadanoff-Baym equations. We can obtain non-equilibrium solutions for the Sachdev-Ye-Kitaev (SYK) models^{180,121,145} with all-to-all and random interactions between q Majorana fermions on N sites. Using conventions from using conventions from Ref.¹⁴⁵, these can be written as

$$H = (i)^{\frac{q}{2}} \sum_{1 \leq i_1 < i_2 < \dots < i_q \leq N} j_{i_1 i_2 \dots i_q} \psi_{i_1} \psi_{i_2} \dots \psi_{i_q} \quad (2.1)$$

where ψ_i are Majorana fermions on sites $i = 1 \dots N$ obeying $\{\psi_i, \psi_j\} = \delta_{ij}$ and $j_{i_1 i_2 \dots i_q}$ are independent Gaussian random variables with zero mean and variance

$$\langle j_{i_1 \dots i_q}^2 \rangle = \frac{J^2 (q-1)!}{N^{q-1}} \quad (2.2)$$

These models are solvable realizations of quantum matter without quasiparticles in equilibrium and here we shall extend their study to non-equilibrium dynamics. In Section 2.2, we begin with an overview of the closed-time-path-integral formalism for Majorana fermions and construct the Kadanoff-Baym equations for a time dependent generalization of the SYK model with time dependent $q = 2$ and $q = 4$ interactions from the disorder averaged partition function. See also the related analysis in¹⁹³. In Section 2.3 we present numerical solutions of the Kadanoff-Baym equations for the fermion Green's function for different quench protocols. These protocols are specified by the time dependence in the $q = 2$ and $q = 4$ interactions in the Hamiltonian. The most interesting results appear for a quench which starts in a thermal state with both $q = 2$ and $q = 4$ interactions in which the $q = 2$ interaction is instantaneously turned off. We find that the numerical solution for the Green's function post quench shows thermalization (of the two-point correlator) in a time of order the inverse final temperature: this thermalization rate is in accord with the fastest possible rate expected in non-quasiparticle systems¹⁷⁷. In Section 2.4, we obtain exact analytic solutions of

the Kadanoff-Baym equation in the limit of large q (no quasiparticles are present in this limit) for a Hamiltonian which contains time dependent q and $2q$ body interactions. The large q solution relies on a remarkable exact $SL(2, \mathbb{C})$ invariance. We find the surprising result that thermalization of the two-point correlator is instantaneous: the interpretation of this result is not fully clear from our results, but it could be an indication of a prethermal state appearing at $q = \infty$. Finally, in Section 2.5 we note connections between the large q solution to quantum gravity on AdS_2 with the Schwarzian effective action ^{175,176,5,121,145,106,59}.

2.2 PRELIMINARIES

2.2.1 KELDYSH FORMALISM FOR MAJORANA FERMIONS

We are interested in studying quenches in which a system starts out in an equilibrium state and is driven out of equilibrium by a change in the Hamiltonian $H(t)$. The density matrix $\widehat{\rho}(t)$ of the system evolves according to the Von Neumann equation and the full time evolution operator is defined as

$$\mathcal{U}(t_1, t_2) = T \exp \left(-i \int_{t_2}^{t_1} dt H(t) \right) \quad (2.3)$$

In the Schrodinger picture, we can write an expectation value of an observable $\mathcal{O}(t)$ as

$$\langle \mathcal{O} \rangle(t) \equiv \frac{\text{Tr}[\mathcal{O} \widehat{\rho}(t)]}{\text{Tr}[\widehat{\rho}(t)]} = \frac{\text{Tr}[\mathcal{U}(-\infty, t) \mathcal{O} \mathcal{U}(t, -\infty) \widehat{\rho}(-\infty)]}{\text{Tr}[\widehat{\rho}(t)]} \quad (2.4)$$

where the trace is over the many-body Hilbert space. The quantity inside the trace involves both forward and backward evolution in time ¹¹⁴.

For an interacting system in equilibrium at zero temperature, we account for the time dependence by starting from the ground state of a non-interacting system $|0\rangle$ in the far past adiabatically turning on the interaction then and adiabatically turning it off in the distant future. Then the final

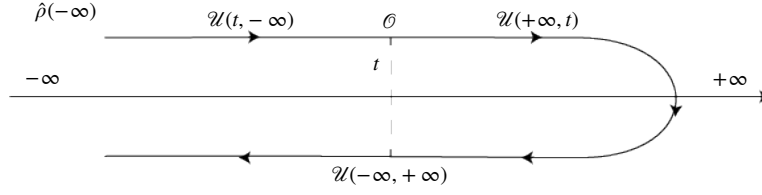


Figure 2.1: Schematic of the closed time contour $\mathcal{C}$ adapted from ¹¹⁴.

state is $\mathcal{U}(\infty, -\infty) |0\rangle = e^{i\varphi} |0\rangle$, which differs from the initial state by only a phase factor. So in equilibrium we only need to consider forward evolution in time as backward evolution is completely contained in the phase factor $e^{i\varphi}$. Out of equilibrium, there is no guarantee that the system will return its initial ground state. In general, the quantity $\mathcal{U}(\infty, -\infty) |0\rangle$ is a superposition of excited states, so backward evolution cannot be eliminated.

To properly account for the backward evolution, one can use the group properties of the unitary time evolution operator and rewrite Eqn. 2.10 as

$$\langle \mathcal{O} \rangle(t) = \frac{\text{Tr}[\mathcal{U}(-\infty, \infty) \mathcal{U}(\infty, t) \mathcal{O} \mathcal{U}(t, -\infty) \hat{\rho}(-\infty)]}{\text{Tr}[\hat{\rho}(-\infty)]} \quad (2.5)$$

where we note that the trace of $\hat{\rho}$ is preserved under unitary operations. Eqn. 2.5 describes evolution along the closed time contour $\mathcal{C}$ where $\mathcal{O}$ is inserted at a time t along the forward branch of the contour. Equivalently

$$\langle \mathcal{O} \rangle(t) = \frac{\text{Tr}[\mathcal{U}(-\infty, t) \mathcal{O} \mathcal{U}(t, \infty) \mathcal{U}(\infty, -\infty) \hat{\rho}(-\infty)]}{\text{Tr}[\hat{\rho}(-\infty)]} \quad (2.6)$$

describes evolution on the backward branch of the contour. In the Schwinger-Keldysh formalism, observables are computed by evolving the initial state both forward and backwards along $\mathcal{C}$ which is most conveniently done by taking half the sum of Eqn. 2.5 and 2.6. We define the time evolution

operator along $\mathcal{C}$ as

$$\mathcal{U}_{\mathcal{C}} = \mathcal{U}(-\infty, \infty)\mathcal{U}(\infty, -\infty) \quad (2.7)$$

For a time dependent Hamiltonian which is different along the top and bottom legs of $\mathcal{C}$, $\mathcal{U}_{\mathcal{C}} \neq 1$.

To calculate observables, we can modify the Hamiltonian on the by adding the source term

$$H'(t) \pm \mathcal{O}J(t) \quad (2.8)$$

where the $+$ ($-$) sign refers to the forward(backward) part of the contour. We define the generating functional as

$$Z[J] \equiv \frac{\text{Tr}[\mathcal{U}_{\mathcal{C}}\widehat{\rho}(-\infty)]}{\text{Tr}[\widehat{\rho}(-\infty)]} \quad (2.9)$$

From equation Eqn. 2.9, we see that $Z[0] = 1$ because $\widehat{\rho}(-\infty)$ is the density matrix in the distant past before interactions are turned out. This allows us to compute expectation values of observables directly from derivatives of $Z[J]$

$$\langle \mathcal{O} \rangle(t) = \frac{i}{2} \frac{\delta Z[J]}{\delta J(t)} \Big|_{J=0} \quad (2.10)$$

Since one can compute disorder averaged quantities directly from the disorder averaged generating function, one does not need to introduce replicas in systems with quenched disorder.

In general we will be interested in using this formalism to study the time evolution of the Green's function of interacting Majorana fermions with quenched disorder out of equilibrium. We start by constructing the path integral for non-interacting Majorana fermions from the familiar complex fermion path integral¹⁸⁶ by expressing the complex fermions Ψ in terms of two real fermions ψ and χ i.e. $\Psi = \psi + i\chi$. Since χ is just a spectator which does not appear in the Hamiltonian, we can

disregard its contribution. Starting from the expression for Z in Eqn. 2.9, we can divide the contour into $2(N - 1)$ discrete time intervals and insert a complete set of fermionic coherent states at each step¹⁴. In the continuum limit, we can write the path integral representation of the generating function as

$$Z = \int D\psi e^{iS[\psi]} \quad (2.11)$$

where the action is given by

$$S[\psi] = \int_C dt \frac{i}{2} \psi \partial_t \psi = \int_{-\infty}^{\infty} dt \left\{ \frac{i}{2} (\psi^+ \partial_t \psi^+ - \psi^- \partial_t \psi^-) \right\} \quad (2.12)$$

On the right hand side of Eqn. 2.12, we have separated the the fields ψ into a component ψ^+ lying on the forward part of the contour and a component ψ^- lying on the backward part of the contour. These fields are not independent due to the presence of terms coupling the time interval from N to $N + 1$ and from $2N$ to 1 . We define the contour-ordered Green's function

$$iG(t_1, t_2) = \langle T_C \psi(t_1) \psi(t_2) \rangle = \int D\psi \psi(t_1) \psi(t_2) e^{iS[\psi]} \quad (2.13)$$

where $S[\psi]$ is the action in Eqn. 2.12. We can also consider two point functions of the fields $\psi^\pm$ fields

$$\langle \psi^\pm(t_1) \psi^\pm(t_2) \rangle = \int D\psi \psi^\pm(t_1) \psi^\pm(t_2) e^{iS[\psi]} \quad (2.14)$$

These define a matrix of Green's functions. Of these, we will be interested in the Majorana fermion

greater(lesser) Green's functions $G^{>(<)}(t_1, t_2)$ defined as follows

$$iG^>(t_1, t_2) \equiv \langle \psi^-(t_1) \psi^+(t_2) \rangle \quad (2.15a)$$

$$iG^<(t_1, t_2) \equiv \langle \psi^+(t_1) \psi^-(t_2) \rangle \quad (2.15b)$$

For free Majorana fermions, the greater and lesser Green's functions are given by

$$G_0^>(t_1, t_2) = -\frac{i}{2} \quad (2.16)$$

$$G_0^<(t_1, t_2) = \frac{i}{2} \quad (2.17)$$

In general Majorana fermions obey the following property

$$G^>(t_1, t_2) = -G^<(t_2, t_1) \quad (2.18)$$

This condition continues to hold even out of equilibrium¹⁴. This conveniently allows us to directly study a single component of the Green's function and then use Eq.(2.18) to construct all other Green's functions.

2.2.2 KADANOFF-BAYM EQUATIONS FROM THE PATH INTEGRAL

The time evolution of the greater Green's function is governed by the Kadanoff-Baym equations derived from the non-equilibrium Dyson equation obtained from the generating function. We will study the non-equilibrium dynamics of the SYK model by solving the Kadanoff-Baym equations for the greater Green's function and tracking its evolution after a quantum quench.

We will consider the following time-dependent generalization of the SYK Hamiltonian

$$H(t) = i \sum_{i < j} j_{2,ij} f_2(t) \psi_i \psi_j - \sum_{i < j < k < l} j_{4,ijkl} f_4(t) \psi_i \psi_j \psi_k \psi_l \quad (2.19)$$

where $j_{2,ij}$ and $j_{4,ijkl}$ are random variables as described in Eq. (32), and $f_2(t)$ and $f_4(t)$ are arbitrary functions of time. We will use $f_2(t)$ and $f_4(t)$ to specify the quench protocol by using them to turn on(off) or rescale couplings. We write the path integral representation for this Hamiltonian of Majorana fermions as

$$Z = \int D\psi \int Dj_{2,ij} \int Dj_{4,ijkl} \mathcal{P}_2(j_{4,ijkl}) \mathcal{P}_1(j_{2,ij}) e^{iS[\psi]} \quad (2.20)$$

with the action

$$S[\psi] = \int_{\mathcal{C}} dt \left\{ \frac{i}{2} \sum_i \psi_i \partial_t \psi_i - i \sum_{i < j} j_{2,ij} f_2(t) \psi_i \psi_j + \sum_{i < j < k < l} j_{4,ijkl} f_4(t) \psi_i \psi_j \psi_k \psi_l \right\} \quad (2.21)$$

Here, the Majorana fields ψ_i live on $\mathcal{C}$, and Z is normalized to one. The probability distributions of $j_{2,ij}$ and $j_{4,ijkl}$ are given by

$$\mathcal{P}_1(j_{2,ij}) = \sqrt{\frac{N}{2j_2^2\pi}} \exp\left(-\frac{N}{2j_2^2} \sum_{i < j} j_{2,ij}^2\right) \quad (2.22)$$

$$\mathcal{P}_2(j_{4,ijkl}) = \sqrt{\frac{N^3}{12j_4^2\pi}} \exp\left(-\frac{N^3}{12j_4^2} \sum_{i < j < k < l} j_{4,ijkl}^2\right) \quad (2.23)$$

The normalization of Z allows us to directly average over disorder by performing the $j_{2,ij}$ and $j_{4,ijkl}$

integrals. This gives us the following path integral

$$\begin{aligned}
Z = \int D\psi \exp \Big\{ & -\frac{1}{2} \int_C dt_1 \sum_i \psi_i \partial_{t_1} \psi_i - \frac{J_2^2}{4N} \sum_{i,j} \int_C dt_1 \int_C dt_2 f_2(t_1) f_2(t_2) \psi_i(t_1) \psi_i(t_2) \psi_j(t_1) \psi_j(t_2) \\
& - \frac{3J_4^2}{4!N^3} \sum_{i,j,k,l} \int_C dt_1 \int_C dt_2 f_4(t_1) f_4(t_2) \psi_i(t_1) \psi_i(t_2) \psi_j(t_1) \psi_j(t_2) \psi_k(t_1) \psi_k(t_2) \psi_l(t_1) \psi_l(t_2) \Big\}
\end{aligned} \tag{2.24}$$

where the action is bi-local in time. We introduce the bi-linear field G (the Green's function)

$$G(t_1, t_2) = -\frac{i}{N} \sum_i \psi_i(t_1) \psi_i(t_2) \tag{2.25}$$

and impose Eq. (2.25) by a Lagrange multiplier Σ (which will turn out to be the self energy). This leads to the following partition function

$$\begin{aligned}
Z = \int D\psi \int DG \int D\Sigma \exp \Big\{ & -\frac{1}{2} \int_C dt_1 \sum_i \psi_i \partial_{t_1} \psi_i + \frac{J_2^2 N}{4} \int_C dt_1 \int_C dt_2 f_2(t_1) f_2(t_2) G(t_1, t_2)^2 \\
& - \frac{3J_4^2 N}{4!} \int_C dt_1 \int_C dt_2 f_4(t_1) f_4(t_2) G(t_1, t_2)^4 + \frac{i}{2} \int_C dt_1 \int_C dt_2 \Sigma(t_1, t_2) \left[G(t_1, t_2) + \frac{i}{N} \sum_i \psi_i(t_1) \psi_i(t_2) \right] \Big\}
\end{aligned}$$

Rescaling integration variables as $\Sigma \rightarrow iN\Sigma$ and integrating out the fermion yields

$$Z = \int DG \int D\Sigma \exp \{ iS[G, \Sigma] \} \tag{2.26}$$

with the action written in terms of G and Σ

$$\begin{aligned}
S[G, \Sigma] = & -\frac{iN}{2} \text{tr} \log[-i(G_0^{-1} - \Sigma)] - \frac{iJ_2^2 N}{4} \int_C dt_1 \int_C dt_2 f_2(t_1) f_2(t_2) G(t_1, t_2)^2 \\
& + \frac{3iJ_4^2 N}{4!} \int_C dt_1 \int_C dt_2 f_4(t_1) f_4(t_2) G(t_1, t_2)^4 + \frac{iN}{2} \int_C dt_1 \int_C dt_2 \Sigma(t_1, t_2) G(t_1, t_2)
\end{aligned} \tag{2.27}$$

where G_0 is the free Majorana Green's function and $G_0^{-1}(t_1, t_2) = i\partial_t \delta_C(t_1, t_2)$. Varying the action with respect to Σ and G yields the Schwinger-Dyson equations for the Green's function and self-energy:

$$G_0^{-1}(t_1, t_2) - G^{-1}(t_1, t_2) = \Sigma(t_1, t_2) \quad (2.28)$$

$$\Sigma(t_1, t_2) = f_2^2(t_1)f_2(t_2)G(t_1, t_2) - f_4^2(t_1)f_4(t_2)G(t_1, t_2)^3 \quad (2.29)$$

Note that the time arguments are with respect to the closed time contour making the matrix structure of the full Green's function implicit. The bare greater and lesser Green's functions for the Majorana fermions are given by Eqns. 2.16 and 2.17. We now use $G^>$ and $G^<$ to define the retarded, advanced and Keldysh Green's functions:

$$G^R(t_1, t_2) \equiv \Theta(t_1 - t_2)[G^>(t_1, t_2) - G^<(t_1, t_2)] \quad (2.30a)$$

$$G^A(t_1, t_2) \equiv \Theta(t_2 - t_1)[G^<(t_1, t_2) - G^>(t_1, t_2)] \quad (2.30b)$$

$$G^K(t_1, t_2) \equiv G^>(t_1, t_2) + G^<(t_1, t_2) \quad (2.30c)$$

Similarly, we introduce retarded and advanced self-energies

$$\Sigma^R(t_1, t_2) \equiv \Theta(t_1 - t_2)[\Sigma^>(t_1, t_2) - \Sigma^<(t_1, t_2)], \quad (2.31)$$

$$\Sigma^A(t_1, t_2) \equiv -\Theta(t_2 - t_1)[\Sigma^>(t_1, t_2) - \Sigma^<(t_1, t_2)] \quad (2.32)$$

For more details on the Schwinger-Keldysh formalism and the saddle point approximation we refer to Refs. ^{14,114,145}.

To obtain the Kadanoff-Baym equations, we rewrite the Schwinger-Dyson equations in (2.28) as

$$\int_{\mathcal{C}} dt_3 G_0^{-1}(t_1, t_3) G(t_3, t_2) = \delta_{\mathcal{C}}(t_1, t_2) + \int_{\mathcal{C}} dt_3 \Sigma(t_1, t_3) G(t_3, t_2) \quad (2.33)$$

$$\int_{\mathcal{C}} dt_3 G(t_1, t_3) G_0^{-1}(t_3, t_2) = \delta_{\mathcal{C}}(t_1, t_2) + \int_{\mathcal{C}} dt_3 G(t_1, t_3) \Sigma(t_3, t_2) \quad (2.34)$$

We can use Eqns. 2.33 and 2.34 to write a integro-differential equations for $G^>(t_1, t_2)$ by first noting that for $t_1 = t_2$ $G(t_1, t_2) = 0$ and for $t_1 \neq t_2$ $G(t_1, t_2)$ is a convolution of two functions on $\mathcal{C}$. For a general function $C(t_1, t_2)$ that can be expressed as a convolution of two functions on $\mathcal{C}$ i.e.

$$C(t_1, t_2) = \int_{\mathcal{C}} dt_3 A(t_1, t_3) B(t_3, t_2) \quad (2.35)$$

the Langreth rules described in Ref. ¹⁹⁷ allow us to write the greater (lesser) component $C^{>(<)}(t_1, t_2)$ along the real axis as

$$C^>(t_1, t_2) \equiv \int_{\mathcal{C}} dt_3 A(t_1^-, t_3) B(t_3, t_2^+) = \int_{-\infty}^{\infty} dt_3 \{A^>(t_1, t_3) B^A(t_3, t_2) + A^R(t_1, t_3) B^>(t_3, t_2)\} \quad (2.36)$$

$$C^<(t_1, t_2) \equiv \int_{\mathcal{C}} dt_3 A(t_1^+, t_3) B(t_3, t_2^-) = \int_{-\infty}^{\infty} dt_3 \{A^<(t_1, t_3) B^A(t_3, t_2) + A^R(t_1, t_3) B^<(t_3, t_2)\} \quad (2.37)$$

where the $+(-)$ superscript indicates that $t_{1,2}^{+(-)}$ lies on the forward (backward) part of the contour.

We can now use 2.36 to rewrite Eq. (2.33) and Eq. (2.34) on the real time axis as

$$\int_{\mathcal{C}} dt_3 \Sigma(t_1^-, t_3) G(t_3, t_2^+) = \int_{-\infty}^{\infty} dt_3 \{ \Sigma^R(t_1, t_3) G^>(t_3, t_2) + \Sigma^>(t_1, t_2) G^A(t_2, t_3) \} \quad (2.38)$$

$$\int_{\mathcal{C}} dt_3 G(t_1^-, t_3) \Sigma(t_3, t_2^+) = \int_{-\infty}^{\infty} dt_3 \{ G^R(t_1, t_3) \Sigma^>(t_3, t_2) + G^>(t_1, t_2) \Sigma^A(t_2, t_3) \} \quad (2.39)$$

The last equation gives us the following equations of motion for $G^>$

$$i\partial_{t_1} G^>(t_1, t_2) = \int_{-\infty}^{\infty} dt_3 \left\{ \Sigma^R(t_1, t_3) G^>(t_3, t_2) + \Sigma^>(t_1, t_3) G^A(t_3, t_2) \right\} \quad (2.40)$$

$$-i\partial_{t_2} G^>(t_1, t_2) = \int_{-\infty}^{\infty} dt_3 \left\{ G^R(t_1, t_3) \Sigma^>(t_3, t_2) + G^>(t_1, t_3) \Sigma^A(t_3, t_2) \right\} \quad (2.41)$$

where

$$\Sigma^>(t_1, t_2) = \tilde{f}_2^2 f_2(t_1) f_2(t_2) G^>(t_1, t_2) - \tilde{f}_4^2 f_4(t_1) f_4(t_2) (G^>(t_1, t_2))^3 \quad (2.42)$$

and Σ^R and Σ^A are defined in Eq. (2.31) and Eq. (2.32). In Section 2.3, we will solve this system of equations numerically. Analytic solutions for the generalization of this model in the large q limit will appear in Section 2.4.

2.3 KADANOFF-BAYM EQUATIONS: NUMERICAL STUDY

All of the quench protocols we considered are of the form

$$\begin{aligned} f_2(t) &= \alpha_1 \Theta(-t) + \alpha_2 \Theta(t) \\ f_4(t) &= \gamma_1 \Theta(-t) + \gamma_2 \Theta(t) \end{aligned} \quad (2.43)$$

where $\Theta(t)$ is the unit step function, and will be denoted by $(J_{2,i}, J_{4,i}) \rightarrow (J_{2,f}, J_{4,f})$. Specifically, we choose the following four protocols

- A) $(J_{2,i}, 0) \rightarrow (J_{2,f}, 0)$ quench: the energy scale of the random hopping model gets suddenly rescaled.
- B) $(0, 0) \rightarrow (0, J_{4,f})$ quench: the system is quenched from an uncorrelated state by switching on the interaction of the SYK model.

- C) $(J_{2,i}, 0) \rightarrow (0, J_{4,f})$ quench: the system is prepared in the ground state of the random hopping model and then quenched to the SYK Hamiltonian.
- D) $(J_{2,i}, J_{4,f}) \rightarrow (0, J_4)$ quench: starting from a ground state with the quadratic and the quartic terms present the quadratic term is switched off.

The numerical results for quenches A-C appear in Appendix B.4, while those for the most interesting case of quench D appear below.

The Kadanoff-Baym equations for the SYK model (Eqs. (2.40) and (2.41)) are solved numerically for each quench protocol described above. The absence of momentum dependence will allow us to explore dynamics at long times after the quench. We set the quench time to $t_1 = t_2 = 0$. For $t_1 < 0$ and $t_2 < 0$, we are in thermal equilibrium and the time dependence of the Green's function is determined by the Dyson equation. When quenching the ground state of the random hopping model, we can use the exact solution for $G^>$ as an initial condition. In all other cases, we solve the Dyson equation self-consistently: for further details we refer to Appendix B.3. Using the boundary conditions given by pre-quench state, we solve Kadanoff-Baym equation for $t_1 > 0$ or $t_2 > 0$, and obtain the post-quench evolution of $G^>$. Integrals in the Kadanoff-Baym equation are computed using the trapezoidal rule. After discretizing the integrals, the remaining ordinary differential equations are solved using a predictor-corrector scheme, where the corrector is determined self-consistently by iteration. For long times after the quench, the numerical effort is equivalent to a second-order Runge-Kutta scheme, because the self-consistency of the predictor-corrector scheme converges very fast. Using this approach right after the quench reduces numerical errors significantly.

To interpret the long time behavior Green's functions, we briefly comment on properties of thermal Green's functions. In thermal equilibrium, all Green's function only depend on the relative time t . The imaginary part of the Fourier transformed retarded Green's function is given by the

spectral function

$$A(\omega) = -2 \text{Im } G^R(\omega) \quad (2.44)$$

In thermal equilibrium, the Kubo-Martin-Schwinger (KMS) condition¹¹⁴ establishes a relationship between $G^>(\omega)$ and $G^<(\omega)$ which is given by

$$G^>(\omega) = -e^{\beta\omega} G^<(\omega) \quad (2.45)$$

Using the definition of G^R , we can write the spectral function in terms of $G^>$ as

$$A(\omega) = G^>(\omega)(1 + e^{-\beta\omega}) \quad (2.46)$$

Similarly the Keldysh component of the fermionic Green's function is related to the spectral function by

$$iG^K(\omega) = iG^>(\omega) + iG^<(\omega) = \tanh(\beta\omega/2)A(\omega) \quad (2.47)$$

As the system approaches equilibrium, it is useful to characterize the correlators in terms of the absolute and relative times given by

$$\mathcal{T} = (t_1 + t_2)/2 \quad , \quad t = t_1 - t_2 \quad (2.48)$$

Out of equilibrium, the Green's we have defined functions depend on both τ and $\mathcal{T}$. However, we can still consider the Fourier transform with respect to t given by

$$G^R(\mathcal{T}, \omega) = \int_0^\infty dt e^{-\delta t} e^{i\omega t} G^R(\mathcal{T} + t/2, \mathcal{T} - t/2) \quad (2.49)$$

known as the Wigner transform. Using Eq. (2.49), we express the spectral function out of equilibrium in terms of $\mathcal{T}$ and ω as

$$(\mathcal{T}, \omega) = -2 \operatorname{Im} G^R(\mathcal{T}, \omega) \quad (2.50)$$

We can compute the spectral function and the Keldysh component of the Green's function by numerically solving for $G^>(t_1, t_2)$. To investigate thermalization behaviour, at each $\mathcal{T}$ we fit

$$\frac{iG^K(\mathcal{T}, \omega)}{A(\mathcal{T}, \omega)} \quad (2.51)$$

to

$$\tanh\left(\frac{\beta_{\text{eff}}(\mathcal{T})\omega}{2}\right) \quad (2.52)$$

at low ω and so determined a time-independent effective inverse temperature $\beta_{\text{eff}}(\mathcal{T})$. By Eq. (2.47), $\beta_{\text{eff}}(\mathcal{T})$ is the actual inverse temperature, β_f , in a thermal state reached as $\mathcal{T} \rightarrow \infty$. The value of β_f is such that the total energy of the system remains the same after the quench at $t = 0^+$.

2.3.1 NUMERICAL RESULTS

We describe the results for quench D above, in which $(J_{2,i}, J_{4,i}) \rightarrow (0, J_{4,f})$ i.e. quenches from a Hamiltonian with a quadratic and a quartic term to a Hamiltonian where the quadratic term has been switched off. In the initial state, the $q = 2$ free fermion term dominates at low energies, and hence we have Fermi liquid behavior at low temperatures. By contrast, the final Hamiltonian describes a non-Fermi liquid even at the lowest energies¹⁸⁰. This quench results in nontrivial dynamics of the two-point function.

To study thermalization, we can check the validity of the fluctuation-dissipation relation and the behavior of the effective temperature $T_{\text{eff}}(\mathcal{T}) = \beta_{\text{eff}}^{-1}(\mathcal{T})$ at times after the quench. The effective temperature can either be obtained from the derivative of $iG^K(\mathcal{T}, \omega)/A(\mathcal{T}, \omega)$ at $\omega = 0$, or as a fit

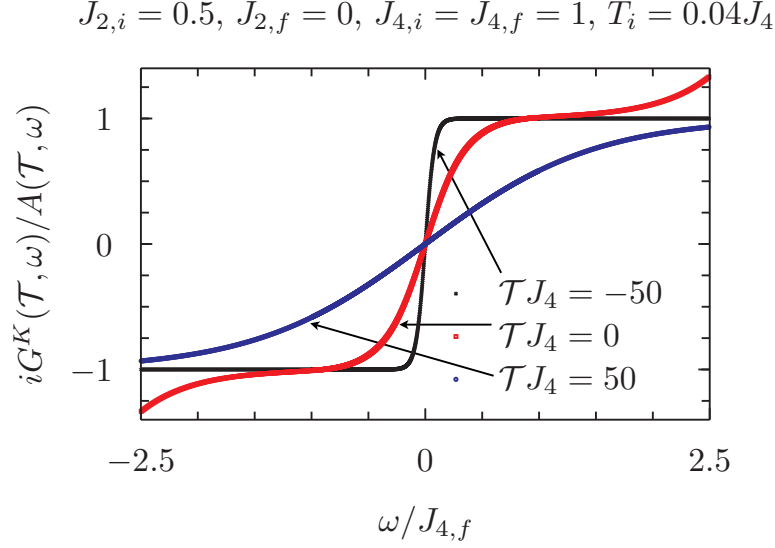


Figure 2.2: The ratio of the Keldysh and spectral function is shown before and after a quench from a $J_2 + J_4$ model to a purely J_4 model. Fits to this data allow determination of the effective temperature as a function of time after the quench, $\beta_{\text{eff}}(\mathcal{T})$ using Eq. (2.52).

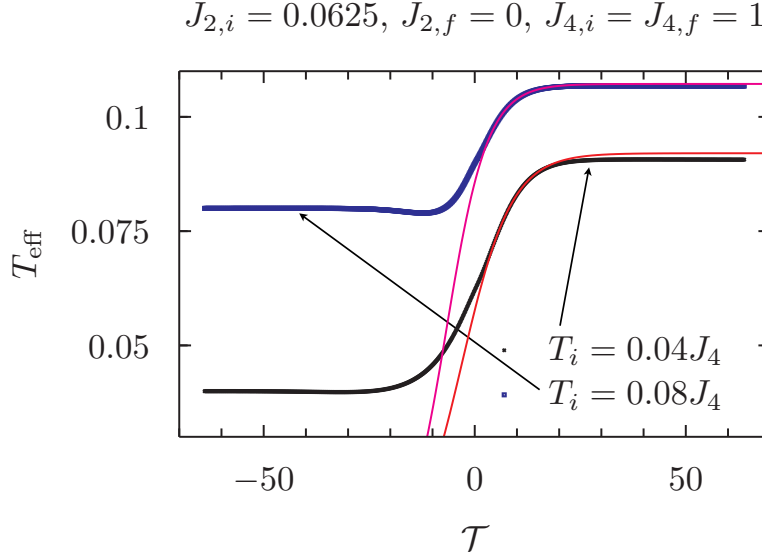


Figure 2.3: Fits to the values of the effective temperature $\beta_{\text{eff}}(\mathcal{T}) = 1/T_{\text{eff}}(\mathcal{T})$ from results as in Fig. 2.2 to Eq. (2.53) to allow determination of the thermalization rate Γ for each quench.

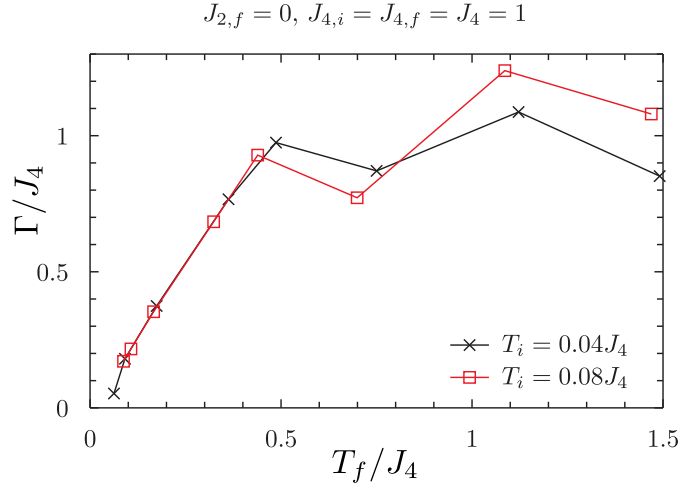


Figure 2.4: The values of the thermalization rate Γ obtained from Fig. 2.3 are shown as a function of the final temperature, T_f , for each quench. One of our main numerical results is the proportionality of Γ to T_f at small $T_f \ll J$. At low temperatures the relaxation towards equilibrium is expected to be controlled by quantum and classical fluctuations and hence we expect an equilibration rate proportional to T_f . At higher temperatures Γ is of order J_4 and since at high temperatures the relaxation is essentially dominated by thermal fluctuations the relaxation rate is dominated by the microscopic interaction energy-scale of the fermions.

of $\tanh[\omega/(2T_{\text{eff}})]$ to that function to some frequency interval. As shown in Fig. 2.2, long after the quench, the numerical results for $iG^K(\mathcal{T}, \omega)/A(\mathcal{T}, \omega)$ are well described by $\tanh[\omega/(2T_{\text{eff}})]$, indicating that the Green's function has thermalized. At intermediate times, the low frequency behaviour is still well described by this function. At high frequencies, deviations are visible, which indicate that $iG^K(\mathcal{T}, \omega)/A(\mathcal{T}, \omega)$ is non-thermal at such times.

Long after the quench, the value of $T_{\text{eff}}(\mathcal{T})$ relaxes exponentially rapidly to T_f . From the numerical analysis we can characterize the late time approach to equilibrium by

$$\beta_{\text{eff}}(\mathcal{T}) = \beta_f + \alpha \exp(-\Gamma \mathcal{T}) \quad (2.53)$$

This defines a thermalization rate, Γ .

For large $\mathcal{T}$ we will use the Green's functions to extract an effective temperature $\beta_{\text{eff}}(\mathcal{T})$ and

In Fig. 2.3 we show the time-dependence of the effective temperature, and a fit of T_{eff}^{-1} to

$$\Gamma = C/\beta_f, \quad \beta_p J_4 \gg 1 \quad (2.54)$$

where C is a numerical constant independent of the initial state. The thermalization rate, Γ obtained from such fits is shown in Fig. 2.4. This rate appears proportional to the final temperature T_f at small final temperatures as noted in Eq. (2.54), and saturates at higher temperatures as is expected for systems without quasiparticle excitations¹⁷⁷. It is difficult to determine Γ numerically at large T_f/J_4 , and this likely leads to the oscillatory behavior present in Fig. 2.4.

2.4 KADANOFF-BAYM EQUATIONS: LARGE q LIMIT

In this section, we obtain exact analytic solutions of the Kadanoff-Baym equation in the limit of large q (no quasiparticles are present in this limit), where q refers to the number of fermions in the interaction in Eq. (30). This limit allows us to study corrections to the low temperature conformal solution and obtain equations that are exactly solvable. The large q solution relies on a remarkable exact $\text{SL}(2, \mathbb{C})$ invariance, and has connections to quantum gravity on AdS_2 with the Schwarzian effective action^{175,176,5,121,145,106,59}. For this limit, we will consider a model in which the $t < 0$ Hamiltonian has both a q -fermion interaction $\mathcal{J}$, and a pq fermion interaction $\mathcal{J}_p$. At $t > 0$, we quench to only a q fermion interaction $\mathcal{J}$. We will show that the non-equilibrium dynamics of this model are exactly solvable in the limit of large q taken at fixed p .

We will consider a model with a q fermion coupling $J(t)$, and a pq fermion coupling $J_p(t)$. For now, we keep the time-dependence of both couplings arbitrary. In the large q limit, the Green's function can be written as¹⁴⁵

$$G^>(t_1, t_2) = -i \langle \psi(t_1) \psi(t_2) \rangle = -\frac{i}{2} \left[1 + \frac{1}{q} g(t_1, t_2) + \dots \right] \quad (2.55)$$

where

$$g(t, t) = 0 \quad (2.56)$$

and the Kadanoff-Baym equations in Eqs. (2.40), (2.41) and (2.42) for $G^>(t_1, t_2)$ now become

$$\begin{aligned} i \frac{\partial}{\partial t_1} G^>(t_1, t_2) &= -i^q \int_{-\infty}^{t_1} dt_3 J(t_1) J(t_3) [(G^>)^{q-1}(t_1, t_3) - (G^<)^{q-1}(t_1, t_3)] G^>(t_3, t_2) \\ &\quad + i^q \int_{-\infty}^{t_2} dt_3 J(t_1) J(t_3) (G^>)^{q-1}(t_1, t_3) [G^>(t_3, t_2) - G^<(t_3, t_2)] \\ &\quad - i^{pq} \int_{-\infty}^{t_1} dt_3 J_p(t_1) J_p(t_3) [(G^>)^{pq-1}(t_1, t_3) - (G^<)^{pq-1}(t_1, t_3)] G^>(t_3, t_2) \\ &\quad + i^{pq} \int_{-\infty}^{t_2} dt_3 J_p(t_1) J_p(t_3) (G^>)^{pq-1}(t_1, t_3) [G^>(t_3, t_2) - G^<(t_3, t_2)] , \\ -i \frac{\partial}{\partial t_2} G^>(t_1, t_2) &= -i^q \int_{-\infty}^{t_1} dt_3 J(t_3) J(t_2) [G^>(t_1, t_3) - G^<(t_1, t_3)] (G^>)^{q-1}(t_3, t_2) \\ &\quad + i^q \int_{-\infty}^{t_2} dt_3 J(t_3) J(t_2) G^>(t_1, t_3) [(G^>)^{q-1}(t_3, t_2) - (G^<)^{q-1}(t_3, t_2)] \\ &\quad - i^{pq} \int_{-\infty}^{t_1} dt_3 J_p(t_3) J_p(t_2) [G^>(t_1, t_3) - G^<(t_1, t_3)] (G^>)^{pq-1}(t_3, t_2) \quad (2.57) \\ &\quad + i^{pq} \int_{-\infty}^{t_2} dt_3 J_p(t_3) J_p(t_2) G^>(t_1, t_3) [(G^>)^{pq-1}(t_3, t_2) - (G^<)^{pq-1}(t_3, t_2)] \end{aligned}$$

where we have defined

$$J_p(t) \equiv J_p f_2(t) , \quad J(t) \equiv J f_4(t) \quad (2.58)$$

Also, recall that Eq. (2.18) relates $G^>$ to $G^<$.

We note a property of the Kadanoff-Baym equations in Eq. (2.57), connected to a comment in Section 2.1 above Eq. (2.48). If we set $J_p(t) = 0$, then all dependence of Eq. (2.57) on $J(t)$ can be scaled away by reparameterizing time via

$$\int^t J(t') dt' \rightarrow t \quad (2.59)$$

This implies that correlation functions remain in thermal equilibrium in the new time co-ordinate.

However, when both $J_p(t)$ and $J(t)$ are non-zero, such a reparameterization is not sufficient, and there are non-trivial quench dynamics, as shown in our numerical study in Section 2.3. Below, we will see that the quench dynamics can also be trivial in the limit $q \rightarrow \infty$, even when both $J_p(t)$ and $J(t)$ are both non-zero. This result arises from fairly non-trivial computations which are described below, and in particular from an $\text{SL}(2, \mathbb{C})$ symmetry of the parameterization of the equations.

In the large q limit, we assume a solution of the form in Eq. (2.55). Inserting Eq. (2.55) into Eq. (2.57), we obtain to leading order in $1/q$

$$\begin{aligned} \frac{\partial}{\partial t_1} g(t_1, t_2) &= 2 \int_{-\infty}^{t_2} dt_3 \mathcal{J}(t_1) \mathcal{J}(t_3) e^{g(t_1, t_3)} - \int_{-\infty}^{t_1} dt_3 \mathcal{J}(t_1) \mathcal{J}(t_3) \left[e^{g(t_1, t_3)} + e^{g(t_3, t_1)} \right] \\ &\quad + 2 \int_{-\infty}^{t_2} dt_3 \mathcal{J}_p(t_1) \mathcal{J}_p(t_3) e^{pg(t_1, t_3)} - \int_{-\infty}^{t_1} dt_3 \mathcal{J}_p(t_1) \mathcal{J}_p(t_3) \left[e^{pg(t_1, t_3)} + e^{pg(t_3, t_1)} \right] \\ \frac{\partial}{\partial t_2} g(t_1, t_2) &= 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}(t_3) \mathcal{J}(t_2) e^{g(t_3, t_2)} - \int_{-\infty}^{t_2} dt_3 \mathcal{J}(t_3) \mathcal{J}(t_2) \left[e^{g(t_3, t_2)} + e^{g(t_2, t_3)} \right] \\ &\quad + 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(t_2) e^{pg(t_3, t_2)} - \int_{-\infty}^{t_2} dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(t_2) \left[e^{pg(t_3, t_2)} + e^{pg(t_2, t_3)} \right] \end{aligned} \quad (2.60)$$

where

$$\mathcal{J}^2(t) = qJ^2(t)2^{1-q}, \quad \mathcal{J}_p^2(t) = qJ_p^2(t)2^{1-pq} \quad (2.61)$$

Remarkably, these non-linear, partial, integro-differential equations are exactly solvable for our quench protocol and all other initial conditions, as we will show in the remainder of this section. The final exact solution appears in Section 2.4.4, and surprisingly shows that the solution is instantaneously in thermal equilibrium at $t = 0^+$.

The key to the exact solution is to take derivatives of either equation in Eq. (2.60) to obtain

$$\frac{\partial^2}{\partial t_1 \partial t_2} g(t_1, t_2) = 2\mathcal{J}(t_1) \mathcal{J}(t_2) e^{g(t_1, t_2)} + 2\mathcal{J}_p(t_1) \mathcal{J}_p(t_2) e^{pg(t_1, t_2)} \quad (2.62)$$

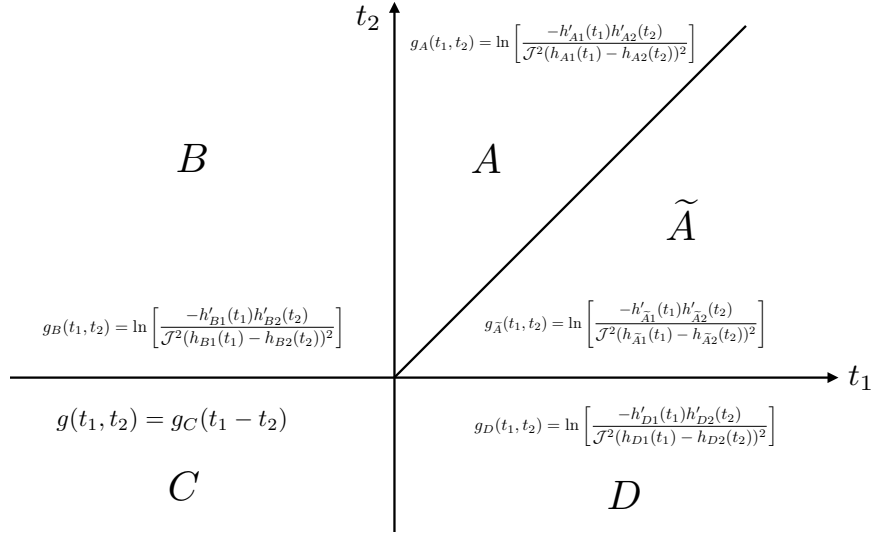


Figure 2.5: The regions in the t_1 - t_2 plane.

We will look at the case where $\mathcal{J}_p(t)$ is non-zero only for $t < 0$, while $\mathcal{J}(t)$ is time independent:

$$\mathcal{J}_p(t) = \mathcal{J}_p \Theta(-t), \quad \mathcal{J}(t) = \mathcal{J} \quad (2.63)$$

This is the analog of the protocol D in Section 2.3. In this case, Eq. (2.62) is the Lorentzian Liouville equation

$$\frac{\partial^2}{\partial t_1 \partial t_2} g(t_1, t_2) = 2\mathcal{J}^2 e^{g(t_1, t_2)} \quad (2.64)$$

in three of the four quadrants of the t_1 - t_2 plane (labeled as in Fig. 2.5). The most general solution of Eq. (2.64) in these quadrants is

$$g_\alpha(t_1, t_2) = \ln \left[\frac{-b'_{\alpha 1}(t_1)b'_{\alpha 2}(t_2)}{\mathcal{J}^2(b_{\alpha 1}(t_1) - b_{\alpha 2}(t_2))^2} \right], \quad \text{for } \alpha = A, \tilde{A}, B, D \quad (2.65)$$

where $b_1(t_1)$ and $b_2(t_2)$ are arbitrary functions of their arguments. Eq. (2.65) will also apply for

$\alpha = C$ when $\mathcal{J}_p = 0$. Eq. (2.56) implies that the functions $b_{A1}(t)$ and $b_{A2}(t)$ obey

$$b'_{A1}(t)b'_{A2}(t) = -\mathcal{J}^2(b_{A1}(t) - b_{A2}(t))^2 \quad (2.66)$$

and similarly

$$b'_{\tilde{A}1}(t)b'_{\tilde{A}2}(t) = -\mathcal{J}^2(b_{\tilde{A}1}(t) - b_{\tilde{A}2}(t))^2 \quad (2.67)$$

A crucial feature of Eq. (2.65) is that $g_\alpha(t_1, t_2)$ remains invariant under the $SL(2, \mathbb{C})$ transformation which maps both $b_{\alpha 1}$ and $b_{\alpha 2}$ by the same transformation

$$b(t) \rightarrow \frac{a b(t) + b}{c b(t) + d} \quad (2.68)$$

where a, b, c, d are complex numbers obeying $ad - bc = 1$. This transformation can be performed for each region α in the t_1, t_2 plane.

Given the symmetries of Eq. (2.60) and the thermal initial conditions, we look for solutions which obey

$$g(t_2, t_1) = [g(t_1, t_2)]^* \quad (2.69)$$

This property can be related to the causality of the Kadanoff-Baym equations as is explained in Appendix B.5. We can satisfy Eq. (2.69) in quadrants A, B, D by the relations

$$\begin{aligned} b_{\tilde{A}1}(t) &= b_{A2}^*(t) \\ b_{\tilde{A}2}(t) &= b_{A1}^*(t) \\ b_{D1}(t) &= b_{B2}^*(t) \\ b_{D2}(t) &= b_{B1}^*(t) \end{aligned} \quad (2.70)$$

The solution in Eq. (2.65) applies in all but the C quadrant of Fig. 2.5. In quadrant C, the system is in thermal equilibrium, and so the Green's function depends only upon time differences. We describe this equilibrium solution in Appendix B.6. In the following subsections, we describe the results of inserting the parameterization in Eq. (2.65) back into Eq. (2.60) to obtain ordinary differential equations for $h_{\alpha 1}$ and $h_{\alpha 2}$ in the remainder of the t_1 - t_2 plane.

2.4.1 QUADRANT B

From Eq. (2.60), we obtain for $t_1 < 0$ and $t_2 > 0$

$$\begin{aligned}
-\frac{\partial}{\partial t_1} g_B(t_1, t_2) &= -\mathcal{J}^2 \int_{-\infty}^{\infty} dt' e^{g_C(t')} \text{sgn}(t' - t_1) - 2\mathcal{J}^2 \int_0^{t_2} dt_3 e^{g_B(t_1, t_3)} \\
&\quad - \mathcal{J}_p^2 \int_{-\infty}^{\infty} dt' \text{sgn}(t' - t_1) e^{p_{g_C}(t')} \\
\frac{\partial}{\partial t_2} g_B(t_1, t_2) &= 2\mathcal{J}^2 \int_{-\infty}^{t_1} dt_3 e^{g_B(t_3, t_2)} - \mathcal{J}^2 \int_{-\infty}^0 dt_3 \left[e^{g_B(t_3, t_2)} + e^{g_D(t_2, t_3)} \right] \\
&\quad - \mathcal{J}^2 \int_0^{t_2} dt_3 \left[e^{g_A(t_3, t_2)} + e^{g_{\bar{A}}(t_2, t_3)} \right]
\end{aligned} \tag{2.71}$$

Inserting Eq. (2.65) in Eq. (2.71) leads to

$$\begin{aligned}
-\frac{b_{B1}''(t_1)}{b_{B1}'(t_1)} &= -\mathcal{J}^2 \int_{-\infty}^{\infty} dt' e^{g_C(t')} \text{sgn}(t' - t_1) - \frac{2b_{B1}'(t_1)}{b_{B1}(t_1) - b_{B2}(0)} \\
&\quad - \mathcal{J}_p^2 \int_{-\infty}^{\infty} dt' \text{sgn}(t' - t_1) e^{p_{g_C}(t')}
\end{aligned} \tag{2.72}$$

$$\begin{aligned}
\frac{b_{B2}''(t_2)}{b_{B2}'(t_2)} &= -\frac{2b_{B2}'(t_2)}{b_{B1}(-\infty) - b_{B2}(t_2)} \\
&\quad - \frac{b_{B2}'(t_2)}{b_{B1}(0) - b_{B2}(t_2)} + \frac{b_{B2}'(t_2)}{b_{B1}(-\infty) - b_{B2}(t_2)} - \frac{b_{B2}^{*'}(t_2)}{b_{B1}^*(0) - b_{B2}^*(t_2)} + \frac{b_{B2}^{*'}(t_2)}{b_{B1}^*(-\infty) - b_{B2}^*(t_2)} \\
&\quad - \frac{b_{A2}'(t_2)}{b_{A1}(t_2) - b_{A2}(t_2)} + \frac{b_{A2}'(t_2)}{b_{A1}(0) - b_{A2}(t_2)} - \frac{b_{A2}^{*'}(t_2)}{b_{A1}^*(t_2) - b_{A2}^*(t_2)} + \frac{b_{A2}^{*'}(t_2)}{b_{A1}^*(0) - b_{A2}^*(t_2)}
\end{aligned} \tag{2.73}$$

We also have the compatibility condition at the boundary between regions B and C, which is

$$g_C(t_1) = \ln \left[\frac{-b'_{B1}(t_1)b'_{B2}(0)}{\mathcal{J}^2(b_{B1}(t_1) - b_{B2}(0))^2} \right] \quad (2.74)$$

taking the derivative of this equation we obtain precisely the first equation in Eq. (2.72) after using Eq. (B.24). This reassures us that the system of equations are not overdetermined. We can easily integrate Eq. (2.74) to obtain

$$\frac{b'_{B2}(0)}{b_{B1}(t_1) - b_{B2}(0)} - \frac{b'_{B2}(0)}{b_{B1}(-\infty) - b_{B2}(0)} = \mathcal{J}^2 \int_{-\infty}^{t_1} dt e^{g_C(t)} \quad (2.75)$$

2.4.2 REGION A

Now $t_2 > t_1 > 0$. From the second Eq. (2.60) we obtain

$$\begin{aligned} \frac{\partial}{\partial t_2} g_A(t_1, t_2) &= 2\mathcal{J}^2 \int_{-\infty}^0 dt_3 e^{g_B(t_3, t_2)} + 2\mathcal{J}^2 \int_0^{t_1} dt_3 e^{g_A(t_3, t_2)} - \mathcal{J}^2 \int_{-\infty}^0 dt_3 \left[e^{g_B(t_3, t_2)} + e^{g_D(t_2, t_3)} \right] \\ &\quad - \mathcal{J}^2 \int_0^{t_2} dt_3 \left[e^{g_A(t_3, t_2)} + e^{\tilde{g}_A(t_2, t_3)} \right] \end{aligned} \quad (2.76)$$

Inserting Eq. (2.65) into Eq. (2.76) results in

$$\begin{aligned} \frac{b''_{A2}(t_2)}{b'_{A2}(t_2)} &= \frac{2b'_{B2}(t_2)}{b_{B1}(0) - b_{B2}(t_2)} - \frac{2b'_{B2}(t_2)}{b_{B1}(-\infty) - b_{B2}(t_2)} - \frac{2b'_{A2}(t_2)}{b_{A1}(0) - b_{A2}(t_2)} \\ &\quad - \frac{b'_{B2}(t_2)}{b_{B1}(0) - b_{B2}(t_2)} + \frac{b'_{B2}(t_2)}{b_{B1}(-\infty) - b_{B2}(t_2)} - \frac{b_{B2}^{*'}(t_2)}{b_{B1}^*(0) - b_{B2}^*(t_2)} + \frac{b_{B2}^{*'}(t_2)}{b_{B1}^*(-\infty) - b_{B2}^*(t_2)} \\ &\quad - \frac{b'_{A2}(t_2)}{b_{A1}(t_2) - b_{A2}(t_2)} + \frac{b'_{A2}(t_2)}{b_{A1}(0) - b_{A2}(t_2)} - \frac{b_{A2}^{*'}(t_2)}{b_{A1}^*(t_2) - b_{A2}^*(t_2)} + \frac{b_{A2}^{*'}(t_2)}{b_{A1}^*(0) - b_{A2}^*(t_2)} \end{aligned} \quad (2.77)$$

The first Eq. (2.60) is given by

$$\begin{aligned}
-\frac{\partial}{\partial t_1} g_A(t_1, t_2) &= -\mathcal{J}^2 \int_{-\infty}^0 dt_3 e^{g_D(t_1, t_3)} - \mathcal{J}^2 \int_0^{t_1} dt_3 e^{\tilde{g}_A(t_1, t_3)} - 2\mathcal{J}^2 \int_{t_1}^{t_2} dt_3 e^{g_A(t_1, t_3)} \\
&\quad + \mathcal{J}^2 \int_{-\infty}^0 dt_3 e^{g_B(t_3, t_1)} + \mathcal{J}^2 \int_0^{t_1} dt_3 e^{g_A(t_3, t_1)}
\end{aligned} \tag{2.78}$$

Inserting Eq. (2.65) into Eq. (2.78) leads to

$$\begin{aligned}
-\frac{b''_{A1}(t_1)}{b'_{A1}(t_1)} &= \frac{2b'_{A1}(t_1)}{b_{A2}(t_1) - b_{A1}(t_1)} \\
&\quad - \frac{b_{B2}'(t_1)}{b_{B1}^*(0) - b_{B2}^*(t_1)} + \frac{b_{B2}^{*'}(t_1)}{b_{B1}^*(-\infty) - b_{B2}^*(t_1)} + \frac{b_{B2}'(t_1)}{b_{B1}(0) - b_{B2}(t_1)} - \frac{b_{B2}'(t_1)}{b_{B1}(-\infty) - b_{B2}(t_1)} \\
&\quad + \frac{b_{A2}'(t_1)}{b_{A1}(t_1) - b_{A2}(t_1)} - \frac{b_{A2}'(t_1)}{b_{A1}(0) - b_{A2}(t_1)} - \frac{b_{A2}^{*'}(t_1)}{b_{A1}^*(t_1) - b_{A2}^*(t_1)} + \frac{b_{A2}^{*'}(t_1)}{b_{A1}^*(0) - b_{A2}^*(t_1)}
\end{aligned} \tag{2.79}$$

From Eqs. (2.77) and (2.79) we conclude

$$\frac{b''_{A1}(t)}{b'_{A1}(t)} + \frac{b''_{A2}(t)}{b'_{A2}(t)} = 2 \left(\frac{b'_{A1}(t) - b'_{A2}(t)}{b_{A1}(t) - b_{A2}(t)} \right) \tag{2.80}$$

This is precisely the logarithmic derivative of the Majorana condition in Eq. (2.66). The compatibility condition at the boundary of region A and region B is

$$\frac{b'_{A1}(0)b'_{A2}(t_2)}{(b_{A1}(0) - b_{A2}(t_2))^2} = \frac{b'_{B1}(0)b'_{B2}(t_2)}{(b_{B1}(0) - b_{B2}(t_2))^2} \tag{2.81}$$

This can be integrated to

$$\frac{b'_{A1}(0)}{b_{A2}(t_2) - b_{A1}(0)} = \frac{b'_{B1}(0)}{b_{B2}(t_2) - b_{B1}(0)} + c_4 \tag{2.82}$$

where c_4 is a constant of integration.

2.4.3 COMBINED EQUATIONS

We adopt a simple choice to solve Eq. (2.82)

$$\begin{aligned} b_{B2}(t) &= b_{A2}(t) \\ b_{B1}(0) &= b_{A1}(0) \\ b'_{B1}(0) &= b'_{A1}(0) \end{aligned} \tag{2.83}$$

Then we find that Eqs. (2.73) and (2.77) are consistent with each other. Collecting all equations, we need to solve

$$\begin{aligned} -\frac{b''_{A1}(t)}{b'_{A1}(t)} &= -\frac{2b'_{A1}(t)}{b_{A1}(t) - b_{A2}(t)} + \frac{b^{*'}_{A2}(t)}{b^*_{B1}(-\infty) - b^*_{A2}(t)} - \frac{b'_{A2}(t)}{b_{B1}(-\infty) - b_{A2}(t)} \\ &+ \frac{b'_{A2}(t)}{b_{A1}(t) - b_{A2}(t)} - \frac{b^{*'}_{A2}(t)}{b^*_{A1}(t) - b^*_{A2}(t)}, \quad t \geq 0 \end{aligned} \tag{2.84}$$

$$b'_{A1}(t)b'_{A2}(t) = -\mathcal{J}^2(b_{A1}(t) - b_{A2}(t))^2, \quad t \geq 0 \tag{2.85}$$

$$\frac{b'_{B2}(0)}{b_{B1}(t) - b_{B2}(0)} = \frac{b'_{B2}(0)}{b_{B1}(-\infty) - b_{B2}(0)} + \mathcal{J}^2 \int_{-\infty}^t dt' e^{g_C(t')}, \quad t \leq 0 \tag{2.86}$$

$$b_{A1}(0) = b_{B1}(0) \tag{2.87}$$

$$b'_{A1}(0) = b'_{B1}(0) \tag{2.88}$$

$$b_{A2}(0) = b_{B2}(0) \tag{2.89}$$

It can be verified that all expressions above are invariant with respect to $\text{SL}(2, \mathbb{C})$ transformations of the $b_{\alpha 1}$ and $b_{\alpha 2}$ fields. Given the values of $b_{B1}(-\infty)$, $b_{B2}(0)$, $b'_{B2}(0)$, and $g_C(t)$, Eqs. (2.86-2.89) determine the values of $b_{A1}(0)$, $b'_{A1}(0)$, and $b_{A2}(0)$. Then Eqs. (2.84, 2.85) uniquely determine $b_{A1}(t)$ and $b_{A2}(t)$ for all $t \geq 0$. Because of the $\text{SL}(2, \mathbb{C})$ invariance, final result for $g_A(t_1, t_2)$ will be independent of the values chosen for $b_{B1}(-\infty)$, $b_{B2}(0)$, $b'_{B2}(0)$.

2.4.4 EXACT SOLUTION

A solution of the form in Eq. (B.38) does not apply to Eqs. (2.84-2.89) in region A because it does not have enough free parameters to satisfy the initial conditions. However, we can use the $SL(2, \mathbb{C})$ invariance of Eqs. (2.84-2.86) to propose the following ansatz:

$$h_{A1}(t) = \frac{a e^{\sigma t} + c}{c e^{\sigma t} + d} \quad (2.90)$$

$$h_{A2}(t) = \frac{a e^{-2i\theta} e^{\sigma t} + b}{c e^{-2i\theta} e^{\sigma t} + d} \quad (2.91)$$

We can now verify that Eq. (2.91) is an exact solution of Eqs. (2.84-2.89). This solution is characterized by 4 complex numbers a, b, c, d and two real numbers θ, σ . These are uniquely determined from the values of $h_{B1}(-\infty)$, $h_{A1}(0)$, $b'_{A1}(0)$, and $h_{A2}(0)$ by the solution of the following 6 equations

$$\begin{aligned} ad - bc &= 1 \\ \sigma &= 2\mathcal{J} \sin(\theta) \\ e^{-4i\theta} &= \frac{(b - d h_{B1}(-\infty))(a^* - c^* h_{B1}^*(-\infty))}{(b^* - d^* h_{B1}^*(-\infty))(a - c h_{B1}(-\infty))} \\ h_{A1}(0) &= \frac{a + b}{c + d} \\ h_{A2}(0) &= \frac{a e^{-2i\theta} + b}{c e^{-2i\theta} + d} \\ b'_{A1}(0) &= \frac{2 \sin(\theta)}{(c + d)^2} \end{aligned} \quad (2.92)$$

Inserting Eq. (2.91) into Eq. (2.65) we obtain the exact solution for the Green's function in region A

$$g_A(t_1, t_2) = \ln \left[\frac{-\sigma^2}{4\mathcal{J}^2 \sinh^2(\sigma(t_1 - t_2)/2 + i\theta)} \right] \quad (2.93)$$

As expected from $SL(2, \mathbb{C})$ invariance, this result is independent of the values of a, b, c, d . Moreover, as stated in Section 2.1, the surprising feature of this result is that it only depends *only* upon the relative time t , and is independent of $\mathcal{T} = (t_1 + t_2)/2$. Indeed Eq. (2.93) obeys a KMS condition^{156,145} for an inverse temperature

$$\beta_f = \frac{2(\pi - 2\theta)}{\sigma} \quad (2.94)$$

we can verify the value in Eq. (2.94) by taking a Fourier transform of Eq. (2.93) and comparing with Eq. (2.45). So we obtain our main result of the large q limit: all of quadrant A is in thermal equilibrium at the temperature β_f . We also numerically integrated Eqs. (2.84-2.89), starting from generic initial conditions, and verified that the numerical solution obeyed the expressions in Eqs. (2.91) and (2.92). The equations in Eq. (2.92) determine the values of σ and θ in the final state, and hence the value of the final temperature via Eq. (2.94). In general, this will be different from the value of the initial temperature in quadrant C.

Eqs. (2.91) and (2.92) also determine the solutions in the other quadrants via the expressions specified earlier. The solutions in region $\tilde{\mathcal{A}}$ follow from the conjugacy property in Eq. (2.70). In quadrant B, we have $h_{B2}(t) = h_{A2}(t)$ in Eq. (2.83), while $h_{B1}(t)$ was specified in Eq. (2.75) using the initial state in quadrant C. Note that the solution in quadrant B is not of a thermal form, as h_{B1} and h_{B2} are not simply related as in Eq. (2.91). The solution in quadrant D follows from that in quadrant B via the conjugacy property in Eq. (2.70). Finally, the initial state in quadrant C was described in Appendix B.6.

Note also that in the limit $\beta_f \mathcal{T} \gg 1$, we have $\theta \ll 1$ and then

$$\sigma = \frac{2\pi}{\beta_f} \quad (2.95)$$

Then Eqs. (2.55) and (2.93) describe the $1/q$ expansion of the low temperature conformal solution¹⁵⁶ describing the equilibrium non-Fermi liquid state (see Appendix B.2). And the value of σ

in Eq. (2.95) is the maximal Lyapunov exponent for quantum chaos¹⁴⁴. This chaos exponent appears in the time evolution of $b_{1,2}(t)$. However, in the large q limit, it does not directly control the rapid thermalization rate. We note that recent studies of Fermi surfaces coupled to gauge fields, and of disordered metals, also found a relaxation/dephasing rate which was larger than the Lyapunov rate^{159,157}.

2.5 CONCLUSIONS

Quantum many-body systems without quasiparticle excitations are expected to locally thermalize in the fastest possible times of order $\hbar/(k_B T)$ as $T \rightarrow 0$, where T is the absolute temperature of the final state¹⁷⁷. This excludes *e.g.* the existence of systems in which the local thermalization rate is given by $\Gamma \sim T^p$ as $T \rightarrow 0$ with $p < 1$, and no counterexamples have been found.

In this chapter we examined SYK models, which saturate the more rigorous bound on the Lyapunov exponent characterizing the rate of growth of chaos in quantum systems¹⁴⁴. Our numerical study of the model with a final Hamiltonian with $q = 4$ showed that this system does thermalize rapidly, and the thermalization rate is consistent with $\Gamma = CT$ at low T where C is dimensionless constant, as indicated in Eq. (2.54) and Fig. 2.4.

We also studied a large q limit of the SYK models, where an exact analytic solution of the non-equilibrium dynamics was possible. Here we found that thermalization of the fermion Green's function was instantaneous. Remarkably, the large q solution given by Eq. (2.65) and (2.91): also has a direct connection with the Schwarzian. The Schwarzian was proposed as an effective Lagrangian for the low energy limit of the equilibrium theory of the SYK model. Consider the Euler-Lagrange equation of motion of a Lagrangian, $\mathcal{L}$, which is the Schwarzian of $b(t)$

$$\mathcal{L}[b(t)] = \frac{b'''(t)}{b'(t)} - \frac{3}{2} \left(\frac{b''(t)}{b'(t)} \right)^2 \quad (2.96)$$

with the equation of motion

$$\left[b'(t)\right]^2 b''''(t) + 3 \left[b''(t)\right]^3 - 4b'(t)b''(t)b'''(t) = 0 \quad (2.97)$$

It can now be verified that the expressions for $b_{1,2}(t)$ in Eq. (2.91) both obey Eq. (2.97), with the parameter σ controlling the exponential growth of $b_{1,2}(t)$. (Note, however, that we did not obtain Eq. (2.91) by solving Eq. (2.97): instead, Eq. (2.91) was obtained by solving the Kadanoff-Baym equations for the large- q SYK model in Eq. (2.60).) The structure of $b_{1,2}(t)$ in Eq. (2.91) is intimately connected to the $SL(2, \mathbb{C})$ invariance and the instantaneous thermalization of the large q limit. The connection with the Schwarzian here indicates that gravitational models^{175,176,5,121,145,106,59} of the quantum quench in AdS_2 , which map to a Schwarzian boundary theory, also exhibit instant thermalization.

Indeed, the equation of motion of the metric in two-dimensional gravity⁵ takes a form identical to that for the two-point fermion correlator in Eq. (2.64). Studies of black hole formation in AdS_2 from a collapsing shell of matter show that the Hawking temperature of the black hole jumps instantaneously to a new equilibrium value after the passage of the shell^{59,58}. These features are strikingly similar to those obtained in our large q analysis. There have been studies of quantum quenches in AdS_2 , either in the context of quantum impurity problems⁶⁰, or in the context of higher-dimensional black holes which have an AdS_2 factor in the low energy limit²²¹; it would be useful to analytically extract the behavior of just AdS_2 by extending such studies.

The final question we must ask is the following: what do our results tell us about the overall thermalization process in the SYK model? In this chapter we have studied the thermalization of the Green's function, which is the simplest object we could have picked. We found that the thermalization was instantaneous in the large q limit, but finite for general q . In order to reconcile the large q limit with numerical results it will be necessary to study higher order corrections in $1/q$ to under-

stand how this connects to the numerical $q = 4$ numerical solution: does the constant $C \rightarrow \infty$ as $q \rightarrow \infty$, or (as pointed to us by Aavishkar Patel) is thermalization at large q a two-step process. In such a two-step scenario, we have a very rapid pre-thermalization of the two point function computed in this article, which is followed by a slower true thermalization of higher order corrections. This pre-thermalization behavior was found in a study of entanglement entropy in an SYK chain for a thermofield double initial state by Gu *et al.*⁸¹. In¹⁹⁴ it was demonstrated that the SYK model obeys eigenstate thermalization.¹²⁴ studied aspects of thermalization of SYK models by very different methods, along with connections to gravity in AdS₂.

The $\frac{1}{q^2}$ correction to the SYK Green's function was computed in²⁰² and was found to improve the approximation significantly. It would be interesting to see if this could be used to provide any insight into the q dependence of the thermalization time. It would also be interesting if the methods used in this work could be adjusted to study non-equilibrium dynamics of observables in the complex SYK model⁷⁹.

3

Thermalization and chaos in a conformal gauge theory

3.1 INTRODUCTION

A remarkable property of closed quantum system exhibiting chaotic dynamics is that they can act as their own heat bath leading to an effective thermal state at late time^{50,195,204,166,227}. One of the great

challenges in condensed matter physics is to understand this process of thermalization in closed many-body systems in the vicinity of interacting quantum critical points. Systems near quantum criticality have fast dynamics and relax on a timescale set by the temperature, $\tau_\phi \sim \frac{\hbar}{k_B T}$. At early times, the timescales of thermalization can be probed via the time dependence of simple local correlation functions. At intermediate times, a process called “scrambling” spreads information about the initial state or perturbation throughout the entire system^{191,94,187,99}. Scrambling after local relaxation has occurred is not easily accessible in local correlation functions, as it must be described by objects that probe degrees of freedom of the entire system.

In quantum systems with many degrees of freedom, the rate of scrambling can be characterized by a quantum Lyapunov exponent which controls the onset of quantum chaos^{126,191,120}. In classical systems, one aspect of chaos is the divergence of initially nearby phase space trajectories. This gives a schematic definition for a classical Lyapunov exponent λ_L as

$$|\delta \mathbf{x}(t)| = e^{\lambda_L t} |\delta \mathbf{x}(0)|. \quad (3.1)$$

An analogous notion in quantum systems is obtained from the behavior of an out-of-time-order correlation function (OTOC)^{126,191,120}. This object has an elegant description in AdS/CFT as the correlation between two entangled black holes connected by a wormhole^{191,168}. In fermionic systems, this object is contained in the regulated squared anti-commutator of two fermionic operators. We denote this object as $\mathcal{F}(t, \mathbf{x})$ which is defined as follows

$$\mathcal{F}(t, \mathbf{x}) = \langle \{ W(t, \mathbf{x}), V(0) \} \{ W(t), V(0) \}^\dagger \rangle_\beta \quad (3.2)$$

where V and W are two fermionic operators separated in space-time.

The squared anti-commutator can be understood as characterizing the effect that one local mea-

surements has on another. For spatially local systems, causality implies that initially separated V and W must have zero anti-commutator (since we are dealing with fermions). However, the squared anti-commutator grows exponentially with time in systems with many local degrees of freedom, and it saturates to its late time value after the “scrambling” time t_* . During this process, the anti-commutator behaves like

$$\mathcal{F}(t, \mathbf{x}) \sim \varepsilon e^{\lambda_L \left(t - \frac{|\mathbf{x}|}{v_B}\right)} \quad (3.3)$$

where λ_L is identified as a Lyapunov exponent in analogy with classical chaos, and v_B is called the butterfly velocity which acts as an effective speed limit. We emphasize that this is expected to hold when there are a large number of local degrees of freedom. More generally, it has been argued recently that the exponential form above is modified by quantum fluctuations in a non-trivial way. Here we focus for simplicity on the spatially averaged object, $\int d^2x \mathcal{F}(\mathbf{x}, t)$, which diagnoses scrambling within the local Hilbert space.

Quantum Lyapunov exponents have by now received intense scrutiny from several different angles. They can be measured experimentally^{198,231,224,228,229,38,225,62,215,129,149}, are related to operator growth^{153,212,223,167,109,151,226,41}, and can be computed numerically or analytically in many model systems^{120,153,212,223,180,82,142,29,96,132,162,117,116,127,192,3,199,78,157,200,57,150,184}. In particular, chaos exponents have been computed for several interesting field theories including scalar matrix theory, the $O(N)$ model, the critical fermi surface, the diffusive metal, and graphene^{196,44,159,157,122}. In this paper, we study many-body chaos in QED_3 with N_f flavors of massless fermions in the large N_f limit. This theory is described by the Lagrangian

$$\mathcal{L} = -\frac{1}{4e^2 N_f} F^2 + \sum_{a=1}^{N_f} i \bar{\psi}_a(x) \gamma^\mu D_\mu \psi_a(x) \quad (3.4)$$

where we define the covariant derivative $D_\mu = \partial_\mu - \frac{iA_\mu}{\sqrt{N_f}}$ and the Lorentzian Dirac matrices are given

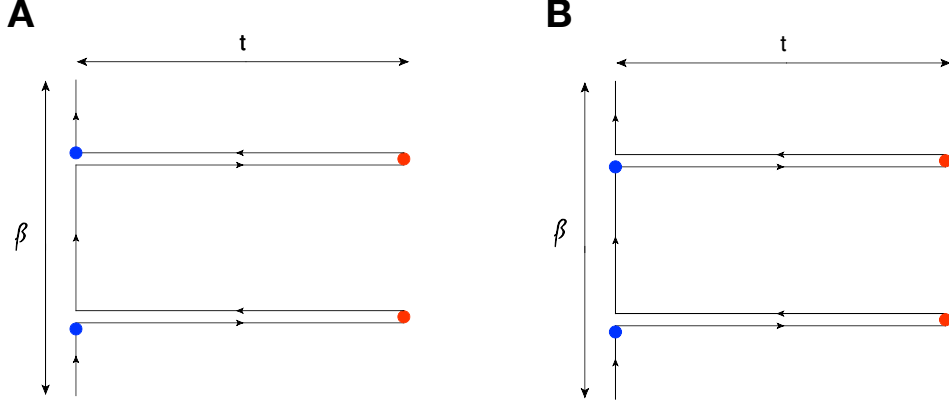


Figure 3.1: A represents the time ordered correlation function on the two fold contour and B represents the OTOC. The red represent A'' and blue represents A' . These are two of the terms in the squared commutator for the photon field. There are two more times, one which is time ordered and one which is out of time order.

by $\gamma^\mu = (\sigma^z, i\sigma^x, i\sigma^y)$ and we work in the $+- -$ signature. QED_3 appears throughout condensed matter as the proposed low energy theory for several lattice systems such as the the Kagome antiferromagnet for $N_f = 2$. It also relevant for a variety of quantum critical points^{115,206}. While these lattice models also contain monopole operators which can drastically modify the long-wavelength physics, for large enough N_f these become irrelevant and the theory flows to a strongly coupled renormalization group fixed point. It is known that the large N_f expansion breaks down below some critical value of N_f , below which the theory is confining^{160,10,85,164}. Below a second critical value of N_f chiral symmetry breaking occurs^{30,105,130}.

Our motivations for studying this model are the following. First, we wanted to understand possible subtleties in the quantum Lyapunov exponent story associated with gauge invariance. In particular, the anti-commutator by itself is not gauge invariant, although the squared anti-commutator is. The anti-commutator can be made gauge invariant by attaching a Wilson line, so this non-locality might conceivably modify the chaos story. Second, it is interesting to understand how to detect failures of the large N_f expansion. One idea is that the chaos bound of Ref.¹⁴⁴ might be in tension with the large N_f result for λ_L , indicating a transition. Of course, the leading $1/N_f$ term is not reliable

away from large N_f , but it might still be suggestive of an issue. Third, we wanted to compare chaos in conventional and unconventional quantum critical points, for example, how does chaos in the $O(N)$ model compare to chaos in QED_3 ?

To analyze chaos in QED_3 , we choose to study the index averaged squared anti-commutator

$$\mathcal{F}(t, \mathbf{x}) = \frac{1}{N_f^2} \sum_{a,b=1}^{N_f} \text{Tr} \left[\sqrt{\hat{\rho}} \{ \psi_a(t, \mathbf{x}), \bar{\psi}_b(0) \} \sqrt{\hat{\rho}} \{ \psi_a(t, \mathbf{x}), \bar{\psi}_b(0) \}^\dagger \right]. \quad (3.5)$$

We restrict attention to the spatial average of $\mathcal{F}$ to avoid issues associated with spatial propagation and focus on chaos within the large local Hilbert space. However, our results could be easily extended to the compute the butterfly velocity at large N_f .

We briefly summarize our main results. We numerically calculate the photon polarization bubble in real time and analyze analytically it in the limit of high temperature and the limit of low frequency. We use it to numerically calculate the fermion decay rate which scales as

$$\Gamma_{\mathbf{k}} \sim \frac{T}{N_f} \quad (3.6)$$

and vanishes as $\mathbf{k} \rightarrow 0$. The Lyapunov exponent takes the form

$$\lambda_L = \frac{2\pi T}{N_f} \mathcal{C} \quad (3.7)$$

with $\mathcal{C} \approx 0.65$. This form is analogous to the form of λ_L in the $O(N)$ model with here the value of $\mathcal{C}$ is slightly higher. Because $\mathcal{C} < 1$, the chaos bound is satisfied for all values of N_f despite the known breakdown of the large N_f expansion. Also, chaos is not significantly stronger in this unconventional quantum critical point as compared to the usual $O(N)$ critical theory, at least at large N .

Notation guide: Throughout we will use lower case letters to denote Lorentz vectors $k^\mu =$

$(k^0, \mathbf{k})$ where the inner product is defined as

$$k^2 = (k^0)^2 - \mathbf{k}^2 \quad (3.8)$$

For Euclidean vectors we will use capital letter $K_\mu = (k_n, \mathbf{k})$ vectors and the Euclidean inner product is defined as

$$K^2 = k_n^2 + \mathbf{k}^2 \quad (3.9)$$

Integrals in Minkowski space are defined as

$$\int_x \equiv \int dt \int_{\mathbf{x}} \quad (3.10a)$$

$$\int_k \equiv \int \frac{dk^0}{2\pi} \int_{\mathbf{k}} \quad (3.10b)$$

Integrals and sums in Euclidean space are defined as are defined

$$\int_X \equiv \int d\tau \int_{\mathbf{x}} \quad (3.11a)$$

$$\oint_K \equiv \frac{1}{\beta} \sum_{k_n} \int_{\mathbf{k}} \quad (3.11b)$$

For Feynman slash notation $\not{k} \equiv \gamma^\mu k_\mu$, we will use the shorthand $\not{k}^0 = \gamma^0 k_0$, $\mathbf{k}\not{\mathbf{k}} = \gamma^i k_i$ and $\not{k} = \gamma^0 k_0 + \gamma^i k_i = \gamma^0 k_0 - \gamma \cdot \mathbf{k}$ where $\gamma = (i\sigma^x, i\sigma^y)$. In Euclidean time $\not{K} \equiv \widetilde{\gamma}^\mu K_\mu$ $\not{K} = \widetilde{\gamma}^0 i k_n + \widetilde{\gamma}^i \mathbf{k}_i$ To go from to $\not{K}$ back to $\not{k}$ we have $i\not{K} \rightarrow \not{k}$. Further conventions are specified in the appendices.

3.2 PRELIMINARIES

In this section, we set up the formalism required to compute $\mathcal{F}(t, \mathbf{x})$ in a $1/N_f$ expansion. The generating function corresponding to Eqn. 3.4 is

$$\mathcal{Z} = \int \mathcal{D}A \mathcal{D}\psi \mathcal{D}\bar{\psi} \exp(iS_M) \quad (3.12)$$

It will also be useful to consider the theory in Euclidean time. Setting $it \rightarrow \tau$ and $A_0 \rightarrow iA_\tau$ we obtain the following Euclidean Lagrangian

$$\mathcal{L}_E = \frac{1}{4e^2 N_f} F^2 - \sum_{a=1}^{N_f} \bar{\psi}_a(X) \tilde{\gamma}_\mu D_\mu \psi_a(X) \quad (3.13)$$

where the Euclidean and Minkowski Dirac matrices are related by $\tilde{\gamma}^0 \equiv \gamma^0$ and $\tilde{\gamma}^j \equiv -i\gamma^j$ and $\tilde{\gamma}^j = \tilde{\gamma}_j$.

We will use repeated lower indices and capital letters when using the Euclidean metric. This gives us the generating functional

$$\mathcal{Z}_E = \int \mathcal{D}A \mathcal{D}\psi \mathcal{D}\bar{\psi} \exp\left(-\int_X \mathcal{L}_E\right) \quad (3.14)$$

where $\int_X \equiv \int_0^\beta d\tau \int_{\mathbf{x}}$. In the limit $e \rightarrow \infty$, we can neglect the Maxwell term in the action 3.14.

To compute $F(t, \mathbf{x})$, we will need the Wightman, Euclidean, and Euclidean propagators for the fermions and the gauge field. For fermions these are defined as

$$\tilde{G}^{ab}(t, \mathbf{x}) = \text{Tr} \left[\sqrt{\tilde{\rho}} \psi_a(t, \mathbf{x}) \sqrt{\tilde{\rho}} \bar{\psi}_b(0, \mathbf{0}) \right] \quad (3.15a)$$

$$G_E^{ab}(\tau, \mathbf{x}) = \text{Tr} \left[\tilde{\rho} \psi_a(\tau, \mathbf{x}) \bar{\psi}_b(0, \mathbf{0}) \right] \quad (3.15b)$$

$$G_R^{ab}(t, \mathbf{x}) = -i \text{Tr} \left[\tilde{\rho} \{ \psi_a(t, \mathbf{x}), \bar{\psi}_b(0, \mathbf{0}) \} \theta(t) \right] \quad (3.15c)$$

In momentum space these propagators can be written in the spectral representation

$$G_W(\not{k}) = \frac{\not{k}}{2 \cosh \frac{\beta k^0}{2}} \quad (3.16a)$$

$$G_E(\not{k}) = \int \frac{d\omega}{2\pi} \frac{\not{k}}{ik_n - \omega} \quad (3.16b)$$

$$G_R(\not{k}) = \int \frac{d\omega}{2\pi} \frac{\not{k}}{k^0 - \omega + i\varepsilon} \quad (3.16c)$$

where we have defined the spectral function as

$$\not{k}(k) = -2 \operatorname{Im}[G_R(\not{k})] \quad (3.17)$$

For free fermions the spectral function is

$$\not{k}(k^0, \mathbf{k}) = \frac{\pi \not{k}}{E_{\mathbf{k}}} [\delta(k^0 - E_{\mathbf{k}}) - \delta(k^0 + E_{\mathbf{k}})] \quad (3.18)$$

which gives us the retarded and Wightman propagators

$$G_W(\not{k}) = \sum_s \frac{s \not{k} \delta(k^0 - s E_{\mathbf{k}})}{2 \cosh \frac{\beta E_{\mathbf{k}}}{2}} \quad (3.19a)$$

$$G_R(\not{k}) = \frac{\not{k}}{(k^0 + i\varepsilon)^2 - E_{\mathbf{k}}^2} \quad (3.19b)$$

Similarly, for the gauge field we define the propagators as

$$\mathcal{D}_W^{\mu\nu}(t, \mathbf{x}) = \operatorname{Tr} \left[\sqrt{\widehat{\rho}} A^\mu(t, \mathbf{x}) \sqrt{\widehat{\rho}} A^\nu(0, \mathbf{0}) \right] \quad (3.20a)$$

$$\mathcal{D}_E^{\mu\nu}(\tau, \mathbf{x}) = i \operatorname{Tr} \left[\widehat{\rho} A^\mu(\tau, \mathbf{x}) A^\nu(0, \mathbf{0}) \right] \quad (3.20b)$$

$$\mathcal{D}_R^{\mu\nu}(t, \mathbf{x}) = -i \operatorname{Tr} \left[\widehat{\rho} [A^\mu(t, \mathbf{x}), A^\nu(0, \mathbf{0})] \theta(t) \right] \quad (3.20c)$$

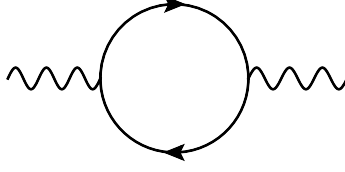


Figure 3.2: The fermion polarization bubble is the $O(1)$ contribution to the gauge field propagator and dominates over the bare term at low energies.

In the spectral representation we can write these as

$$\mathcal{D}_W^{\mu\nu}(k) = \frac{\mathcal{A}^{\mu\nu}(k)}{2 \sinh \frac{\beta k^0}{2}} \quad (3.21a)$$

$$\mathcal{D}_E^{\mu\nu}(K) = \int \frac{d\omega}{2\pi} \frac{\mathcal{A}^{\mu\nu}(\omega, \mathbf{k})}{ik_n - \omega} \quad (3.21b)$$

$$\mathcal{D}_R^{\mu\nu}(k) = \int \frac{d\omega}{2\pi} \frac{\mathcal{A}^{\mu\nu}(\omega, k)}{k^0 - \omega + i\varepsilon} \quad (3.21c)$$

$$(3.21d)$$

where we define the spectral function as

$$\mathcal{A}^{\mu\nu} = -2 \text{Im} \mathcal{D}_R^{\mu\nu}(k) \quad (3.22)$$

The gauge field propagator $\mathcal{D}^{\mu\nu}$ is $O(1)$ in N_f and hence a non-perturbative object. Since the self energy dominates at low energies and $\mathcal{D}^{\mu\nu}$ completely in terms of the self energy.

At finite temperature, the choice of frame breaks Lorentz invariance. As a result, longitudinal and transverse become distinct. In $2 + 1d$, the photon has one longitudinal and one transverse polarization with associated polarization vectors $\varepsilon_L^\mu(k)$ $\varepsilon_T^\mu(k)$. A thermal mass can appear in the longitudinal component of the longitudinal part without violating gauge invariance. To make calculation simpler we will use Coulomb gauge throughout with gauge fixing condition $\nabla \cdot \mathbf{A} = 0$. In this gauge, we define the longitudinal polarization vector $\varepsilon_L^\mu(k) = u^\mu = (1, 0, 0)$, where u^μ is the rest frame of

the medium. For $\mathbf{k}$ lying along the x axis we can write the transverse polarization vector as following

$$\varepsilon_T^\mu(k) = \frac{1}{|\mathbf{k}|} (0, -k^y, k^x) \quad (3.23)$$

In this gauge, we can write the bare propagator in the following form

$$i\mathcal{D}^{\mu\nu}(k^0, \mathbf{k}) = -\frac{ie^2 N_f}{k^2} u^\mu u^\nu - \frac{ie^2 N_f}{k^2} P_T^{\mu\nu}(k) \quad (3.24)$$

where $P_T^{\mu\nu}(k)$ is a transverse projection tensor defined as

$$P_T^{\mu\nu}(k) = \left(\delta^{ij} - \frac{k^i k^j}{|\mathbf{k}|^2} \right) g_i^\mu g_j^\nu \quad (3.25)$$

We can also write $P_T^{\mu\nu}(k)$ as a tensor product of the transverse polarization vectors

$$P_T^{\mu\nu}(k) = \varepsilon_T^\mu(\mathbf{k}) \varepsilon_T^\nu(\mathbf{k}) \quad (3.26)$$

For the self energy, current conservation requires

$$k_\mu \Pi^{\mu\nu}(k) = 0 \quad (3.27)$$

This allows us to write it with the following tensor structure.

$$\Pi^{\mu\nu}(k) = \frac{k^2}{\mathbf{k}^2} \Pi^{00}(k) P_L^{\mu\nu}(k) + \Pi^{\gamma\gamma} P_T^{\mu\nu}(k) \quad (3.28)$$

where longitudinal projector is defined as

$$P_L^{\mu\nu}(k) = \left(-g^{\mu\nu} + \frac{k^\mu k^\nu}{k^2} \right) - P_T^{\mu\nu}(k) \quad (3.29)$$

Since the tensor 3.28 has only two independent components, we can calculate everything in terms of the spatial components of the self energy $\Pi^{xx}(k^0, \mathbf{k})$, $\Pi^{yy}(k^0, \mathbf{k})$ and write $\Pi^{00}(k^0, \mathbf{k})$ in terms of these functions.

We start with the following expression for $\Pi_{\mu\nu}(K)$ in Euclidean time

$$\Pi_{\mu\nu}(ik_n, \mathbf{k}) = \oint_P \text{Tr} \left[\tilde{\gamma}_\mu \frac{-i\cancel{P}}{P^2} \tilde{\gamma}_\nu \frac{-i(\cancel{K} - \cancel{P})}{(K - P)^2} \right] \quad (3.30)$$

For Euclidean gamma matrices we have

$$\text{Tr}[\tilde{\gamma}_\mu \cancel{P} \tilde{\gamma}_\nu \cancel{Q}] = -2(P_\mu(K_\nu - P_\nu) + P_\nu(K_\mu - P_\mu) - \delta_{\mu\nu}(P \cdot (K - P))) \quad (3.31)$$

which gives us

$$\Pi_{\mu\nu}(K) = -2 \oint_{\{P\}} \left[\frac{P_\mu(K - P_\nu) + P_\nu(K - P_\mu) - \delta_{\mu\nu}P \cdot (K - P)}{P^2(K - P)^2} \right] \quad (3.32)$$

We can simplify the $K^\mu P^\nu + K^\nu P^\mu$ terms by shifting $P \rightarrow K - P$ in half of this term to get

$$\Pi_{\mu\nu}(K) = - \oint_{\{P\}} \left[\frac{2\delta_{\mu\nu}}{P^2} + \frac{-K^2\delta_{\mu\nu} + 2K_\mu K_\nu - 4P_\mu P_\nu}{P^2(K - P)^2} \right] \quad (3.33)$$

We leave the rest of the calculation of the polarization bubble to appendix C.1.

We write the full photon propagator in terms of $\Pi^{00}(k^0, \mathbf{k})$ and $\Pi^{yy}(k^0, \mathbf{k})$ the as

$$i\mathcal{D}^{\mu\nu}(k) = -\frac{i}{\frac{1}{c^2 N_f} k^2 - \Pi^{00}(k^0, \mathbf{k})} u^\mu u^\nu - \frac{i}{\frac{1}{c^2 N_f} k^2 - \Pi^{yy}(k^0, \mathbf{k})} P_T^{\mu\nu}(k) \quad (3.34)$$

Since the self energy is $O(1)$, it will always dominate over the bare term. Thus we drop the bare term

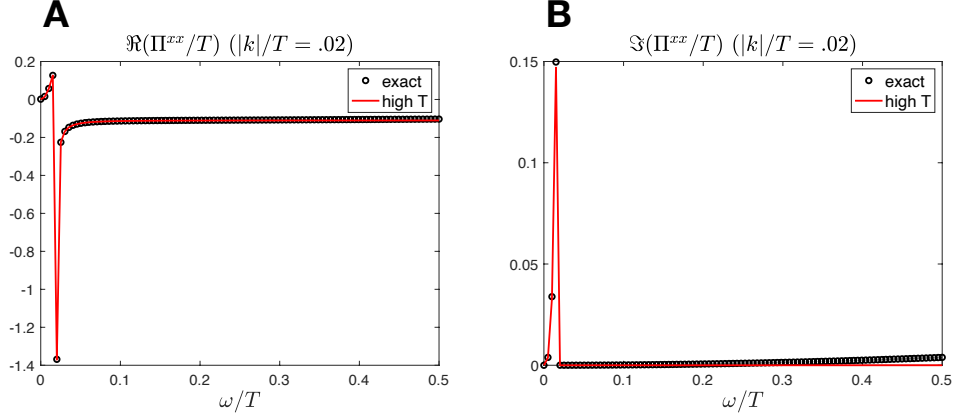


Figure 3.3: The imaginary and real parts of the full polarization component $\Pi^{xx}(k^0, \mathbf{k})$ are plotted against the high temperature limits in A and B respectively.

and write

$$i\mathcal{D}^{\mu\nu}(k) = \frac{i}{\Pi^{00}(k^0, \mathbf{k})} u^\mu u^\nu + \frac{i}{\Pi^{\gamma\gamma}(k^0, \mathbf{k})} P_T^{\mu\nu}(k) \quad (3.35)$$

At finite temperature, a thermal mass $m_{thermal} = \frac{\sqrt{eT \log 2}}{\pi}$ will appear in the longitudinal part of the propagator Π^{00} . This does not violate gauge invariance and is due to the Debye screening of the gauge field.

Next we consider the first correction to the fermion self energy. Since the fermion mass is $1/N_f$ suppressed, to leading order the physical effects are controlled by the imaginary. This introduces a damping rate $\Gamma_{\mathbf{k}}$ leading to a finite lifetime to the fermions. In terms of the photon and free fermion spectral functions, we can write the self energy as

$$\Sigma(\not{k}) = -\frac{1}{N_f} \not{\mathcal{F}}_{\{P\}} \int \frac{d\nu}{2\pi} \int \frac{d\omega}{2\pi} \tilde{\gamma}_\mu \not{p}_n - \omega \tilde{\gamma}_\nu \frac{\mathcal{A}_{\mu\nu}(\nu, \mathbf{k} - \mathbf{p})}{ik_n - ip_n - \nu} \quad (3.36)$$

Doing the Matsubara sum gives us

$$\Sigma(\not{k}) = \frac{1}{N_f} \int_{\mathbf{p}} \int \frac{d\nu}{2\pi} \int \frac{d\omega}{2\pi} \tilde{\gamma}_\mu \not{p} \tilde{\gamma}_\nu \mathcal{A}_{\mu\nu}(\nu, \mathbf{k} - \mathbf{p}) \left(\frac{1 - n_f(\omega) + n_b(\nu)}{ik_n - \omega - \nu} \right) \quad (3.37)$$

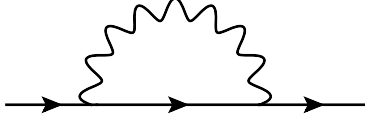


Figure 3.4: The leading contribution to the fermion self energy is $O(\frac{1}{N_f})$ so the correction to the mass can be neglected.

Analytically continuing back to real time and inserting the free fermion spectral function gives us

$$\begin{aligned}\Sigma_R(\not{k}) &= \frac{1}{N_f} \int_{\mathbf{p}} \int \frac{d\nu}{2\pi} \int \frac{d\omega}{2\pi} \sum_{s=\pm} \frac{\pi s \delta(\omega - sE_{\mathbf{p}}) \gamma_{\mu} \not{\epsilon} \gamma_{\nu}}{E_{\mathbf{p}}} \mathcal{A}^{\mu\nu}(\nu, \mathbf{k} - \mathbf{p}) \frac{1 - n_f(\omega) + n_b(\nu)}{i k_n - \omega - \nu} \\ &= \frac{1}{N_f} \int_{\mathbf{p}} \frac{d\nu}{2\pi} \sum_{s=\pm} \frac{s}{2E_{\mathbf{p}}} \gamma_{\mu} \not{\epsilon} \gamma_{\nu} \mathcal{A}^{\mu\nu}(\nu, \mathbf{k} - \mathbf{p}) \frac{1 - n_f(sE_{\mathbf{p}}) + n_b(\nu)}{k^0 - sE_{\mathbf{p}} - \nu + i\varepsilon}\end{aligned}\quad (3.38)$$

where $p_s = (sE_{\mathbf{p}}, \mathbf{p})$ and $E_{\mathbf{p}} = |\mathbf{p}|$. Defining $\Gamma_{\mathbf{k}}$ as the imaginary part of the pole in the propagator, $\Gamma_{\mathbf{k}}$ can be written in terms of the self energy as

$$\Gamma_{\mathbf{k}} = -\frac{1}{2E_{\mathbf{k}}} \text{Im Tr}[\not{k} \Sigma_R(\not{k})] \Big|_{k^0=E_{\mathbf{k}}} \quad (3.39)$$

Taking the imaginary part of Eq. 3.38 we have

$$\begin{aligned}\text{Im } \Sigma_R(\not{k}) &= -\frac{1}{N_f} \int_{\mathbf{p}} \int \frac{d\nu}{2\pi} \sum_{s=\pm} \frac{\pi s}{2E_{\mathbf{p}}} \gamma_{\mu} \not{\epsilon} \gamma_{\nu} \mathcal{A}^{\mu\nu}(\nu, \mathbf{k} - \mathbf{p}) \delta(k^0 - sE_{\mathbf{p}} - \nu) \\ &\quad (1 - n_f(sE_{\mathbf{p}}) + n_b(\nu)) \\ &= -\frac{1}{N_f} \int_{\mathbf{p}} \frac{s}{4E_{\mathbf{p}}} \sum_{s=\pm} \left[\gamma_{\mu} \not{\epsilon} \gamma_{\nu} \mathcal{A}^{\mu\nu}(k - p_s) (1 - n_f(sE_{\mathbf{p}}) + n_b(k^0 - sE_{\mathbf{p}})) \right]\end{aligned}\quad (3.40)$$

Expanding the photon propagator into its longitudinal and transverse components, this becomes

$$\begin{aligned}
\Gamma_{\mathbf{k}} &= -\frac{1}{2E_{\mathbf{k}}} \text{Im Tr}[\not{k}\Sigma_R(\not{k})] \Big|_{k^0=E_{\mathbf{k}}} \\
&= \frac{1}{N_f} \int_{\mathbf{p}} \sum_{s=\pm} \sum_{\lambda=L,T} \frac{s}{4E_{\mathbf{k}}E_{\mathbf{p}}} \mathcal{A}_{\lambda}(k-p_s) (2(k \cdot \varepsilon_{\lambda}(k-p_s))(p_s \cdot \varepsilon_{\lambda}(k-p_s)) - k \cdot p_s (\varepsilon_{\lambda}(k-p_s) \cdot \varepsilon_{\lambda}(k-p_s))) \\
&\quad (1 - n_f(sE_{\mathbf{p}}) + n_b(k^0 - sE_{\mathbf{p}})) \Big|_{k^0=E_{\mathbf{k}}} \tag{3.41}
\end{aligned}$$

We can evaluate Eq. 3.41 in Coulomb gauge by choosing the longitudinal polarization vector $\varepsilon_L^{\mu} = u^{\mu}$. The complete expression is given in Appendix C.2. We also note that this expression is consistent with the finite temperature optical theorem. Implementing $\Gamma_{\mathbf{k}}$ as a shift in the pole, we can write the retarded propagator as

$$G_R(\not{k}) = \frac{1}{2E_{\mathbf{k}}} \left(\frac{\gamma^0 E_{\mathbf{k}} - \gamma \cdot \mathbf{k}}{k^0 - E_{\mathbf{k}} + i\Gamma_{\mathbf{k}}} - \frac{-\gamma^0 E_{\mathbf{k}} + \gamma \cdot \mathbf{k}}{k^0 + E_{\mathbf{k}} + i\Gamma_{\mathbf{k}}} \right). \tag{3.42}$$

Note that the fermion 2-point function is not gauge invariant, so this decay rate is not manifestly gauge invariant and is defined in the Coulomb gauge which we use. It is consistently used below in the $1/N_f$ expansion to compute gauge-invariant observables, and it presumably manifests in other gauge-invariant response functions.

3.3 DIAGRAMMATIC EXPANSION FOR THE SQUARED FERMION ANTI-COMMUTATOR

We now consider the diagrammatic expansion of the squared anti-commutator of two fermion operators by expanding $\mathcal{F}(t, \mathbf{x})$ in the interaction on both time folds and keeping only diagrams to order $1/N_f$ which are uncrossed. The rules are very similar to those summarized in⁷ except all of the propagators and hence rungs have a matrix structure. In summary:

- i. Vertex insertions are restricted to lie on real time folds

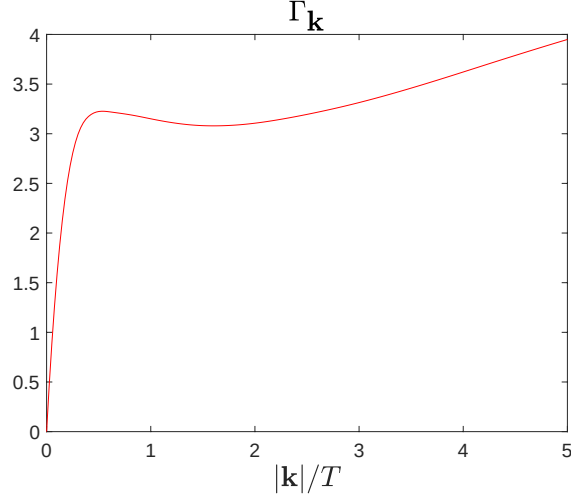


Figure 3.5: A plot of the fermion decay rate Γ_k in units of temperature which vanishes as $k \rightarrow 0$.

2. Horizontal lines correspond to retarded or advanced photon or fermions propagators including self energy corrections
3. Vertical lines correspond to full photon propagators or bare fermion Wightman propagators
4. Each rung or pair of rails is a matrix in spinor space and can be written in terms of a tensor product basis of gamma matrices. A rung or pair of rails is added to the ladder by matrix multiplication.

We now consider the two rungs at this order. The first rung involves a single photon Wightman function sandwiched between two gamma matrices. The whole rung can be giving us the following expression

$$\begin{aligned}
 \mathcal{R}_1(k-k') &= \frac{1}{N_f} \gamma_\mu \mathcal{D}_W^{\mu\nu}(k-k') \gamma_\nu \\
 &= \frac{1}{N_f} \gamma_\mu \mathcal{Q}_b(k^0 - k'^0) \mathcal{A}^{\mu\nu}(k-k') \gamma_\nu
 \end{aligned} \tag{3.43}$$

where we define $\mathcal{Q}_b(\omega) = (2 \sinh(\frac{\beta\omega}{2}))^{-1}$ for bosons and $\mathcal{Q}_f(\omega) = (2 \cosh(\frac{\beta\omega}{2}))^{-1}$ for fermions.

We can write the longitudinal and transverse contributions to $\mathcal{R}_1(k - k')$ in terms of a tensor products of polarization vectors. In $2 + 1d$, we can express the transverse projector $P_T^{\mu\nu}(k)$ as the tensor product of a single transverse polarization vector

$$P_T^{\mu\nu}(k) = \varepsilon_T^\mu(\mathbf{k}) \varepsilon_T^\nu(\mathbf{k}) \quad (3.44)$$

where we define $\varepsilon_T^\nu(\mathbf{k})$ lying along the x axis as

$$\varepsilon_T^\mu(\mathbf{k} - \mathbf{k}') = \frac{1}{|\mathbf{k}|} (0, -k^y, k^x) \quad (3.45)$$

Defining $\bar{k} = k - k'$ as the relative momentum, we can write a general expression for $\mathcal{R}_1(\bar{k})$

$$\mathcal{R}_1(\bar{k}) = \frac{1}{N_f} \sum_{\lambda=L,T} \mathcal{Q}_b(\bar{k}^0) \mathcal{A}_\lambda(\bar{k}) (\not{\epsilon}_\lambda(\bar{k}) \not{\epsilon}_\lambda(\bar{k})) \quad (3.46)$$

where we have separated the longitudinal and transverse parts of the rung. Using the definitions of the longitudinal and transverse polarization vectors this becomes

$$\mathcal{R}_1(\bar{k}) = \frac{\mathcal{Q}_b(\bar{k}^0)}{N_f} \left(\mathcal{A}_L(\bar{k}) (\gamma_0 \gamma_0) + \mathcal{A}_T(\bar{k}) (\gamma_i \gamma_j) (\delta^{ij} - \widehat{\mathbf{k}}^i \widehat{\mathbf{k}}^j) \right) \quad (3.47)$$

The second rung is a box structure consisting of two fermion Wightman functions and two retarded photon propagators.

We can write this as

$$\mathcal{R}_2(k, k') = \frac{1}{N_f} \int_p \gamma_\rho G_W(\not{p} - \not{k}') \gamma_\sigma \gamma_\nu G_W(\not{k} - \not{p}) \gamma_\mu \mathcal{D}_R^{\sigma\nu}(p) \mathcal{D}_R^{\mu\rho}(q - p) \quad (3.48)$$

where $p^\mu = (p^0, \mathbf{p})$ Inserting the free fermion spectral functions, we get

$$\mathcal{R}_2(k, k') = \frac{1}{N_f} \int_p \sum_{ss'=\pm} \frac{\pi^2 ss' \delta((p^0 - k'^0) - s' E_{\mathbf{k}' - \mathbf{p}}) \delta((k^0 - p^0) - s E_{\mathbf{k} - \mathbf{p}})}{E_{\mathbf{k}' - \mathbf{p}} E_{\mathbf{k} - \mathbf{p}}} \mathcal{Q}_f(p^0 - k'^0) \mathcal{Q}_f(k^0 - p^0) (\gamma_\rho (\not{p} - \not{k}') \gamma_\sigma \gamma_\nu (\not{k} - \not{p}) \gamma_\mu) \mathcal{D}_R^{\mu\rho}(q - p) \mathcal{D}_R^{\sigma\nu}(p) \quad (3.49)$$

defining the absolute Lorentz momentum $K = \frac{k+k'}{2}$ corresponding to the Lorentz vector $K^\mu = (K^0, \mathbf{K})$ and the relative momentum $\bar{k} = k - k'$ corresponding to the Lorentz vector $\bar{k}^\mu = (\bar{k}^0, \bar{\mathbf{k}})$.

Shifting $p \rightarrow p + K$, we define $E_\pm = E_{\bar{\mathbf{k}}/2 \pm \mathbf{p}}$ given by

$$E_\pm = \sqrt{|\bar{\mathbf{k}}|^2/4 + |\mathbf{p}|^2 \pm |\bar{\mathbf{k}}||\mathbf{p}| \cos \theta} \quad (3.50)$$

where θ is the angle between $\bar{\mathbf{k}}$ and $\mathbf{p}$

This gives us

$$\mathcal{R}_2(k, k') = \frac{1}{N_f} \int_p \sum_{ss'=\pm} \frac{\pi^2 ss' \delta(\bar{k}^0/2 + p^0 - s E_+) \delta(\bar{k}^0/2 - p^0 - s' E_-)}{E_+ E_-} \mathcal{Q}_f(-\bar{k}^0/2 - p^0) \mathcal{Q}_f(\bar{k}^0/2 - p^0) (\gamma_\rho (\bar{\mathbf{k}}/2 + \not{p}) \gamma_\sigma \gamma_\nu (\bar{\mathbf{k}}/2 - \not{p}) \gamma_\mu) \mathcal{D}_R^{\mu\rho}(q - p - K) \mathcal{D}_R^{\sigma\nu}(p + K) \quad (3.51)$$

performing the p^0 integral in Eq. 3.49, we get

$$\mathcal{R}_2(\bar{k}, K) = \frac{1}{N_f} \int_{\mathbf{p}} \sum_{ss'=\pm} \frac{\pi ss' \delta(\bar{k}^0 - s E_+ - s' E_-)}{2 E_+ E_-} \mathcal{Q}_f(s' E_-) \mathcal{Q}_f(\bar{k}^0 - s' E_-) (\gamma_\rho (\bar{\mathbf{k}}/2 + \not{\tilde{p}}) \gamma_\sigma \gamma_\nu (\bar{\mathbf{k}}/2 - \not{\tilde{p}}) \gamma_\mu) \mathcal{D}_R^{\sigma\nu}(\tilde{p} + K) \mathcal{D}_R^{\mu\rho}(-\tilde{p} - K) \quad (3.52)$$

where $\tilde{p}^\mu = (\bar{k}^0/2 - s' E_-, \mathbf{p})$. In three dimensions $\gamma^\mu \gamma^\nu \gamma^\rho = g^{\mu\nu} \gamma^\rho - g^{\mu\rho} \gamma^\nu + g^{\nu\rho} \gamma^\mu$ which allows us to

write Eqn. 3.52 in terms of the polarization vectors as follows

$$\begin{aligned}
\mathcal{R}_{2,\lambda\lambda'}(\bar{k}, K) &= \frac{1}{N_f} \int_{\mathbf{p}} \sum_{\lambda\lambda'=L,T} \sum_{ss'=\pm} \frac{\pi ss' \delta(\bar{k}^0 - sE_+ - s'E_-)}{2E_+E_-} \mathcal{Q}_f(s'E_-) \mathcal{Q}_f(\bar{k}^0 - s'E_-) \\
&\quad ((\bar{k} + 2\tilde{\mathbf{p}}) \cdot \varepsilon_{\lambda'}(-\tilde{\mathbf{p}} - K) \not{\epsilon}_{\lambda}(\tilde{\mathbf{p}} + K) - (\bar{k}/2 + \tilde{\mathbf{p}}) \varepsilon_{\lambda}(\tilde{\mathbf{p}} + K) \cdot \varepsilon_{\lambda'}(-\tilde{\mathbf{p}} - K) \\
&\quad ((\bar{k} - 2\tilde{\mathbf{p}}) \cdot \varepsilon_{\lambda}(\tilde{\mathbf{p}} + K) \not{\epsilon}_{\lambda'}(-\tilde{\mathbf{p}} - K) - (\bar{k}/2 - \tilde{\mathbf{p}}) \varepsilon_{\lambda}(\tilde{\mathbf{p}} + K) \cdot \varepsilon_{\lambda'}(-\tilde{\mathbf{p}} - K)) \\
&\quad \mathcal{D}_{R\lambda}(\tilde{\mathbf{p}} + K) \mathcal{D}_{R\lambda'}(-\tilde{\mathbf{p}} - K)
\end{aligned} \tag{3.53}$$

Since the external momentum $\mathbf{q}$ is $O\left(\frac{1}{N_f}\right)$, the momenta of the two photon propagators are approximately anti-parallel. Therefore, we only need to consider terms in the sum over λ where $\lambda = \lambda'$.

We can now write the whole rung in the tensor product form

$$\begin{aligned}
\mathcal{R}_2(\bar{k}, K) &= \frac{1}{N_f} \int_{\mathbf{p}} \sum_{ss'=\pm} \sum_{\lambda=L,T} \frac{\pi ss' \delta(\bar{k}^0 - sE_+ - s'E_-)}{2E_+E_-} \mathcal{Q}_f(s'E_-) \mathcal{Q}_f(\bar{k}^0 - s'E_-) \\
&\quad \mathcal{D}_{R,\lambda}(\tilde{\mathbf{p}} + K) \mathcal{D}_{R,\lambda}(-\tilde{\mathbf{p}} - K) \left(\mathcal{R}_{2,\lambda}^{(1)}(\bar{k}, K, \mathbf{p})^\mu \gamma_\mu \mathcal{R}_{2,\lambda}^{(2)}(\bar{k}, K, \mathbf{p})^\nu \gamma_\nu \right)
\end{aligned} \tag{3.54}$$

where we define the coefficients of $\gamma_\mu \otimes \gamma_\nu$ as

$$\begin{aligned}
\mathcal{R}_2^{\mu\nu}(\bar{k}, K) &= \frac{1}{N_f} \int_{\mathbf{p}} \sum_{ss'=\pm} \sum_{\lambda=L,T} \frac{ss' \delta(\bar{k}^0 - sE_+ - s'E_-)}{2E_+E_-} \mathcal{Q}_f(s'E_-) \mathcal{Q}_f(\bar{k}^0 - s'E_-) \\
&\quad \mathcal{D}_{R,\lambda}(\tilde{\mathbf{p}} + K) \mathcal{D}_{R,\lambda}(-\tilde{\mathbf{p}} - K) \mathcal{R}_{2,\lambda}^{(1)}(\bar{k}, K, \mathbf{p})^\mu \mathcal{R}_{2,\lambda}^{(2)}(\bar{k}, K, \mathbf{p})^\nu
\end{aligned} \tag{3.55}$$

The expressions for the matrix elements defined in 3.55 are contained in appendix.

We can do the integral over $|\mathbf{p}|$ where the zeroes of the delta function are given by

$$|\mathbf{p}|_0 = \frac{\bar{k}^0}{2} \sqrt{\frac{(\bar{k}^0)^2 - |\bar{\mathbf{k}}|^2}{(\bar{k}^0)^2 - |\bar{\mathbf{k}}|^2 \cos^2 \theta_{\mathbf{p}}}} \tag{3.56}$$

where $\theta_{\mathbf{p}}$ is the angle between the loop momenta $\mathbf{p}$ and the relative momenta $\bar{\mathbf{k}}$.

For the root to be positive, we have condition $(k^0)^2 < |\mathbf{k}|^2 \cos^2 \theta$ or $(k^0)^2 > |\mathbf{k}|^2$, which occurs for $(s, s') = (1, 1)$ and $(s, s') = (\pm 1, \mp 1)$.

3.4 BETHE SALPETER EQUATION AND THE LYAPUNOV EXPONENT

Our diagrammatic expansion of the squared fermion anti-commutator will give us an integral equation we will solve to obtain the Lyapunov exponent and butterfly velocity. This equation is reminiscent of the original Bethe-Salpeter equations first used to study bound states in quantum electrodynamics in ^{64,182,183}.

In matrix form, the equation for the ladder is basis dependent. However it, turns out that at long times, we can take advantage of the matrix structure of this equation and turn it into two scalar equations for components of a “wave-function” To illustrate this, we start define the Lorentz vectors $q^\mu = (\nu, \mathbf{q})$ $k^\mu = (k^0, \mathbf{k})$ and the following on-shell projectors

$$\Lambda_{\pm}^{(i)}(\mathbf{k}) = \frac{E_{\mathbf{k}} \pm \gamma^0 \gamma \cdot \mathbf{k}}{2E_{\mathbf{k}}} \quad (3.57)$$

where $i = 1, 2$ refers to the particle index and $\gamma \cdot \mathbf{k} = -\gamma^i k_i$. These have the following properties

$$\Lambda_{\pm}^{(i)}(\mathbf{k}) \Lambda_{\pm}^{(i)}(\mathbf{k}) = \Lambda_{\pm}^{(i)}(\mathbf{k}), \quad \Lambda_{\pm}^{(i)}(\mathbf{k}) \Lambda_{\mp}^{(i)}(\mathbf{k}) = 0, \quad \Lambda_{+}^{(i)}(\mathbf{k}) + \Lambda_{-}^{(i)}(\mathbf{k}) = 1 \quad (3.58)$$

The eigenvectors of $\Lambda_{\pm}^{(i)}(\mathbf{k})$ are

$$u_{+}(k) = \left(\frac{E_{\mathbf{k}}}{ik^x + k^y}, 1 \right) \quad (3.59a)$$

$$u_{-}(k) = \left(-\frac{E_{\mathbf{k}}}{ik^x + k^y}, 1 \right) \quad (3.59b)$$

From Eqns. 3.59a and 3.59b we see $u_{-}(\mathbf{k}) = u_{+}(-\mathbf{k})$. Acting with the projectors on $u_{\pm}(\mathbf{k})$ we get

$$\Lambda_{\pm}(\mathbf{k})u_{\pm}(\mathbf{k}) = u_{\pm}(\mathbf{k}) \quad \Lambda_{\mp}(\mathbf{k})u_{\pm}(\mathbf{k}) = 0 \quad (3.60a)$$

From this, we conclude that $u_{\pm}(\mathbf{k})$ are eigenvectors of $\Lambda_{\pm}(\mathbf{k})$ with eigenvalue one and $u_{\pm}(\mathbf{k})$ are an eigenvectors of $\Lambda_{\mp}(\mathbf{k})$ with eigenvalue zero. We define $\Lambda_s^{(1)}(\mathbf{k})\Lambda_{s'}^{(2)}(\mathbf{k})$ as the tensor product of two projection operators. Eq. 3.58 gives us the additional four properties

$$(\Lambda_{+}^{(1)}(\mathbf{k})\Lambda_{+}^{(2)}(\mathbf{k}) + (\Lambda_{-}^{(1)}(\mathbf{k})\Lambda_{+}^{(2)}(\mathbf{k}) + (\Lambda_{+}^{(1)}(\mathbf{k})\Lambda_{-}^{(2)}(\mathbf{k})) + (\Lambda_{-}^{(1)}(\mathbf{k})\Lambda_{-}^{(2)}(\mathbf{k}))) = 1 \quad (3.61)$$

The eigenvectors of $u_{\pm}^{(1,2)}$. $\Lambda_s^{(1)}(\mathbf{k})\Lambda_{s'}^{(2)}(\mathbf{k})$ can be written as tensor products of the form

$$u_{ss'}^{(1,2)}(\mathbf{k}) = u_s^{(1)}(\mathbf{k})u_{s'}^{(2)}(\mathbf{k}) \quad (3.62)$$

We can use the free fermion spectral function to write the retarded Green's function in the form $\not{k}g_R(k)$.

$$G_R^*(\not{k} - \not{q})G_R(\not{k}) = (\not{k} - \not{q})\not{k}g_R^*(k - q)g_R(k) \quad (3.63)$$

Consider the product $g_R^*(q - k)g_R(k)$. Writing it explicitly in terms of momentum and frequency arguments we have

$$g_R^*(k - q)g_R(k) = \sum_{ss'=\pm} \frac{ss'}{4E_{\mathbf{k}}E_{\mathbf{k}-\mathbf{q}}} \left(\frac{1}{k^0 - \nu - sE_{\mathbf{k}-\mathbf{q}} - i\varepsilon} \right) \left(\frac{1}{k^0 - s'E_{\mathbf{k}} + i\varepsilon} \right) \quad (3.64)$$

Evaluating the k^0 integral we have four poles: $k^0 = \nu - sE_{\mathbf{k}-\mathbf{q}} + i\varepsilon$ and $k^0 = s'E_{\mathbf{k}} - i\varepsilon$ which give residues proportional to $\sim (\nu + sE_{\mathbf{k}-\mathbf{q}} - s'E_{\mathbf{k}})^{-1}$. To get the longtime behavior, we keep only the contribution from the most singular terms $\sim (\nu \pm E_{\mathbf{k}-\mathbf{q}} \mp E_{\mathbf{k}})^{-1}$ where we close the contour clockwise on the lower half plane

In the end we have

$$g_R^*(k-q)g_R(k) \rightarrow \sum_{s=\pm} -\frac{\pi i \delta(k^0 - sE_{\mathbf{k}})}{2E_{\mathbf{k}-\mathbf{q}}E_{\mathbf{k}}} \left[\frac{1}{\nu - s(E_{\mathbf{k}} - E_{\mathbf{k}-\mathbf{q}}) + i\varepsilon} \right] \quad (3.65)$$

Note that for $\mathbf{q} = 0$, these terms become proportional to ν^{-1} . Assuming $\Gamma_{\mathbf{q}-\mathbf{k}} \approx \Gamma_{\mathbf{q}}$ we can shift the poles in 3.65 to get

$$g_R^*(k-q)g_R(k) \rightarrow \sum_{s=\pm} \frac{\pi i \delta(k^0 - sE_{\mathbf{k}})}{2E_{\mathbf{k}-\mathbf{q}}E_{\mathbf{k}}} \left[\frac{1}{\nu - s(E_{\mathbf{k}} - E_{\mathbf{k}-\mathbf{q}}) + 2i\Gamma_{\mathbf{k}}} \right] \quad (3.66)$$

For small $\mathbf{q}$, we can express Eqn. 3.65 in terms of projection operators as follows

$$G_R^*(\not{k}-\not{q})G_R(\not{k}) \approx \sum_{s=\pm} s 2\pi i \delta(k^0 - sE_{\mathbf{k}}) \frac{\Lambda_s^{(1)}(\mathbf{k})\gamma^0\Lambda_s^{(2)}(\mathbf{k})\gamma^0}{\nu - s(E_{\mathbf{k}} - E_{\mathbf{k}-\mathbf{q}}) + 2i\Gamma_{\mathbf{k}}} \quad (3.67)$$

where we used using $(\gamma^0)^2 = 1$. Consider the first term in the ladder, which is the tensor product of two retarded Green's functions

$$\mathcal{F}_0(t, \mathbf{x}) = \frac{1}{N_f} [G_R(t, \mathbf{x})G_R^*(t, \mathbf{x})] \quad (3.68)$$

In momentum space we can define the object $f(q, k)$ as

$$\mathcal{F}(q) = \frac{1}{N_f} \int_k f(q, k) \quad (3.69)$$

And write the first term as

$$\mathcal{F}_0(q) = \frac{1}{N_f} \int_k G_R^*(\not{k}-\not{q})G_R(\not{k}) \quad (3.70)$$

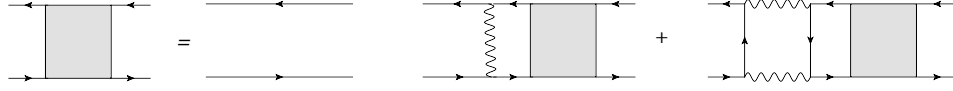


Figure 3.6: A diagrammatic picture of the square of the fermion anticommutator. The lower rails and retarded propagators and upper rails are advanced propagators and the rails are Wightman functions.

We can now write the infinite ladder as

$$\mathcal{F}(q) = \frac{1}{N_f} \int_k G_R^*(\not{k} - \not{q}) G_R(\not{k}) \left(1 + \int_{k'} \mathcal{R}(k, k') f(k, k', q) \right) \quad (3.71)$$

Where $q^\mu = (\nu, \mathbf{q})$, $k^\mu = (k^0, \mathbf{k})$ and $k'^\mu = (k'^0, \mathbf{k}')$, and $\mathcal{R}(k, k')$ is the sum of the two rungs. It is important to note that we have computed the two rungs in a different basis, and we would need to use the Fierz identities to explicitly write $\mathcal{R}(k, k')$ in the correct matrix form.

Assuming exponential growth at long times, the ladder will remain invariant under the addition of one rung. For $\mathbf{q}$ small, we can make the approximation $E_{\mathbf{k}} - E_{\mathbf{k}-\mathbf{q}} = \partial E_{\mathbf{q}} \approx \mathbf{q} \cdot \mathbf{v}_{\mathbf{k}}$ where $\mathbf{v}_{\mathbf{k}} = \nabla_{\mathbf{k}} E_{\mathbf{k}}$ is the group velocity. Dropping the constant term and inserting Eqn. 3.67 into 3.71, we get

$$(-i\nu + i\partial E_{\mathbf{q}} + 2\Gamma_{\mathbf{k}})f(q, k) = \frac{1}{N_f} \int_{k'} \sum_{s=\pm} (\Lambda_s^{(1)}(\mathbf{k}) \gamma^0 \Lambda_s^{(2)}(\mathbf{k}) \gamma^0) \delta(k^0 - E_{\mathbf{k}}) \mathcal{R}(k, k') f(q, k') \quad (3.72)$$

Acting on both sides of Eqn. 3.72 with $(\Lambda_s^{(1)}(\mathbf{k}) \gamma^0 \Lambda_s^{(2)}(\mathbf{k}) \gamma^0)$, we see from Eqn. 3.61 that Eqn. 3.72 is just multiplied by one or zero. So instead of considering the matrix $f(q, k)$, we can consider an eigenvalue equation for a vector $v(q, k)$ which is an eigenvector of the projection operators. We can now propose the following on-shell form of $v(q, k)$

$$v(\nu, \mathbf{q}, \mathbf{k}) = \sum_{s=\pm} \Lambda_s^{(1)}(\mathbf{k}) \Lambda_s^{(2)}(\mathbf{k}) \delta(k^0 - sE_{\mathbf{k}}) v_s(\nu, \mathbf{q}, \mathbf{k}) \quad (3.73)$$

Since $v_{\pm}(\nu, \mathbf{q}, \mathbf{k})$ are eigenstates of $\Lambda_{\pm}^{(1)}(\mathbf{k}) \Lambda_{\pm}^{(2)}(\mathbf{k})$ for small q we can write $v_{\pm}^{(i)}(\nu, \mathbf{q}, \mathbf{k})$ in the

form

$$v_{\pm}(\nu, \mathbf{q}, \mathbf{k}) = u_{\pm}^{(1,2)}(\mathbf{k}) \psi_{\pm}(\nu, \mathbf{q}, \mathbf{k}) \quad (3.74)$$

where $\psi_{\pm}(\nu, \mathbf{q}, \mathbf{k})$ is a scalar function. To write equations for $\psi_{\pm}(\nu, \mathbf{q}, \mathbf{k})$ we take the inner product of both sides, defining a scalar kernel written in terms of both rungs and as

$$K_{ss'}(\mathbf{k}, \mathbf{k}') = \frac{\langle u_s^{(1,2)}(\mathbf{k}) | \gamma_{(1)}^0 \gamma_{(2)}^0 (\mathcal{R}_1(s\mathbf{k}, s'\mathbf{k}') + \mathcal{R}_2(s\mathbf{k}, s'\mathbf{k}')) | u_{s'}^{(1,2)}(\mathbf{k}') \rangle}{\langle u_s^{(1,2)}(\mathbf{k}) | u_s^{(1,2)}(\mathbf{k}) \rangle} \quad (3.75)$$

Because inner product is basis independent, it does not matter that the rungs are written in different bases.

$$(-i\nu + is\partial E_{\mathbf{q}} + 2\Gamma_{\mathbf{k}}) \psi_s(\nu, \mathbf{q}, \mathbf{k}) = \sum_{s'=\pm} \frac{1}{N_f} \int_{\mathbf{k}'} K_{ss'}(\mathbf{k}, \mathbf{k}') \psi_{s'}(\nu, \mathbf{q}, \mathbf{k}') \quad (3.76)$$

Eqns. 3.76 now looks like two coupled effective Dirac equations for a scalar wave function. We can now define a kernel matrix in the ss' indices as

$$\widehat{K}_{ss'} = K_{ss'} - 2N_f \Gamma_{\mathbf{k}} \delta_{ss'} \quad (3.77)$$

and $\Psi(\nu, \mathbf{q}, \mathbf{k}) = (\psi_+(\nu, \mathbf{q}, \mathbf{k}), \psi_-(\nu, \mathbf{q}, \mathbf{k}))^T$

$$(-i\nu \widehat{\sigma}_0 + i\partial E_{\mathbf{q}} \widehat{\sigma}_z) \Psi(\nu, \mathbf{q}, \mathbf{k}) = \frac{1}{N_f} \int_{\mathbf{k}'} \widehat{K}(\mathbf{k}, \mathbf{k}') \Psi(\nu, \mathbf{q}, \mathbf{k}') \quad (3.78)$$

In the time domain, this becomes

$$(\partial_t \widehat{\sigma}_0 + i\mathbf{q} \cdot v_{\mathbf{k}} \widehat{\sigma}_z) \Psi(t, \mathbf{q}, \mathbf{k}) = \frac{1}{N_f} \int_{\mathbf{k}'} \widehat{K}(\mathbf{k}, \mathbf{k}') \Psi(t, \mathbf{q}, \mathbf{k}') \quad (3.79)$$

3.5 NUMERICS

We define the Lyapunov exponent to be the largest eigenvalue of $\widehat{K}$ at $\mathbf{q} = 0$ with eigenvector ψ_1 . We assume ψ_1 is rotationally symmetric so the eigenvalue equation can be simplified by projecting both sides onto states with angular momentum $l = 0$. The resulting eigenvalue equation involves a one dimensional integral transform, involving only the magnitude of the momentum. Below we summarize the computational procedure, with further details in Appendix C.5.

First, we compute Π^{xx} and Π^{yy} numerically on a fine grid in $(\omega, |\mathbf{k}|)$ space and use these to construct the full photon spectral function matrix. We then use these to calculate the fermion decay rate and the matrix components of the rung functions. Since these quantities effectively vanish when the norms of their arguments greatly exceed T , we use a simple linearly spaced grid with a hard momentum cutoff of order $10 T$. The temperature is the only scale in the problem, so the entire numerical computation is set up in terms of the scaled variables $\mathfrak{k} = k/T$ and $\mathfrak{w} = \omega/T$.

We can write λ_L in the form

$$\lambda_L = \frac{2\pi T}{N_f} \mathcal{C} \quad (3.80)$$

Numerics give $\mathcal{C} \approx 0.65$. λ_L takes the same form for the $O(N)$ model with $\mathcal{C} = 0.51$. For nonzero $\mathbf{q}$ we can also calculate a butterfly velocity v_B which we expect to be $v_B \sim 1$. We do not give a precise number as computing v_B is more numerically involved and the spatial propagation is expected to be significantly modified by higher order in $\frac{1}{N_f}$ corrections²²³.

3.6 DISCUSSION

In this work we studied thermalization and chaos in QED_3 at finite temperature in a large N_f expansion. We calculated the photon polarization tensor in real time numerically and also obtained exact expressions in the limits of high temperature and high frequency. We numerically computed the

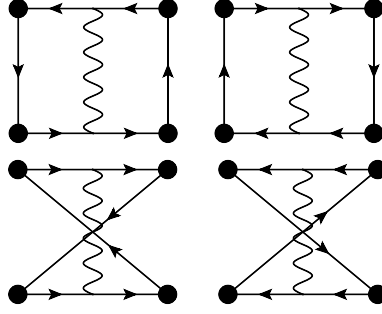


Figure 3.7: We get four types of terms from expanding the squared current-current commutator to first order. The first nontrivial diagrams can be divided into two groups and are shown above. The black dots are current insertions. Each group contains a diagram corresponding to each direction of particle flow. Only the diagrams with uncrossed rungs will contribute to chaos.

fermion decay rate and obtained the Lyapunov exponent from a two-body Dirac-like equation for the squared anti-commutator.

While the object we have computed, $\mathcal{F}(t, \mathbf{x})$, is gauge invariant, the fermion anti-commutator by itself is not. Furthermore, because of their anti-commutation relations, fermions are inherently non-local, thus the squared anti-commutator we are computing is actually a non-local object even though the growth is causal.

We can compare the growth of the squared fermion anti-commutator with the growth of local operators. The most natural local operator in QED₃ is the fermion current. This is defined as

$$j_a^\mu(t, \mathbf{x}) = \bar{\psi}_a(t, \mathbf{x}) \gamma^\mu \psi_a \quad (3.81)$$

We can consider the squared commutator of two current operators

$$\mathcal{C}(t, \mathbf{x}) = \frac{1}{N_f} \delta_{\mu\rho} \delta_{\nu\sigma} \text{Tr} \left[\sqrt{\hat{\rho}} [j_a^\mu(t, \mathbf{x}), j_b^\nu(0, 0)] \sqrt{\hat{\rho}} [j_a^\rho(t, \mathbf{x}), j_b^\sigma(0, 0)]^\dagger \right] \quad (3.82)$$

Another object we could have considered is the squared commutator for the gauge field. Due to current conservation, we know that the photon and current correlation functions are related up to

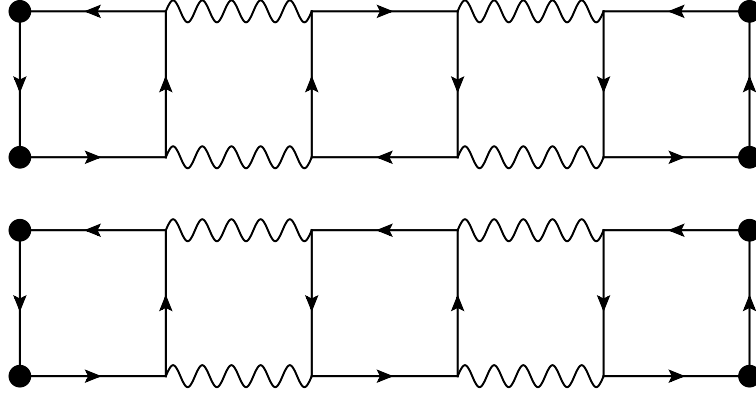


Figure 3.8: Terms contributing to the current current commutator at higher order.

contact terms which vanish on shell.

We expect composite operators will grow as fast as elementary operators⁴⁴. However, it is not guaranteed that these grow at the same rate. We can compare $\mathcal{C}(t, \mathbf{x})$ to $\mathcal{F}(t, \mathbf{x})$ by looking at the structure of the ladder diagrams. The technical details are left to appendix C.4). We see that all of the rungs in $\mathcal{F}(t, \mathbf{x})$ are also contained in $\mathcal{C}(t, \mathbf{x})$ as well as additional structures with crossed legs and particle arrows oriented in opposite directions. The internal structure of this ladder contains the same terms in the squared fermion anti-commutator as well as other pieces. If we expand $\mathcal{C}(t, \mathbf{x})$ further, we obtain diagrams containing particle flows in opposing directions as shown in Fig. 3.8. In principle, these can also occur in the ladder expansion for $\mathcal{F}(t, \mathbf{x})$. However, these terms are suppressed by $\frac{1}{N_f}$, so they do not contribute to chaos to leading order. Thus, we argue that the diagrams in $\mathcal{F}(t, \mathbf{x})$ are all contained in $\mathcal{C}(t, \mathbf{x})$, so chaos must grow at least as fast as the rate obtained from $\mathcal{F}(t, \mathbf{x})$.

At small N_f , the fermions are expected to develop a dynamical mass through a chiral phase transition. In this situation, we expect an exponentially dilute gas of particles with an exponentially long scattering time at temperature below the mass gap. This would correspond to a Lyapunov exponent $\lambda_L \sim e^{-m/T}$. It has also been proposed that QED₃ exhibits an intermediate phase which

is dominated by fluctuations in a vector channel and is characterized by a Lorentz-breaking vector condensate^{30,105}. It would be interesting to see how λ_L scales in this regime.

In the future it would be interesting to see if one could adapt a kinetic method such as described in⁷⁸ to simplify the calculation of chaos exponents in other gauge theories. It would also be interesting to improve numerics so that the Lyapunov exponent can be calculated to higher accuracy and get an estimate for the butterfly velocity. Our work also serves as a starting point to compute other dynamical quantities in finite temperature QED₃ such as the charge diffusion coefficient in systems with disorder. While the decay rate we computed is not gauge invariant, it is the first step in the computation of physical response functions.

4

Bounding Transport in Disordered Holographic Matter

4.1 INTRODUCTION

In conventional metals such as iron or aluminum, Fermi liquid theory predicts that the conductivity $\sigma \sim T^{-2}$, and indeed at low temperatures this scaling can be confirmed for most metals¹³. However,

there is a long-standing experimental puzzle in which certain “strange metals” instead have $\sigma \sim T^{-1}$ ³³. Unlike ordinary Fermi liquids, these strange metals are believed to be strongly correlated, making it difficult to find controllable theoretical models for them^{201,179}.

A few years ago, it was noted that one can use the classic Drude formula for quasiparticle transport¹³ to define a relaxation time τ :

$$\tau \equiv \frac{m}{ne^2}\sigma, \quad (4.1)$$

with n the carrier density and m a quasiparticle effective mass (measured by quantum oscillations). One finds that for many strange metals³³:

$$\tau \sim \frac{\hbar}{k_B T}. \quad (4.2)$$

However, it is hard to take such a result seriously as a quasiparticle scattering time, because this time scale has been argued for some time to be the fastest possible time scale characterizing the dynamics in any interacting quantum system^{171,173,177}. This has recently been revisited in¹⁴⁴. If we take τ seriously as some type of “relaxation time”, this time scale should not be faster than the “interaction time scale” (4.2). However,⁸⁸ pointed out that it would not be unreasonable to demand

$$\tau \gtrsim \frac{\hbar}{k_B T}, \quad (4.3)$$

In the absence of quasiparticles,⁸⁸ further noted that it is easier to postulate that diffusion constants of charge and energy are related to a relaxation time τ :

$$D \gtrsim v^2 \tau \gtrsim v^2 \frac{\hbar}{k_B T}, \quad (4.4)$$

where v is a velocity scale. This formula is better than (4.1) since it does not depend on the existence

of quasiparticles to make sense. If v was independent of temperature, then since $D\chi = \sigma$, with χ the charge susceptibility,⁸⁸ proposed that $\sigma \sim T^{-1}$ in strange metals because their charge diffusion constant saturates the universal lower bound (4.4). The universality of such a bound is appealing, but without a precise conjecture for how to compute v , such a bound is not useful. For example, if v and/or χ has temperature-dependence, then (4.4) can be satisfied while σ does not scale as T^{-1} .

Assuming a conjecture for what v is, one way to check such a bound rigorously is to use gauge-gravity duality^{230,92}. This technique allows us to access the physics of certain strongly interacting finite temperature and density quantum systems by mapping their dynamics onto classical gravity. Indeed, early results¹²⁵ out of gauge-gravity duality suggested a universal viscosity bound relating the shear viscosity η to the entropy density s :

$$\eta \gtrsim \frac{\hbar s}{k_B}. \quad (4.5)$$

A precise coefficient was found for the simplest holographic models¹²⁵ and was conjectured to be universal. Such a viscosity bound can be understood by the following chain of logic:

$$\eta \sim \varepsilon \tau \gtrsim \frac{\hbar \varepsilon}{k_B T} \sim \frac{\hbar s}{k_B}. \quad (4.6)$$

with ε the energy density. More recently, it has been shown that it is possible to parametrically violate these viscosity bounds^{32,31,165,147,104,93,2,34,133}. Nonetheless, (4.5) qualitatively holds for a large number of theories, and has not been meaningfully violated experimentally¹. Hence, there is hope that the original conjecture (4.5) is still an excellent frame for thinking about the viscosity of strongly interacting quantum systems, even if it is not precise.

Given a sharp conjecture for what the velocity scale v is in (4.4) and the zoo of holographic models which are now routinely studied²³⁰, it is natural to use holographic approaches to test (4.4). For

the remainder of the chapter, we will set $c = \hbar = k_B = 1$ for simplicity; such units can straightforwardly be restored with dimensional analysis.

4.1.1 CHARGE DIFFUSION AND THE BUTTERFLY EFFECT

The recent papers^{23,24} have proposed that v should be interpreted as a velocity called the butterfly velocity v_B . Let us specialize to the study of charge diffusion in a locally charge neutral quantum field theory. In this case,²³ computed D in a large set of spatially homogeneous low temperature scaling theories and, in many cases, found the simple relation:

$$D = \mathcal{C} \frac{v_B^2}{T}, \quad (4.7)$$

with $\mathcal{C}$ an $\mathcal{O}(1)$ constant which depends on certain low energy scaling dimensions of the theory, but not on further details. If $v = v_B$, then (4.4) suggests we should find

$$D \gtrsim \frac{v_B^2}{T} \quad (4.8)$$

in more generic situations.

Before we discuss the extent to which (4.8) holds, let us introduce what the butterfly velocity actually is. Let us consider a generic quantum field theory with operator V , localized around $x = 0$ and $t = 0$, and W , localized around $x = x_0$ and $t = t_0$. If $t_0 = 0$, then causality implies that V and W must be uncorrelated. At later times, this need not be the case. In strongly interacting systems, one will generically find^{191,168}

$$\frac{\langle V(0,0) W(x_0,t_0) V(0,0) W(x_0,t_0) \rangle_T}{\langle V(0,0) V(0,0) \rangle_T \langle W(x_0,t_0) W(x_0,t_0) \rangle_T} = 1 - e^{\lambda_L(t_0 - t_* - |x_0|/v_B)}, \quad (4.9)$$

where t_* is a time scale called the “scrambling time”, and v_B is a velocity scale called the butterfly

velocity. When $x_0 = 0$, this equation is reminiscent of chaos theory. In a chaotic system, a small perturbation can grow exponentially large at late times, and so two nearly identical initial conditions can lead to dramatically different outcomes: this is called the butterfly effect. λ_L , identified as a Lyapunov exponent, tells us the rate at which the quantum system can become “scrambled” and lose memory of its initial state. Recently, ¹⁴⁴ has pointed out that under plausible physical assumptions,

$$\lambda_L \leq 2\pi T. \quad (4.10)$$

This gives a precise meaning to (4.3). When $x_0 \neq 0$, there is a further spatial delay in scrambling – this is what is captured by the butterfly velocity. One can crudely think of it as the speed at which quantum information can propagate through the system. More precisely, ¹⁶⁹ has suggested identifying the butterfly velocity with an effective Lieb-Robinson velocity – this latter velocity scale has been of great importance in quantum information theory for many years. For our purposes, the precise quantum information theoretic interpretation of v_B is not important – as emphasized in ²³, we focus on v_B since it is a natural velocity scale to define in a strongly-coupled quantum system.

4.1.2 BREAKING TRANSLATION SYMMETRY

We will test (4.8) in charge neutral theories without translation symmetry. There are two reasons why such a test is important. Firstly, to the extent one can even define viscosity in such systems, translation symmetry breaking parametrically spoils the celebrated viscosity bound (4.5)^{93,2,34,133}. D is well-defined in such theories as long as charge is conserved, but one might expect that translation symmetry breaking can spoil any bound. Secondly, and more importantly, in a typical metal (which is at finite charge density) the only reason that the charge diffusion constant is not infinite is because impurities or umklapp processes break translation invariance. When translation invariance is weak^{91,136,141}, σ and D are both parametrically large. Hence, translation symmetry breaking must be

a non-perturbatively strong effect in order for D to have any chance of saturating a universal lower bound such as (4.8). So although we will be studying charge neutral systems for computational simplicity, where σ and D are finite even in homogeneous systems, it is crucial that (4.8) be robust to translation symmetry breaking in order for it to be a sensible proposal.

There are some reasons to be optimistic about (4.8), at least in holographic models. It has recently been shown that some classes of holographic models (in particular, the Einstein-Maxwell theory in four bulk spacetime dimensions) admit sharp conductivity bounds^{75,76,100}.^{*} Such conductivity bounds can be saturated by the predictions of simpler models of disorder such as massive gravity²⁸ or Q-lattice/linear axion models^{51,8,52,69}. Hence, it may be the case that (4.8) is robust to translation symmetry breaking. On the other hand, simple counterexamples to the conductivity bounds suggested⁶³ by these simple models can readily be generated¹⁰⁰ even by simply changing the spacetime dimension.

In this chapter, we compute D and v_B in inhomogeneous holographic matter coupled to a charge-neutral scalar operator $\mathcal{O}$. If the field H which linearly couples to $\mathcal{O}$ varies on long (hydrodynamic) spatial length scales ξ , we can construct the dual geometry and compute D and v_B analytically. Our computation demonstrates that there is no sharp lower bound for the charge diffusion constant, even in our simple class of models. In fact, we find that in the same low temperature scaling regimes studied in²³:

$$D \leq \mathcal{C} \frac{v_B^2}{2T}, \quad (4.11)$$

with equality only holding in homogeneous backgrounds where H is constant. Hence, (4.8) does not generally hold with a precise constant prefactor. While we do not present an explicit example

^{*}Such conductivity bounds must always be interpreted within the context of a specific holographic action: namely, for any given *theory*, the disorder profile cannot reduce the conductivity below a minimal value. It is possible to, in addition, modify the boundary theory by modifying the bulk action, to decrease the conductivity further^{15,71}. Since (4.8) contains theory dependent quantities on both sides of the inequality, there is hope that it may be more general.

where the temperature dependence of D differs from the temperature dependence of v_{B}^2/T at low temperature, we also cannot rule this possibility out in exotic holographic matter.

The outline of this chapter is as follows. In Section 4.1.2. In Section 4.2, we review the fluid-gravity correspondence and describe the construction of the striped black holes we use to compute v_{B} and D . Sections 4.3 and 4.4 compute D and v_{B} respectively. We find that both D and v_{B} admit, at leading order in ξ^{-1} , a simple hydrodynamic interpretation. In the low-temperature limit of ²³, we demonstrate (4.11) in Section 4.5. Appendix D contains further technical details.

4.2 STRIPED BLACK HOLES

We will study asymptotically anti-de Sitter geometries in $d+2$ spacetime dimensions, dual to systems with a UV conformal fixed point:

$$ds^2(r \rightarrow 0) = g_{MN} dx^M dx^N = \frac{dr^2 - dt^2 + d\mathbf{x}^2}{r^2} \quad (4.12)$$

r is the bulk radial coordinate, and t and $\mathbf{x}$ represent time and space in the boundary field theory directions. Without (yet) specifying the bulk matter content, we suppose that there is a scalar field Φ , dual to a relevant operator of dimension Δ in the UV conformal field theory. We consider states deformed by a spatially inhomogeneous source for the operator dual to Φ . The geometric interpretation of this is that the boundary conditions for Φ are:

$$\Phi(x, r \rightarrow 0) = \frac{1}{L^{d/2}} H(x) r^{d+1-\Delta} + \dots \quad (4.13)$$

We wish to find a background geometry dual to a state of the theory at finite temperature T , assuming charge neutrality (at least for the charge whose diffusion constant we are computing). For simplicity, we will suppose that the function $H(x)$ only depends on a single coordinate, which we

will label as (italic) x . To this charge neutral background, we then add a $U(1)$ gauge field A , so the total action of the bulk theory is:

$$S = S_{\text{matter}} - \int d^{d+2}x \sqrt{-g} Z(\Phi) \frac{F^2}{4e^2}, \quad (4.14)$$

with $F = dA$. The diffusion constant D we compute will be the diffusion constant associated with the dual conserved charge.

In general, the construction of such inhomogeneous geometries cannot be done analytically. However, suppose that the function $H(x)$ varies on long wavelengths ξ : i.e.

$$\frac{\partial_x H}{H} \sim \frac{1}{\xi} \quad (4.15)$$

and that the dimensionless parameter

$$\frac{1}{\xi T} \rightarrow 0. \quad (4.16)$$

In what follows, we will often discuss the perturbative expansion in $1/\xi$, and this will imply that $1/\xi T$ is the small dimensionless parameter governing the expansion. In this case, we expect that the field theory – deformed by a very long wavelength source – reaches local thermal equilibrium as if it was in a homogeneous medium. This “hydrodynamic” disorder^{137,139} is particularly elegant to treat holographically: the fluid-gravity correspondence^{22,21} allows us to analytically construct inhomogeneous black hole backgrounds as a perturbative expansion in $1/\xi$; see e.g.⁴². This will allow us to analytically compute the charge diffusion constant D and butterfly velocity v_B .

At leading order in $1/\xi$, it is particularly simple to write down the inhomogeneous background geometry, with our specified boundary conditions for Φ and g_{MN} . To do so, one must find a coordinate system which is regular at the horizon. In such a regular coordinate system, we first construct the background geometry assuming that $H(x) = \text{constant}$. One such coordinate system is

Eddington-Finkelstein coordinates, where

$$ds^2 = -2\tilde{a}(r; H)dvdr - a(r; H)dv^2 + b(r; H)d\mathbf{x}^2, \quad (4.17a)$$

$$\Phi = \phi(r; H). \quad (4.17b)$$

We have replaced here the standard time coordinate t , with a coordinate v which is constant along in-falling light rays.[†] The functions $a(r; H)$, $\tilde{a}(r; H)$ and $b(r; H)$ are functions of H , the boundary condition on the scalar Φ , and also the temperature T of the horizon. Since black hole theorems imply that T must be uniform across the horizon^{2.14}, we only denote the explicit dependence on H , which is allowed to be spatially varying shortly. Note that the coordinate r may be reparameterized: this implies that $\tilde{a}$ is not a true degree of freedom. For future reference, we note that the inverse metric has components

$$g^{rr} = \frac{a}{\tilde{a}^2}, \quad g^{rv} = \frac{1}{\tilde{a}}, \quad g^{ij} = \frac{ij}{b}, \quad (4.18)$$

with all other components vanishing.

Given a solution at temperature $T > 0$ to the bulk equations of motion, the fluid-gravity correspondence asserts that given the inhomogeneous boundary conditions above, the bulk fields are:

$$ds^2 = -2\tilde{a}(r; H(x))dvdr - a(r; H(x))dv^2 + b(r; H(x))d\mathbf{x}^2 + \mathcal{O}\left(\frac{1}{\xi}\right), \quad (4.19a)$$

$$\Phi = \phi(r; H(x)) + \mathcal{O}\left(\frac{1}{\xi}\right). \quad (4.19b)$$

The leading order solution can be found by simply gluing together a locally homogeneous solution.

[†]A more standard metric choice might have been $ds^2 = gdr^2 - fdt^2 + bdx^2$. These are often called Fefferman-Graham coordinates. One can get from this coordinate system to Eddington-Finkelstein coordinates by defining $dv = dt - \sqrt{f/g}dr$. However, such metrics are not well-suited for the fluid-gravity correspondence, as we will explain shortly.

For our purposes, we can understand this result as follows. The metric is completely regular near the horizon – namely, no component of g_{MN} or g^{MN} diverges near the horizon. Hence, one can check that (for any reasonable norm)

$$\left| R_{MN}[g_0] - \frac{R[g_0]}{2}(g_0)_{MN} - T_{MN}^{\text{matter}}[g_0, \Phi_0, \dots] \right| = \mathcal{O}\left(\frac{1}{\xi}\right), \quad (4.20)$$

where g_0, Φ_0 etc. are the metric and matter content given in (4.19). Furthermore, we can systematically correct g and Φ , order by order in ξ^{-1} , in a local fashion. In contrast to Eddington-Finkelstein coordinates (4.19), the standard Fefferman-Graham coordinates, in which the background metric is diagonal, have a singular metric at the horizon, and so are not well-suited for the fluid gravity correspondence.

For the purposes of this chapter, we will find it sufficient to work with the fluid-gravity expansion at leading order – the key point is that this expansion is controllable and the perturbative limit $\xi \rightarrow \infty$ is well-behaved.

4.3 CHARGE DIFFUSION

As in ²³ we will compute the charge diffusion constant via the ratio

$$D_{xx} \equiv \frac{\sigma_{xx}}{\chi}. \quad (4.21)$$

Here σ and χ are the globally defined electrical conductivity and charge susceptibility, respectively; these quantities are easier to compute holographically than D . As our striped fluids are anisotropic, both D and σ are tensors, but for the remainder of the chapter we suppress their explicit indices and focus on diffusion in the striped direction. (4.21) is called an Einstein relation, and its origin is well understood in homogeneous media ¹¹³. The common derivation of (4.21) in homogeneous

media exploits translation invariance and so cannot be applied directly to inhomogeneous media. Nonetheless, in Appendix D, we show that (4.21) remains correct in the inhomogeneous fluids of interest in this chapter. Hence, we turn to the holographic computation of σ and χ .

4.3.1 CONDUCTIVITY

We begin with the computation of the conductivity σ . To accomplish this, we employ the holographic membrane paradigm approach first developed in ¹⁰², and expanded in ^{55,54,16} to generalize to inhomogeneous geometries. This technique becomes especially simple in our case, where the translation symmetry is broken only in a single direction ⁵⁵.

In our model, this technique proves especially simple. The Maxwell equations associated with the bulk gauge field A in (4.14) are

$$\partial_M (\sqrt{-g} Z(\Phi) F^{MN}) = 0. \quad (4.22)$$

To compute σ , we turn on a perturbation

$$A = -E v dx + \tilde{A}. \quad (4.23)$$

The first contribution above imposes a time-independent electric field at the boundary, and the second contribution records the bulk response. Importantly, since only $F = dA$ enters (4.22), $\tilde{A}$ is v -independent. As the background is charge neutral, A will not couple to perturbations of any other bulk fields. By symmetry, the only possible non-vanishing components of F are F_{rv} , F_{rx} and F_{vx} , and they are functions of r and x alone. Hence we may write (4.22) when $N = r$ and $N = x$ respectively

as:

$$\partial_x (\sqrt{-g} Z F^{xr}) = 0, \quad (4.24a)$$

$$\partial_r (\sqrt{-g} Z F^{rx}) = 0, \quad (4.24b)$$

which immediately implies that

$$J = \frac{1}{e^2} \sqrt{-g} Z F^{xr} \quad (4.25)$$

is a constant. The standard AdS/CFT dictionary may be employed in the UV, and J can be recognized as the expectation value $\langle J_x \rangle$. From Ohm's Law,

$$\sigma = \frac{J}{E}. \quad (4.26)$$

At the horizon, g^{rr} vanishes and so employing our leading order fluid-gravity metric (4.19):

$$J = \frac{Z}{e^2} \sqrt{-g} g^{xx} g^{rv} F_{xv} = \frac{Z}{e^2} b^{\frac{d-2}{2}} (E - \partial_x \tilde{A}_v). \quad (4.27)$$

We now divide both sides of this equation by the object outside the parentheses, and perform a spatial average, which we denote as $\mathbb{E}[\dots] = \frac{1}{L_x} \int dx \dots$, with L_x denoting the “length” of the theory in the x -direction (possibly infinite). We find

$$\sigma = \frac{1}{\mathbb{E} \left[\frac{e^2}{Z} b^{-\frac{d-2}{2}} \right]}. \quad (4.28)$$

This equation can be understood through hydrodynamics¹³⁷. The conductivity of a striped fluid

with an inhomogeneous local conductivity $\tilde{\sigma}(x)$ is simply

$$\sigma = \frac{1}{\mathbb{E}\left[\frac{1}{\tilde{\sigma}(x)}\right]}. \quad (4.29)$$

The object inside the spatial average in (4.28) can be interpreted as the inverse conductivity of a fluid, but if it was homogeneous. In our particular fluid-gravity limit, of course, a hydrodynamic interpretation of σ is quite natural, but remarkably a hydrodynamic interpretation persists even beyond the fluid-gravity limit^{75,139,54,16}.

4.3.2 SUSCEPTIBILITY

We define the net charge susceptibility as

$$\chi \equiv \frac{\partial \mathbb{E}[n]}{\partial \mu}, \quad (4.30)$$

where $n = \langle \mathcal{J}^t \rangle$ is the charge density of the boundary theory, and μ is the (spatially homogeneous) chemical potential. To compute this holographically, we turn on an infinitesimal gauge field A as before, but this time set the boundary conditions in the UV to be

$$A(r=0, x) = \mu dv. \quad (4.31)$$

In general, there is no elegant membrane paradigm technique to compute χ in terms of a horizon quantity. Indeed, as we will see, χ will depend on details of the full bulk geometry. Still, let us press ahead, working in a gauge where $A_r = 0$. The v -component of (4.22) reads

$$\partial_r (\sqrt{-g} Z F^{rv}) + \partial_x (\sqrt{-g} Z F^{xv}) = 0. \quad (4.32)$$

As we will justify more carefully soon, the second term can be neglected in the fluid-gravity limit.

Hence, in our gauge, we obtain an “effective” radially conserved quantity at each x

$$n = -\frac{1}{e^2} \sqrt{-g} Z F^{vr} = -\frac{1}{e^2} b^{\frac{d}{2}} \tilde{a}^{-1} Z \partial_r A_v, \quad (4.33)$$

which the AdS/CFT dictionary identifies as the boundary theory expectation value of $\langle \mathcal{I} \rangle$. This justifies why we have labeled this radially conserved quantity n , as it is the local charge density. Now, we employ the following trick²³: if the horizon is located at $r = r_+$, since $A_v(r = r_+) = 0$ in order for A to be well-defined,

$$n(x) \int_0^{r_+} dr \frac{e^2 \tilde{a}}{Z b^{\frac{d}{2}}} = \int_0^{r_+} dr (-\partial_r A_v) = \mu, \quad (4.34)$$

where we have employed the boundary conditions in the last step. We note that $b \sim \tilde{a} \sim r^{-2}$ as $r \rightarrow 0$, and that both b and $\tilde{a}$ are regular at the horizon; thus A_v will be finite everywhere. Hence, upon applying the uniform chemical potential μ , we see that

$$\chi = \frac{\partial \mathbb{E}[n]}{\partial \mu} = \mathbb{E} \left[\left(\int_0^{r_+} dr \frac{e^2 \tilde{a}}{Z b^{\frac{d}{2}}} \right)^{-1} \right]. \quad (4.35)$$

To confirm that this is the correct answer at leading order in ξ^{-1} , we note that (4.33), together with the value for $n(x)$ in (4.34), gives us an approximate solution for A_v at all r and x , which is smooth everywhere and will not exhibit any singular behavior at finite T . By simply plugging this “guess” into (4.32), we see that it is correct to leading order: regularity of A_v , and the geometry, ensures that all corrections in ξ^{-1} are subleading.

In the limit of hydrodynamic disorder, because $n(\mu, H, \dots)$ is a local quantity we expect that

$$\chi = \mathbb{E} [\tilde{\chi}(x)], \quad (4.36)$$

with $\tilde{\chi}(x) = \partial n(x)/\partial \mu$ a local susceptibility. Our holographic computation in the fluid-gravity geometries confirms this explicitly.

To summarize, the diffusion coefficient can now be written in terms of spatial averages as

$$D = \frac{1}{\mathbb{E}\left[b^{\frac{2-d}{2}}Z(r;H(x))^{-1}\right]\mathbb{E}\left[\left(\int_0^{r_+} dr b^{-\frac{d}{2}}\tilde{\alpha}Z(r;H(x))^{-1}\right)^{-1}\right]}. \quad (4.37)$$

This result can essentially be understood through classical hydrodynamics. The reason that it is nonetheless useful to do the computation holographically is that we now have an explicit formula for D in terms of bulk quantities. It is not obvious that the butterfly velocity v_B – which by definition probes *quantum* chaos – admits any classical interpretation. We will now be able to compare v_B to D .

4.4 BUTTERFLY VELOCITY

Recall that the butterfly effect is a phenomena in chaotic systems in which an initially small perturbation can grow exponentially large at later times. In the introduction, we captured this in terms of a peculiar 4-point correlation function (4.9). Several recent works have pointed out that the holographic interpretation of the butterfly effect is a geometric shock wave, propagating along the horizon^{191,168}. We will not fully justify why here, only explain the basic idea. Black holes such as the ones we have constructed can be maximally extended to “double-sided” geometries, which contain two copies of the field theory, where time runs in opposite directions, along with a black hole and a white hole²¹⁴. This maximally extended geometry can be interpreted as two entangled copies of a field theory in a particular excited and entangled state. We now imagine slightly perturbing the left field theory (where we take time to run backwards) at a “late” time $t = t_0$, by adding a small amount of massless particles. This should set off the butterfly effect, and the geometry at time $t = 0$ should

be very different than what it was without the perturbation.

To leading order, we might expect that this perturbation does nothing to the classical geometry in the limit $G_N \rightarrow 0$. However, if t_0 is large, then these massless particles will follow null trajectories very close to the (past) horizon, where they become exponentially blue-shifted, analogous to particles falling into the classic Schwarzschild black hole. We then must solve Einstein's equations in the presence of a thin shell of energy associated with this in-falling blue-shifted matter^{56,190}. The resulting geometry will contain a shock at the horizon (a sudden shift in the geometry), which allows us to capture both λ and v_B , as defined in (4.9).

4.4.1 KRUSKAL COORDINATES

We have now outlined the computation of v_B in words. The first thing to do is now to adopt a coordinate system which completely covers the maximally extended black holes of interest, called Kruskal coordinates. For simplicity, we begin with homogeneous black holes (corresponding to the right side field theory in the maximally extended case), where

$$ds^2 = 2A(UV; H)dUdV + B(UV; H)d\mathbf{x}^2. \quad (4.38)$$

In terms of the null in-going coordinate v , and a null out-going coordinate

$$du \equiv -\frac{2\tilde{a}}{a}dr - dv, \quad (4.39)$$

the Kruskal coordinates U and V are defined as

$$U = Le^{2Tu}, \quad (4.40a)$$

$$V = Le^{2Tv}. \quad (4.40b)$$

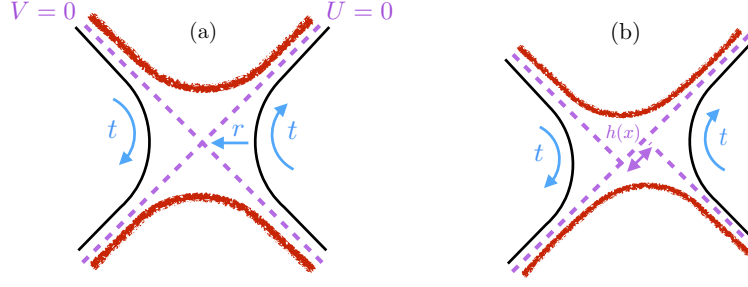


Figure 4.1: (a) The global Kruskal geometry. The smooth black lines on the left/right side of the diagram denote the “AdS” boundaries of the left/right field theories. The rough red lines at the top and bottom denote the singularities inside of a black hole and a white hole respectively. The dashed purple lines denote event horizons at $U = 0$ and $V = 0$. Blue arrows denote the direction as well as the standard “bulk radial” coordinate r . Eddington-Finkelstein coordinates only cover the region $V \geq 0$ in this diagram. The properties of this black hole vary smoothly in the x direction, which runs out of the page. (b) The effect of the “geometric shockwave” is a shift $h(x)$ between the horizons as one crosses $U = 0$.

Hence,

$$A = \frac{2a}{(4T)^2 UV}, \quad (4.41a)$$

$$B = b. \quad (4.41b)$$

The event horizons are located at $U = 0$ and/or $V = 0$. Note that A and B are functions only of the product UV , and that despite appearances, A is in fact regular at $UV = 0$. Figure 4.1(a) shows the global spacetime in Kruskal coordinates.

Now, we return to the case where $H(x)$ is not a constant. In fact, to leading order in ξ^{-1} , the inhomogeneous black hole in Kruskal coordinates is given by:

$$ds^2 = 2A(UV; H(x))dUdV + B(UV; H(x))dx^2 + \mathcal{O}\left(\frac{1}{\xi^2}\right), \quad (4.42a)$$

$$\Phi = \varphi(UV; H(x)). \quad (4.42b)$$

We justify this claim more carefully in Appendix D. Kruskal coordinates are not commonly em-

ployed for a “fluid-gravity” correspondence. In Eddington-Finkelstein coordinates, given *time-dependent* fluid flows in the boundary theory, the bulk geometry may be constructed “tube-wise”: namely, given data at a boundary spacetime point $(t, \mathbf{x})$, one can construct the local geometry for all r . In Kruskal coordinates, U and V both relate to time t , and so one would need dynamics for all t to construct the bulk geometry on for any given $\mathbf{x}$. However, since we are looking at static geometries, we trivially know the boundary “fluid dynamics” for all times t . Furthermore, Kruskal coordinates share the same key regularity property of Eddington-Finkelstein coordinates: both g_{MN} and g^{MN} are completely regular at $U = 0$ or $V = 0$. Thus, it is guaranteed that (4.42) indeed solves the equations of motion at leading order in $1/\xi$, and that the effects of perturbations are not singular. Hence, we can use a *local* fluid-gravity expansion to construct an inhomogeneous static black hole, order-by-order in $1/\xi$, directly in Kruskal coordinates. We explicitly discuss the construction in Appendix D. Our main result can be obtained from the leading order solution (4.42).

4.4.2 SHOCKWAVE GEOMETRY

As we mentioned previously, we consider a small amount of matter thrown into the left-half of our Kruskal black hole at an “early” time $t = t_0$. For simplicity, we consider a thin shell of matter, located at $x = 0$, and uniform in the $d - 1$ spatial directions perpendicular to x . As this matter falls towards the horizon, it creates a very sharp distribution of energy which we approximate as

$$T_{UU}^{\text{shock}} = \mathcal{E} e^{2Tt}(x)(U); \quad (4.43)$$

all other components of the stress tensor vanish. The (x) is not a true function, but can be approximated as such for studying long distance physics¹⁶⁸ – at the end of this section, we will provide a few more comments on this assumption. Here $\mathcal{E}$ is the (proper) energy density of the infalling shell of matter, and e^{2Tt} is a blue-shift factor as it falls towards the horizon. The solution to Einstein’s

equations in the background of such a source has been known for some time in a homogeneous geometry¹⁹⁰. The presence of this pulse of energy located at the $U = 0$ horizon creates a mismatch in the location of the horizon as one passes from the left to the right side (see Figure 4.1(b)):

$$V \rightarrow V + \Theta(U)b(x), \quad U \rightarrow U, \quad \mathbf{x} \rightarrow \mathbf{x}. \quad (4.44)$$

Using (4.44), we see this results in the following backreacted metric

$$ds^2 = 2A(UV; H(x))dUdV + B(UV; H(x))d\mathbf{x}^2 - 2A(UV; H(x))\delta(U)dU^2 + \mathcal{O}\left(\frac{1}{\xi}\right). \quad (4.45)$$

The stress tensor is modified to

$$T_{MN} = T_{MN}^{(0)} + T_{MN}^{(1)} \quad (4.46)$$

where $T_{MN}^{(0)}$ is the stress tensor without the shock wave, and the only non-zero component of $T_{MN}^{(1)}$ is (at leading order in ξ^{-1})

$$T_{UU}^{(1)} = \mathcal{E}e^{2Tt_0}(U)(x) - 2b(x, t)(U)T_{UV}^0. \quad (4.47)$$

We have used the fact that on the background, $T_{VV}^{(0)} = 0$ ¹⁹⁰ in deriving this result. We discuss higher order corrections to this shock wave geometry in Appendix D.2.

We now follow and compute v_b by solving Einstein's equations for $b(x)$. This computation – in the homogeneous case $H(x) = \text{constant}$ – is well-understood^{190,23,169}, and so we focus on the new effects coming from inhomogeneity. It is instructive to see how the shock wave affects the Ricci tensor R_{MN} . The only shock-dependent contributions to R_{MN} coming from the metric (4.45) are in the UU component, (keep in mind that further corrections are possible at $\mathcal{O}(\xi^{-2})$ due to fluid-

gravity corrections to (4.45), as discussed in Appendix D.2):

$$R_{UU}^{(1)} = (U) \left[\frac{A}{B} \partial_x^2 b + \frac{1}{B} \partial_x (\partial_x A b) + \frac{d-2}{2} \frac{\partial_x B}{B^2} \partial_x (A b) + 2b \frac{\partial_{UV} A}{A} + \frac{d}{2} b \frac{\partial_{UV} B}{B} \right]. \quad (4.48)$$

To simplify these results, we have used the following equalities, which hold in the sense of distributions:[‡]

$$U(U) = U^2(U) = U^2(U)^2 = 0, \quad (U) = -U'(U). \quad (4.49)$$

From (4.48) we see that x -derivatives do not enter the expression for $R_{UU}^{(1)}$ in a singular way near $U = 0$; similar calculations confirm this result for the remainder of Einstein's equations. Assuming that the matter content is also well-behaved (we give an explicit example where this is so in Appendix D.2), we may take the $\xi \rightarrow \infty$ limit smoothly. Using that, at order ξ^0 , the background Einstein equations give¹⁹⁰

$$8G_N T_{UV}^{(0)} = \frac{\partial_{UV} A}{A} + \frac{d}{2} \frac{\partial_{UV} B}{B}, \quad (4.50)$$

one can show that the coefficient of (U) in the UU component of Einstein's equations (with lowered indices) only vanishes when $b(x)$ obeys the differential equation¹⁹⁰

$$(\partial_x^2 - m(x)^2) b(x) = 8G_N \frac{B(0,0)}{A(0,0)}(x) \mathcal{E} e^{2Tr}, \quad (4.51)$$

with the “mass” m is defined as

$$m(x)^2 \equiv \frac{d}{2A(0; H(x))} \partial_{UV} B(UV; H(x)) \Big|_{U=0}. \quad (4.52)$$

We now solve for the profile $b(x)$. This can be done using a standard WKB-like argument. Let us

[‡]A transparent way to see the last equality is to approximate $(u) = \varepsilon^{-1} f(u/\varepsilon)$ with $\varepsilon \rightarrow 0$ and f a suitably chosen function: e.g. $f(x) = \frac{1}{2} e^{-|x|}$. One finds $\int du (u(u))^2 \sim \varepsilon$ which vanishes in the $\varepsilon \rightarrow 0$ limit.

define

$$b(x) \equiv \exp \left[- \left| \int_0^x dx' m(x') \right| \right] b_0(x), \quad (4.53)$$

where $b_0(x)$ obeys the differential equation

$$\partial_x^2 b_0 - 2m(x) \partial_x b_0 - b_0 \partial_x m(x) = 8G_N \frac{B(0,0)}{A(0,0)}(x) \mathcal{E} e^{2Tt}. \quad (4.54)$$

If we demand, as is physically sensible, that $b(x)$ vanish at $x = \pm\infty$, then (4.54) implies that $\partial_x b_0 = \mathcal{O}(\xi^{-1})$. Thus, we conclude that at leading order in ξ , we may approximate (at large x):

$$b \approx 8G_N \frac{B(0,0)}{A(0,0)} \mathcal{E} e^{\lambda(t-|x|/v_B)}, \quad (4.55)$$

with λ given by (4.10) and

$$\frac{1}{v_B} \equiv \mathbb{E} \left[\frac{1}{\tilde{v}_B} \right] \quad (4.56)$$

with

$$\frac{1}{\tilde{v}_B} \equiv \frac{1}{2T} \sqrt{\frac{d}{2A(0; H(x))} \partial_{UV} B(0; H(x))} \quad (4.57)$$

Interestingly, the butterfly velocity in this inhomogeneous background can be interpreted in a very simple classical picture. Imagine a “butterfly” moving through a medium with local velocity $\tilde{v}_B(x)$. What would the effective velocity of the butterfly be? It is easiest to compute this effective velocity v_B by measuring the time it takes to travel a distance ℓ :

$$t(\ell) = \int dt = \int \frac{dx}{\tilde{v}_B(x)} = \ell \mathbb{E} \left[\frac{1}{\tilde{v}_B} \right] \equiv \frac{\ell}{v_B}, \quad (4.58)$$

in agreement with our (leading order) holographic calculation (4.56). Although this simple classical picture is intuitive, it is worth keeping in mind that the butterfly velocity is a manifestation of

quantum chaos.

The derivation above assumed that the influence of the “pulse” at the horizon was localized. However, our derivation of the butterfly velocity is not particularly sensitive to this assumption. As we have seen, $b(x)$ obeys a linear differential equation. Even if $b(x)$ is sourced by a source of finite width a , for distances $x \gg a$, $b(x) \sim e^{-\lambda x/v_b}$ by linearity, and so our derivation of the butterfly velocity is unchanged.

4.5 SCALING GEOMETRIES

We now compare D and v_b . It is instructive to begin with a simple Einstein-Maxwell-dilaton (EMD) holographic models, with bulk action

$$S = \int d^{d+2}x \sqrt{-g} \left[\frac{1}{16G_N} \left(R - \frac{1}{2}(\partial\Phi)^2 - \mathcal{V}(\Phi) \right) - \frac{Z(\Phi)}{4} F^2 \right]. \quad (4.59)$$

We have carefully checked the validity of the fluid-gravity expansion in Kruskal coordinates in Appendix D.1, as well as subleading corrections to v_b in Appendix D.2, for this model.

Let us now consider the low-temperature scaling limit of our gravity theory^{40,92}. Let us temporarily assume that $H(x) = \text{constant}$. For “generic” Liouville-like potentials:

$$\mathcal{V}(\Phi \rightarrow \infty) = \mathcal{V}_0 e^{\beta\Phi}, \quad (4.60)$$

one finds, at $T = 0$, the emergent IR geometry (in Eddington-Finkelstein coordinates, in a certain gauge) is a hyperscaling-violation geometry:

$$ds^2 = \frac{L^2}{r^2} \left(H^{\frac{1}{d+1-\Delta}} r \right)^{\frac{2\theta}{d}} c_0 \left(2dvdr - dv^2 + d\mathbf{x}^2 \right), \quad (4.61)$$

so long as $\theta < 0$ (or $\theta > d$, though this case may be unphysical[?]). This requirement follows from the formula^{40?}

$$\beta^2 = \frac{2\theta}{d(d-\theta)}, \quad (4.62)$$

and the fact that β must be real. We emphasize that (4.61) is not the full geometry, and that this approximation will fail for $r \lesssim r_{\text{uv}} \sim H^{\frac{1}{d+1-\Delta}}$ – for smaller values of r , there is a UV completion to an asymptotically AdS geometry, the details of which are unimportant to us. At low temperatures

$$\frac{T}{H^{\frac{1}{d+1-\Delta}}} \ll 1, \quad (4.63)$$

the IR geometry is only modified from (4.61) in a simple manner. The IR emergent hyperscaling geometry remains, but now with a planar black hole horizon. The metric is approximately given by

$$ds^2 = \frac{L^2}{r^2} \left(H^{\frac{1}{d+1-\Delta}} r \right)^{\frac{2\theta}{d}} c_0 \left(2dvdr - f_T(r)dv^2 + d\mathbf{x}^2 \right), \quad (4.64)$$

with emblackening factor

$$f_T(r) = 1 - \left(\frac{r}{r_+} \right)^{d+1-\theta}, \quad (4.65)$$

where

$$r_+ = \frac{d+1-\theta}{4T}. \quad (4.66)$$

In this geometry, $\Phi \sim \log r$ in the IR scaling region. Hence, we expect, for canonical choices $Z(\Phi) \sim \exp[\beta'\Phi]$, that

$$Z = Z_0 \left(H^{\frac{1}{d+1-\Delta}} r \right)^\gamma, \quad (4.67)$$

for some scaling exponent γ . Again, this IR geometry is only valid for $r \gtrsim r_{\text{uv}}$.

Of course, we wish to study geometries where $H(x)$ is not constant. However, as we have discussed in Section 4.2, so long as (4.16) is satisfied, these geometries can be constructed by simply

“gluing” together the homogeneous geometries point-by-point in x , using the local value for $H(x)$ in (4.64). It is critical that (4.16) is obeyed – if we take the limit $T \rightarrow 0$ with ξ fixed, then we expect such a perturbation to decay in the IR geometry (see e.g. ⁴²). Because we further assume (4.63), the homogeneous geometries which we must glue together take a particularly simple form (4.64) in the IR, at leading order in $1/\xi T$ and $T/H^{\frac{1}{d+1-\Delta}}$, point-by-point in x . We also remind the reader that the validity of the fluid-gravity expansion is not sensitive to the assumption (4.63) – indeed, (4.19) holds more generally so long as (4.16) holds. Nevertheless, we will be interested in geometries where the near-horizon limit of metrics such as (4.19) can be approximated by (4.64), with an x -dependent H .

Using the results of the previous sections, it is simple to compute D and v_B . We begin by computing σ, χ and D . Using (4.28), along with the near-horizon geometry (4.64) we obtain a local effective conductivity

$$\tilde{\sigma} = \frac{1}{e^2} b(r_+)^{\frac{d}{2}-1} Z(r_+) = \frac{Z_0}{e^2} c_0^{\frac{d}{2}-1} H^{\frac{1}{d+1-\Delta}(\frac{2\theta}{d}(\frac{d}{2}-1)+\gamma)} r_+^{\gamma+(\frac{2\theta}{d}-2)(\frac{d}{2}-1)}. \quad (4.68)$$

Next, we use (4.35) to compute $\tilde{\chi}$. At $T \sim H^{\frac{1}{d+1-\Delta}}$, $\tilde{\chi}$ will be a complicated function of T :

$$\tilde{\chi} \approx \left(\int_{r_{\text{uv}}}^{r_+} \frac{e^2 dr}{Z_0 c_0^{\frac{d}{2}-1} H^{\frac{1}{d+1-\Delta}(\frac{2\theta}{d}(\frac{d}{2}-1)+\gamma)} r^{\gamma+(\frac{2\theta}{d}-2)(\frac{d}{2}-1)}} + \int_0^{r_{\text{uv}}} dr \frac{e^2 \tilde{a}}{b^{d/2} Z} \right)^{-1} \quad (4.69)$$

Recall r_{uv} is the scale at which (4.64) fails to be a good approximation. Since this scale is set by H and not T , in the limit (4.63), we note that only the first term of (4.69) is T -dependent. We focus on the limit where this is the dominant term in $\tilde{\chi}$ in the limit $r_+ \gg r_{\text{uv}}$,[§] which occurs when

$$\left(1 - \frac{\theta}{d}\right)(d-2) - \gamma > -1 \quad (4.70)$$

[§]If this term is subleading, then $\tilde{\chi}$ will be sensitive to the UV geometry, whereas $\tilde{\sigma}$ and $\tilde{v}_B$ both depend only on the near-horizon geometry. Hence there can be no universal relation between D and v_B , a point noted in ²³.

In ²³, it was noted that this will happen if the IR scaling dimension for the charge susceptibility is positive.[¶] In this case, (4.69) can be approximated as

$$\begin{aligned}\tilde{\chi} &= \left(1 + 2\left(1 - \frac{\theta}{d}\right)\left(\frac{d}{2} - 1\right) - \gamma\right) Z_0 c_0^{\frac{d}{2}-1} H^{\frac{1}{d+1-\Delta}\left(\frac{2\theta}{d}\left(\frac{d}{2}-1\right)+\gamma\right)} r_+^{\gamma-1+\left(\frac{2\theta}{d}-2\right)\left(\frac{d}{2}-1\right)} \\ &= \left(1 + 2\left(1 - \frac{\theta}{d}\right)\left(\frac{d}{2} - 1\right) - \gamma\right) \tilde{\sigma} \frac{4T}{d+1-\theta}.\end{aligned}\quad (4.71)$$

Combining (4.37), (4.68) and (4.71) we obtain D :

$$D = \frac{d+1-\theta}{1 + 2\left(1 - \frac{\theta}{d}\right)\left(\frac{d}{2} - 1\right) - \gamma} \frac{1}{4T\mathbb{E}[H^\eta]\mathbb{E}[H^{-\eta}]} \quad (4.72)$$

where the power η is generically not zero:

$$\eta = \frac{1}{d+1-\Delta} \left(\frac{\theta}{d}(d-2) + \gamma \right). \quad (4.73)$$

If the integral for χ is dominated away from the horizon, then the diffusion constant becomes parametrically large ²³.

Let us now turn to the computation of v_B . Using (4.41), we find that

$$\frac{1}{v_B^2} = \frac{(4T)^2 dUV}{(2T)^2 4a} \partial_{UV} b \approx \frac{(r_+ - r)d}{a} (-\partial_r b). \quad (4.74)$$

In the latter step, we have taken the near-horizon limit. For the geometries (4.64), this leads to the simple constant result:

$$\tilde{v}_B^2 = \frac{d+1-\theta}{2(d-\theta)}, \quad (4.75)$$

in agreement with ^{23,169}, though now in more generic striped backgrounds. Interestingly, in this

[¶]This assumes that we interpret generic exponents γ as anomalous dimensions for the charge density, as in ^{70,90}.

special case where $v_{\text{B}} \sim T^0$, this velocity has also been found to characterize entanglement entropy growth during thermalization⁴.

If we assume that H is a constant, we find that the constant $\mathcal{C}$ defined in (4.11) is

$$\mathcal{C} = \frac{d - \theta}{2(1 - \frac{\theta}{d})(\frac{d}{2} - 1) - \gamma}. \quad (4.76)$$

However, if $H(x)$ is not constant, then D is not related to v_{B} in a simple manner, regardless of the value of γ . Using the Cauchy-Schwarz inequality (see e.g.¹⁰⁰), we find that

$$\mathbb{E}[H^\eta] \mathbb{E}[H^{-\eta}] \geq 1. \quad (4.77)$$

Hence, we conclude that the inequality of (4.11) holds: namely, $2TD \leq \mathcal{C}v_{\text{B}}^2$. In principle, there is no bound on the ratio Dv_{B}^{-2} – the diffusion constant can be parametrically small compared to the butterfly velocity. As we emphasized in the introduction, this inequality is the “wrong sign” – an incoherent metal should have diffusion constants bounded from below, and so the butterfly velocity evidently cannot always serve as the velocity scale in (4.8) in any sharp sense. However, (4.8) continues to hold in a qualitative sense, as in these simple models $v_{\text{B}} \sim T^0$ and $D \sim 1/T$, in agreement with the scaling noted in²³.

We also expect our conclusions to hold in more complicated bulk models where we have not explicitly checked that the fluid-gravity expansion is well-behaved in Kruskal coordinates. In more general scaling regimes where the bounds of²³ hold in homogeneous systems, we find *locally*

$$\frac{\tilde{\sigma}}{\tilde{\chi}} = \frac{\mathcal{C}}{2T} \tilde{v}_{\text{B}}^2. \quad (4.78)$$

Then

$$\frac{D}{v_{\text{B}}^2} \frac{2T}{\mathcal{C}} = \frac{\mathbb{E}[\tilde{v}_{\text{B}}^{-1}]^2}{\mathbb{E}[\tilde{\chi}]\mathbb{E}[\tilde{\sigma}^{-1}]} \frac{2T}{\mathcal{C}} = \frac{\mathbb{E}[\tilde{\sigma}^{-1/2}\tilde{\chi}^{1/2}]^2}{\mathbb{E}[\tilde{\chi}]\mathbb{E}[\tilde{\sigma}^{-1}]} \leq 1, \quad (4.79)$$

again by the Cauchy-Schwarz inequality, as $\tilde{\sigma}$ and $\tilde{\chi}$ are positive everywhere. Our result that the charge diffusion constant is upper bounded by the butterfly velocity in these striped models is not peculiar to the Einstein-dilaton model we studied above. The inequality (4.11) may change beyond the hydrodynamic limit, or when the disorder breaks translation symmetry in multiple directions.

4.6 CONCLUSIONS

In this chapter, we have shown that in striped charge neutral holographic quantum matter, the butterfly velocity cannot generally be used to quantitatively provide a lower bound for the charge diffusion constant. This implies that, to the extent that (4.8) should hold exactly in any incoherent metal, the velocity scale v in (4.8) either cannot be the butterfly velocity. It would be interesting if there is a slower velocity scale for which (4.8) remains true, even in the striped geometries we have constructed in this chapter. We also note that we did not present an explicit example where DT/v_B^2 had non-trivial T -dependence as $T \rightarrow 0$, though we cannot rule this possibility out. In general, bounds on transport coefficients, which are robust to the specific nature of disorder, are known to exist only in a handful of holographic systems^{75,76,100}. It is always the case that complicating the bulk models sufficiently can lead to the violation of any naive bound.

Finally, we comment that in this charge-neutral hydrodynamic limit, although charge diffusion is very weak, the energy diffusion constant will be very large¹⁶. Our work cannot rule out that similarly to⁷⁶, there are holographic models where a v_B -based bound on energy diffusion is very robust. Since both D and v_B do not seem bounded in the incoherent limit in “mean-field” disordered holographic metals²⁴, this is not entirely implausible. It was found in $D_E \sim v_B \tau_L$ is a generic low temperature property of homogeneous holographic lattice models that flow to infra-red fixed points including the homogeneous SYK chain^{26,82,49}. However, this relation becomes an upper for disorder averaged quantities in inhomogeneous SYK chains⁸⁰.

It was later realized that chaos can be related to transport through the phenomena of “pole-skipping” of the retarded energy density two-point function which tightly constrains the behavior of the dispersion relation of longitudinal (sound) hydrodynamic excitations^{78,25,27,77,74}. If the hydrodynamic gradient expansion of this two-point function for $|\omega|/T \sim |k|/T \ll 1$ truncates at first order, then it is possible to directly relate the butterfly velocity to the energy diffusion constant. However, in general, this series goes beyond first order which suggests that chaos, charge, and energy diffusion are only related in the absence of other characteristic scales.

Even beyond the hydrodynamic limit, the direct current transport coefficients of most holographic models can be computed in terms of an emergent hydrodynamic theory on the black hole horizon (at finite temperature)^{55,54,16,137}. It would be interesting if, for arbitrary black holes, there was some simple partial differential equation governing the shift function b , defined in Section 4.4. This may lead to fundamental bounds on the butterfly velocity, analogously to how conductivity bounds may be found^{75,76,100}.



Appendix for Chapter I

A.1 EVALUATION OF $e^{-3\Delta/T}$ CONTRIBUTION

The expression for $c_2(T, \eta)$ in Eq. (1.38) can be written in the limit $T \rightarrow 0$ as

$$c_2(T, \eta) = -\frac{2\pi}{2!3!T} \int_{-\pi}^{\pi} \frac{dk_1 dk_2 dq_1 dq_2 dq_3}{(2\pi)^5} |_{NS} \langle k_1 k_2 | \sigma_0^z | q_1 q_2 q_3 \rangle_R|^2 e^{-(\varepsilon_{q_1} + \varepsilon_{q_2} + \varepsilon_{q_3})/T} \\ \times \delta(\varepsilon_{k_1} + \varepsilon_{k_2} - \varepsilon_{q_1} - \varepsilon_{q_2} - \varepsilon_{q_3}), \quad (\text{A.1})$$

where the factorials are combinatoric factors from converting the sums to integrals. From Eq. (1.14), the form factor is

$$\begin{aligned}
|_{NS}\langle k_1 k_2 | \sigma_0^z | q_1 q_2 q_3 \rangle_R|^2 &= \frac{(4f^2 b)^{\frac{1}{2}} |1 - b^2|^{\frac{1}{4}}}{\varepsilon_{k_1} \varepsilon_{k_2} \varepsilon_{q_1} \varepsilon_{q_2} \varepsilon_{q_3}} \\
&\times \left(\frac{\sin\left(\frac{k_1 - k_2}{2}\right) \sin\left(\frac{q_1 - q_2}{2}\right) \sin\left(\frac{q_1 - q_3}{2}\right) \sin\left(\frac{q_2 - q_3}{2}\right)}{\sin\left(\frac{k_1 - p_1}{2}\right) \sin\left(\frac{k_1 - q_2}{2}\right) \sin\left(\frac{k_1 - q_3}{2}\right) \sin\left(\frac{k_2 - q_1}{2}\right) \sin\left(\frac{k_2 - q_2}{2}\right) \sin\left(\frac{k_2 - q_3}{2}\right)} \right)^2 \\
&\times \left(\frac{(\varepsilon_{k_1} + \varepsilon_{q_1})(\varepsilon_{k_1} + \varepsilon_{q_2})(\varepsilon_{k_1} + \varepsilon_{q_3})(\varepsilon_{k_2} + \varepsilon_{q_1})(\varepsilon_{k_2} + \varepsilon_{q_2})(\varepsilon_{k_2} + \varepsilon_{q_3})}{4(\varepsilon_{k_1} + \varepsilon_{k_2})(\varepsilon_{q_1} + \varepsilon_{q_2})(\varepsilon_{q_1} + \varepsilon_{q_3})(\varepsilon_{q_2} + \varepsilon_{q_3})} \right)^2. \quad (\text{A.2})
\end{aligned}$$

We now attempt to take the $T \rightarrow 0$ limit of Eq. (A.1) in a manner similar to the analysis below Eq. (1.28). The integral is dominated by small $q_{1,2,3} \sim \sqrt{T}$. This allows us to make the following approximations

$$\begin{aligned}
\varepsilon_{q_i} &\approx \frac{q_i^2}{2m} + \Delta \\
\sin\left(\frac{q_i - q_j}{2}\right) &\approx \frac{q_i - q_j}{2} \\
\varepsilon_{q_1} + \varepsilon_{q_2} &= 2\Delta
\end{aligned} \quad (\text{A.3})$$

We can now write Eqn. (A.2) as a product of a term containing the q_1, q_2 , and q_3 dependence, with one containing the k_1 and k_2 dependence. The $q_{1,2,3}$ integral is sharply peaked about momenta $\sim \sqrt{T}$ in the q_1, q_2, q_3 plane, so we can extend the limit of integration over these momenta out to

infinity giving us the following

$$\begin{aligned}
c_2(T, \eta) \approx & -\frac{e^{-3\Delta/T} 4\pi J(b^2|1-b^2|)^{\frac{1}{4}}}{3!2!\Delta^9 2^{16} T} \int_{-\pi}^{\pi} \frac{dk_1 dk_2}{(2\pi)^2} \frac{((\varepsilon_{k_1} + \Delta)^3 (\varepsilon_{k_2} + \Delta)^3)^2}{\varepsilon_{k_1} \varepsilon_{k_2} (\varepsilon_{k_1} + \varepsilon_{k_2})^2} \\
& \times \left(\frac{\sin\left(\frac{k_1 - k_2}{2}\right)}{\sin^3\left(\frac{k_1}{2}\right) \sin^3\left(\frac{k_2}{2}\right)} \right)^2 \delta(\varepsilon_{k_1} + \varepsilon_{k_2} - 3\Delta) \\
& \times \int_{-\infty}^{\infty} \frac{dq_1 dq_2 dq_3}{(2\pi)^3} ((q_1 - q_2)(q_1 - q_3)(q_2 - q_3))^2 e^{-(q_1^2 + q_2^2 + q_3^2)/2mT}. \quad (\text{A.4})
\end{aligned}$$

The $q_{1,2,3}$ integrals evaluate to

$$\int_{-\infty}^{\infty} \frac{dq_1 dq_2 dq_3}{(2\pi)^3} (q_1 - q_2)^2 (q_1 - q_3)^2 (q_2 - q_3)^2 e^{-(q_1^2 + q_2^2 + q_3^2)/2mT} = 12(2\pi)^{-\frac{3}{2}} (mT)^{\frac{9}{2}}, \quad (\text{A.5})$$

which yields

$$\begin{aligned}
c_2(T, \eta) \approx & -\frac{e^{-3\Delta/T} 2J(b^2|1-b^2|)^{\frac{1}{4}} (mT)^{\frac{9}{2}}}{\Delta^9 (2\pi)^{\frac{1}{2}} 2^{16} T} \\
& \times \int_{-\pi}^{\pi} \frac{dk_1 dk_2}{(2\pi)^2} \frac{((\varepsilon_{k_1} + \Delta)^3 (\varepsilon_{k_2} + \Delta)^3)^2}{\varepsilon_{k_1} \varepsilon_{k_2} (\varepsilon_{k_1} + \varepsilon_{k_2})^2} \frac{\sin^2\left(\frac{k_1 - k_2}{2}\right)}{\sin^6\left(\frac{k_1}{2}\right) \sin^6\left(\frac{k_2}{2}\right)} \delta(\varepsilon_{k_1} + \varepsilon_{k_2} - 3\Delta) \quad (\text{A.6})
\end{aligned}$$

Now we have to perform the final integrals over $k_{1,2}$. Because of the singularities in the form factors at small momenta, the integrals have infrared divergencies which need to be treated differently depending upon the value of b .

For $3 < b < 5$, the energy conservation delta function in Eq. (A.6) prevents a divergence. The argument of the delta function does not vanish when either $k_1 = 0$ or $k_2 = 0$. Consequently, the $k_{1,2}$ integrals are finite, and we obtain

$$c_2(T, \eta) \sim T^{7/2} e^{-3\Delta/T}, \quad T \ll \Delta, \quad 3 < b < 5, \quad (\text{A.7})$$

so that the contribution to $1/T_1$ is $\sim T^{9/2}e^{-3\Delta/T}$.

However, for $1 < b < 3$, there are divergences in Eq. (A.6). The divergences are present when either $k_1 = 0$ or $k_2 = 0$. So let us consider the form of Eq. (A.6) when both k_1 and k_2 are small. After suitable rescaling of momenta, we obtain an expression of the form

$$c_2(T, \eta) \sim T^{7/2}e^{-3\Delta/T} \int dk_1 dk_2 \frac{(k_1 - k_2)^2}{k_1^6 k_2^6} \delta\left((k_1^2 + 1)^{1/2} + (k_2^2 + 1)^{1/2} - 3\right). \quad (\text{A.8})$$

This integral has an effective divergence $\sim \int dk/k^6$. As we discussed below Eq. (1.38), this divergence will be cancelled by the other terms in Eq. (1.35). A conjectured estimate is obtained by cutting off the divergence at $k \sim \sqrt{T}$, which leads to $c_2(T) \sim Te^{-3\Delta/T}$ and therefore a contribution to $1/T_1$ which is $\sim T^2e^{-3\Delta/T}$.

B

Appendix for Chapter 2

B.1 SPECTRAL FUNCTION AND GREEN'S FUNCTIONS OF RANDOM HOPPING MODEL

The random hopping model of Majorana fermions, the SYK Hamiltonian for $q = 2$, is exactly solvable^{[145](#)}. The Matsubara Green's function follows from the solution of the Dyson equation

$$G(i\omega_n)^{-1} = i\omega_n - \Sigma(i\omega_n) = i\omega_n + J_2^2 G(i\omega_n). \quad (\text{B.1})$$

This can be rewritten as a quadratic equation for $(G(i\omega_n))^{-1}$. Matching the branches with the correct high-frequency asymptotics and symmetry yields the propagator

$$G(i\omega_n) = \frac{2}{i\omega_n + i \operatorname{sgn}(\omega_n) \sqrt{\omega_n^2 + 4J_2^2}}. \quad (\text{B.2})$$

Analytically continuing to the real frequency axis from the upper half plane yields the retarded Green's function,

$$G^R(\omega + i\delta) = \frac{2}{\omega + i\delta + i\sqrt{4J_2^2 - (\omega + i\delta)^2}}, \quad (\text{B.3})$$

where δ can be set to zero due to the presence of the finite imaginary part and $\operatorname{sgn} \omega_n \rightarrow 1$ in the analytic continuation from the upper half plane. For the spectral function, we obtain

$$A(\omega) = -2 \operatorname{Im} G^R(\omega + i\delta) = \frac{2}{J_2} \sqrt{1 - \left(\frac{\omega}{2J_2}\right)^2} \quad \text{for } |\omega| < 2J_2, \quad (\text{B.4})$$

which is the well-known semicircular density of states.

This yields the following expression for $G^>(t)$

$$iG^>(t) = \frac{1}{2J_2 t} (J_1(2J_2 t) - iH_1(2J_2 t)), \quad (\text{B.5})$$

where J_1 is the Bessel function of the first kind (`BesselJ` in `Mathematica`) and H_1 is the Struve function (`StruveH` in `Mathematica`).

B.2 SPECTRAL FUNCTION IN THE CONFORMAL LIMIT

In the scaling limit at non-zero temperature the retarded Green's function is given by the following expression

$$iG^R(t) = 2b(\cos \pi \Delta) \left(\frac{\pi}{\beta \sinh \frac{\pi t}{\beta}} \right)^{2\Delta} \theta(t), \quad (\text{B.6})$$

where $\Delta = 1/q$ is the fermion scaling dimension. At $q = 4$ we obtain

$$iG_c^R(t) = \sqrt{2}b \left(\frac{\pi}{\beta \sinh \frac{\pi t}{\beta}} \right)^{\frac{1}{2}} \theta(t) \quad (\text{B.7})$$

The Wigner transform of the retarded Green's functions is given by

$$\begin{aligned} iG^R(\omega) &= \sqrt{2}b \int_0^\infty dt e^{i\omega t} \left(\frac{\pi}{\beta \sinh \frac{\pi t}{\beta}} \right)^{\frac{1}{2}} \\ &= b \left(\frac{\pi}{\beta} \right)^{-\frac{1}{2}} B\left(\frac{1}{2}; \frac{1}{4} - \frac{i\beta\omega}{2\pi}\right), \end{aligned} \quad (\text{B.8})$$

where $b = (4\pi J_4^2)^{-1/4}$. The associated spectral function is

$$A(\mathcal{T}, \omega) = 2b \left(\frac{\pi}{\beta} \right)^{-\frac{1}{2}} \text{Re} \left[B\left(\frac{1}{2}; \frac{1}{4} - \frac{i\beta\omega}{2\pi}\right) \right]. \quad (\text{B.9})$$

B.3 DETAILS ON THE NUMERICAL SOLUTION OF KADANOFF-BAYM EQUATION

The Green's function is typically determined on two-dimensional grids in (t, t') space with 8000×8000 or 12000×12000 points, where the quench happens after half of the points in each direction. When starting from initial states in which only J_2 is finite, the Green's function decays algebraically in time. This leads to significant finite size effects in Fourier transforms. In order to reduce the latter, most numerical results were obtained by starting from a thermal state in which J_2 and J_4 , or only J_4 , are finite. In these cases, the Green's function decays exponentially as a function of the relative time dependence.

We checked the quality and consistency of the results by monitoring the conservation of energy, the normalization of the spectral function and the real and imaginary part of the retarded propagator are Kramers-Kronig consistent for long times after the quench.

In order to time-evolve the Kadanoff-Baym equations we have to determine $G^<(t_1, t_2)$ for t_1, t_2 . When quenching the ground state of the random hopping model, we can use the exact solution for $G^>$ as initial condition. When we do not have an analytic expression the Green's function we solve the Dyson equation self-consistently according to the following scheme:

1. Prepare $iG^>$ with an initial guess, for example the propagator of the random hopping model.
2. Computation of retarded self-energy in time domain:

$$i\Sigma^R(t) = \Theta(t)(i\Sigma^>(t) + i\Sigma^>(-t)) \quad (\text{B.10})$$

3. Fourier transformation

$$i\Sigma^R(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} i\Sigma^R(t) \quad (\text{B.11})$$

4. Solve the Dyson equation:

$$G^R(\omega) = \frac{1}{\omega - \Sigma^R(\omega)} \quad (\text{B.12})$$

5. Determine spectral function

$$A(\omega) = -2 \text{Im } G^R(\omega) \quad (\text{B.13})$$

6. Determine $iG^>(\omega)$ from spectral function,

$$iG^>(\omega) = (1 - n_F(\omega))A(\omega) \quad (\text{B.14})$$

Note that this is the only step in the self-consistency procedure where the temperature β^{-1} enters through the Fermi function n_F .

7. Fourier transformation to the time domain

$$iG^>(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} iG^>(\omega) \quad (\text{B.15})$$

8. Continue with step 2 until convergence is reached.

B.4 NUMERICAL QUENCHES: SIMPLER CASES

This appendix describes the quenches A-C discussed in Section 2.3.

B.4.1 $(J_{2,i}, 0) \rightarrow (J_{2,f}, 0)$ QUENCH: RESCALING OF THE RANDOM-HOPPING MODEL

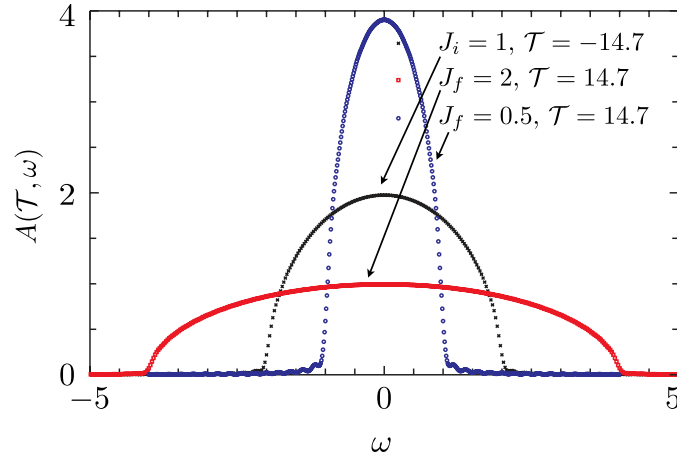


Figure B.1: The spectral function of the random hopping model A long before ($\mathcal{T} = -14.7$) and after ($\mathcal{T} = 14.7$) a parameter quench from $J_{2,i} = 1$ to $J_{2,f} = 0.5$ and 2.0 . The width depends only on the value of J_f , becoming wider if $J_f > J_i$ and narrower if $J_f < J_i$.

We start by discussing the simplest quench protocol A. In Fig. B.1 we show the spectral function long before and long after a parameter quench. These results were obtained from a numerical Fourier transformation of the retarded Green's function as described by Eq. (2.49) with a broadening of $\delta = 0.025$. In Fig. B.2, we show the frequency dependence of the Keldysh component of the

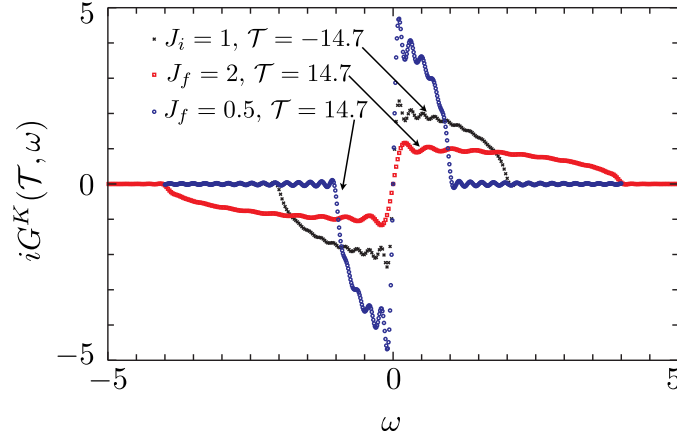


Figure B.2: The Keldysh component of the Green's function of the random hopping model A long before ($\mathcal{T} = -14.7$) and after ($\mathcal{T} = 14.7$) a parameter quench from $J_{2,i} = 1$ to $J_{2,f} = 0.5$ and 2.0 .

fermionic Green's function for the same quench protocol as in Fig. B.1. We determined the inverse temperature β using Eq. (2.52), and found that βJ_2 was roughly constant across the quench. All three results of this $(J_{2,i}, 0) \rightarrow (J_{2,f}, 0)$ quench are consistent with a rescaling of energy scales for all quantities. This is expected because the random hopping model has only one energy scale J_2 , and the analog of the reparameterization in Eq. (2.59) applies here. This also applies to the complementary case $(0, J_{4,i}) \rightarrow (0, J_{4,f})$ (as was pointed out to us by J. Maldacena). For analytic expressions of the spectral functions we refer to Appendix B.1.

B.4.2 $(0, 0) \rightarrow (0, J_4)$ QUENCH: FROM BARE MAJORANA FERMIONS TO THE SYK MODEL

Next we consider the quench B. In Fig. B.3, we show the spectral function long after suddenly switching on the quartic interaction in the SYK model, starting from bare, noninteracting Majorana fermions. When interpreting results of this quench protocol, it is important to keep in mind that the Hamiltonian before the quench is zero, so that any finite $J_{4,f}$ is an arbitrarily strong perturbation. We found that the total energy is zero both before and after the quench, the quench does not pump energy into the system, and that the Keldysh component of the Green's function vanishes be-

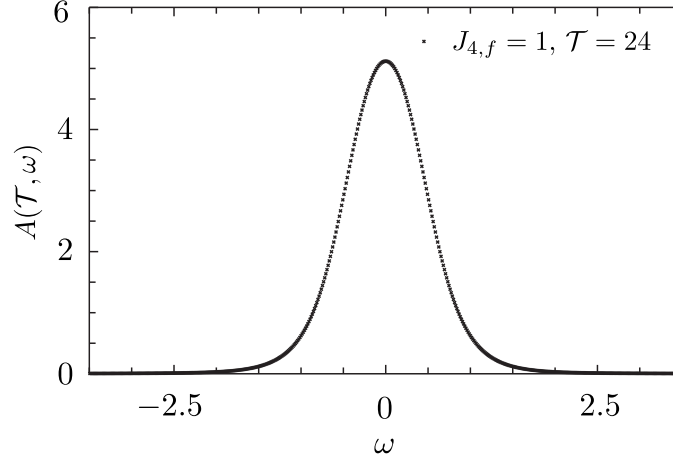


Figure B.3: The spectral function for Majorana fermions is shown long after a quench starting with free Majorana fermions and switching on a $q = 4$ SYK interaction term. Here $J_{4,f} = 1$.

fore and after the quench. The latter is to be expected for free Majorana fermions, and is consistent with the post-quench temperature of the system being infinite, $\beta_f = 0$. Indeed the spectral function in Fig. B.3 agrees with that of the SYK model at infinite temperature.

B.4.3 $(J_{2,i}, 0) \rightarrow (0, J_{4,f})$: FROM A QUADRATIC TO A QUARTIC MODEL

We now consider C, the model containing both $q = 2$ and $q = 4$ interactions. The quench from the purely quadratic to the purely quartic model decouples the regions $t_1, t_2 < 0$ from $t_1, t_2 > 0$: this can be seen from the structure of the Kadanoff-Baym equations, and of the self-energy for this quench protocol, which reads

$$\Sigma(t_1, t_2) = \Theta(-t_1)\Theta(-t_2)J_2^2 G(t_1, t_2) - \Theta(t_1)\Theta(t_2)J_4^2 G(t_1, t_2)^3. \quad (\text{B.16})$$

Inserting this into the Kadanoff-Baym equations for $t_1 > 0$ and $t_2 > 0$, we can see that all time integrals are restricted to positive times, and the initial condition does not matter. Although $G^>(t_1, t_2)$ shows some time evolution for $t_1 \gtrless 0$ and $t_2 \lesseqgtr 0$ due to integrals involving the region with $t_1 < 0$ and

$t_2 < 0$, $G^>$ in this region does not influence the time evolution at positive t_1, t_2 because $\Sigma(t_1, t_2) = 0$ when $t_1 \gtrless 0$ and $t_2 \lesseqgtr 0$. Thus, the relevant initial condition for the time evolution of $G^>(t_1, t_2)$ is $iG^>(t_1 = 0, t_2 = 0) = 1/2$. Hence propagating $iG^>$ forward in time using the self-energy of the quartic model, we obtain the same time evolution at positive times as we did when starting from bare Majorana fermions in case B. So the final state here is also the SYM model at infinite temperature.

B.5 CONJUGACY PROPERTY OF $g(t_1, t_2)$

In this appendix we illustrate that the causality structure of the Kadanoff-Baym equations eq. (2.60) leads to the property

$$[g(t_1, t_2)]^* = g(t_2, t_1) \quad (\text{B.17})$$

in all four quadrants of Fig. B.4. Since the system is thermal in quadrant C the conjugacy property

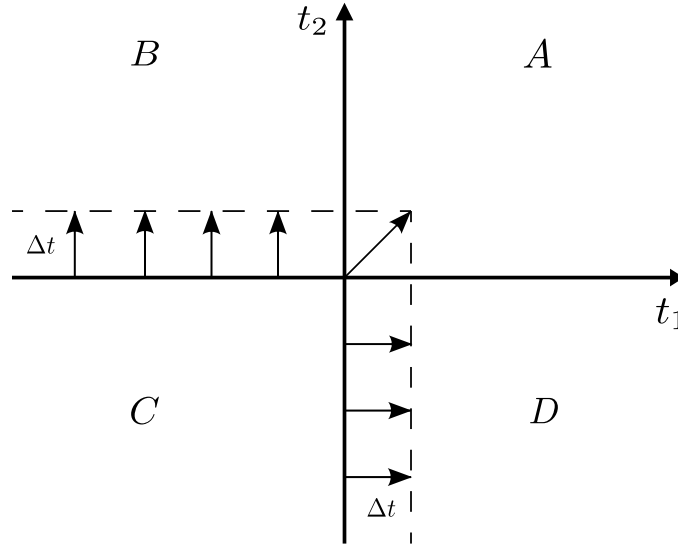


Figure B.4: The propagation of the conjugacy property in the t_1 - t_2 plane.

$[g(t_1, t_2)]^* = g(t_2, t_1)$ can be read off the thermal solution for $t_1, t_2 \leq 0$. Next, we consider the propagation from the line $\{(t_1, t_2) \in \mathbb{R}^2 | t_2 = 0, t_1 \leq 0\}$ for an infinitesimal time Δt in the t_2 direction to the line $\{(t_1, t_2) \in \mathbb{R}^2 | t_2 = \Delta t, t_1 \leq 0\}$. We discretize equation (2.60) as follows

$$\begin{aligned} \frac{1}{\Delta t} [g(t_1, \Delta t) - g(t_1, 0)] &= 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}(t_3) \mathcal{J}(0) e^{g(t_3, 0)} - \int_{-\infty}^0 dt_3 \mathcal{J}(t_3) \mathcal{J}(0) [e^{g(t_3, 0)} + e^{g(0, t_3)}] \\ &\quad + 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(0) e^{pg(t_3, 0)} - \int_{-\infty}^0 dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(0) [e^{pg(t_3, 0)} + e^{pg(0, t_3)}] \end{aligned} \quad (\text{B.18})$$

and solve the resulting equation for $g(t_1, \Delta t)$. Taking the complex conjugate leads to

$$\begin{aligned} g(t_1, \Delta t)^* &= g(t_1, 0)^* + \Delta t \left\{ 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}(t_3) \mathcal{J}(0) e^{g(t_3, 0)^*} - \int_{-\infty}^0 dt_3 \mathcal{J}(t_3) \mathcal{J}(0) [e^{g(t_3, 0)^*} + e^{g(0, t_3)^*}] \right. \\ &\quad \left. + 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(0) e^{pg(t_3, 0)^*} - \int_{-\infty}^0 dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(0) [e^{pg(t_3, 0)^*} + e^{pg(0, t_3)^*}] \right\}. \end{aligned} \quad (\text{B.19})$$

On the right hand side we are allowed to use the property $g(t_1, t_2)^* = g(t_2, t_1)$ since all g 's are still living in the C quadrant. We obtain

$$\begin{aligned} g(t_1, \Delta t)^* &= g(0, t_1) + \Delta t \left\{ 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}(t_3) \mathcal{J}(0) e^{g(0, t_3)} - \int_{-\infty}^0 dt_3 \mathcal{J}(t_3) \mathcal{J}(0) [e^{g(0, t_3)} + e^{g(t_3, 0)}] \right. \\ &\quad \left. + 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(0) e^{pg(0, t_3)} - \int_{-\infty}^0 dt_3 \mathcal{J}_p(t_3) \mathcal{J}_p(0) [e^{pg(0, t_3)} + e^{pg(t_3, 0)}] \right\}. \end{aligned} \quad (\text{B.20})$$

Next we let the system propagate from the line $\{(t_1, t_2) \in \mathbb{R}^2 | t_2 \leq 0, t_1 = 0\}$ for an infinitesimal time Δt in the t_1 direction to the line $\{(t_1, t_2) \in \mathbb{R}^2 | t_1 = \Delta t, t_2 \leq 0\}$. We discretize equation (2.57) as

follows

$$\begin{aligned}
-\frac{1}{\Delta t} [g(\Delta t, t_2) - g(0, t_2)] &= \int_{-\infty}^0 dt_3 \mathcal{J}(0) \mathcal{J}(t_3) \left[e^{g(0, t_3)} + e^{g(t_3, 0)} \right] - 2 \int_{-\infty}^{t_2} dt_3 \mathcal{J}(0) \mathcal{J}(t_3) e^{g(0, t_3)} \\
&\quad + \int_{-\infty}^0 dt_3 \mathcal{J}_p(0) \mathcal{J}_p(t_3) \left[e^{pg(0, t_3)} + e^{pg(t_3, 0)} \right] - 2 \int_{-\infty}^{t_2} dt_3 \mathcal{J}_p(0) \mathcal{J}_p(t_3) e^{pg(0, t_3)}
\end{aligned} \tag{B.21}$$

We set $t_2 = t_1$ in the last equation and solve for $g(\Delta t, t_1)$ leading to

$$\begin{aligned}
g(\Delta t, t_1) &= g(0, t_1) - \Delta t \left\{ \int_{-\infty}^0 dt_3 \mathcal{J}(0) \mathcal{J}(t_3) \left[e^{g(0, t_3)} + e^{g(t_3, 0)} \right] - 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}(0) \mathcal{J}(t_3) e^{g(0, t_3)} \right. \\
&\quad \left. + \int_{-\infty}^0 dt_3 \mathcal{J}_p(0) \mathcal{J}_p(t_3) \left[e^{pg(0, t_3)} + e^{pg(t_3, 0)} \right] - 2 \int_{-\infty}^{t_1} dt_3 \mathcal{J}_p(0) \mathcal{J}_p(t_3) e^{pg(0, t_3)} \right\}. \tag{B.22}
\end{aligned}$$

Comparing the right hand side of (B.19) with (B.22) we conclude $g(t_1, \Delta t)^* = g(\Delta t, t_1)$. Furthermore, the point $g(\Delta t, \Delta t)$ fulfills the conjugate property trivially. In total we propagated the conjugate property one time slice. Repeating this argument for every time slice of size Δt , the property will hold in all four quadrants of Fig. B.4.

B.6 INITIAL STATE IN THE LARGE q LIMIT

In quadrant C in Fig. 2.5, all Green's functions are dependent only on time differences, so we can write

$$g_C(t_1, t_2) \equiv g_C(t_1 - t_2) \tag{B.23}$$

and Eq. (2.60) as

$$\frac{dg_C(t)}{dt} = \mathcal{J}^2 \int_{-\infty}^{\infty} dt' e^{g_C(t')} \text{sgn}(t' - t) + \mathcal{J}_p^2 \int_{-\infty}^{\infty} dt' e^{pg_C(t')} \text{sgn}(t' - t). \tag{B.24}$$

This implies the second order differential equation (also obtainable from Eq. (2.62))

$$-\frac{d^2 g_C}{dt^2} = 2\mathcal{J}^2 e^{g_C} + 2\mathcal{J}_p^2 e^{p g_C}. \quad (\text{B.25})$$

Eq. (B.25) turns out to be exactly solvable at $p = 2$ (pointed out to us by Wenbo Fu, following¹⁰⁷). Inserting $g_C(t) = \ln(-1/f(t))$ into Eq. (B.25) we get

$$\frac{1}{f} \frac{d^2 f}{dt^2} - \frac{1}{f^2} \left(\frac{df}{dt} \right)^2 = -\frac{2\mathcal{J}^2}{f} + \frac{2\mathcal{J}_2^2}{f^2} \quad (\text{B.26})$$

The solution of this differential equation yields

$$g_C(t) = \ln \left[\frac{-\sigma^2}{\sqrt{4\mathcal{J}^4 + 2\mathcal{J}_2^2 \sigma^2} \cosh(\sigma t - 2i\theta) - 2\mathcal{J}^2} \right], \quad (\text{B.27})$$

with

$$\cos(2\theta) = \frac{2\mathcal{J}^2 - \sigma^2}{\sqrt{4\mathcal{J}^4 + 2\mathcal{J}_2^2 \sigma^2}}. \quad (\text{B.28})$$

By evaluating the Fourier transform of the Green's function we find $G^>(-\omega) = e^{-\beta_i \omega} G^>(\omega)$, or by analytically continuing to imaginary time, we find that the initial inverse temperature is

$$\beta_i = \frac{2(\pi - 2\theta)}{\sigma}. \quad (\text{B.29})$$

We will ultimately be interested in the scaling limit in which $\theta \rightarrow 0$ and $\sigma \ll \mathcal{J}, \mathcal{J}_2$.

Eq. (B.25) is also exactly solvable at $p = 1/2$ by

$$g_C(t) = 2 \ln \left[\frac{\sigma^2/2}{i\sqrt{4\mathcal{J}_{1/2}^4 + \mathcal{J}^2 \sigma^2} \sinh(\sigma t/2 - i\theta) + 2\mathcal{J}_{1/2}^2} \right], \quad (\text{B.30})$$

where now

$$\sin(\theta) = \frac{\sigma^2/2 - 2\mathcal{J}_{1/2}^2}{\sqrt{4\mathcal{J}_{1/2}^4 + \mathcal{J}^2\sigma^2}}. \quad (\text{B.31})$$

The value of the inverse initial temperature remains as in Eq. (B.29).

Note that both solutions in Eqs. (B.27) and (B.30) obey

$$g_C(-t) = g_C^*(t) \quad (\text{B.32})$$

It is also useful to recast the solution in quadrant C for the case $\mathcal{J}_p = 0$ in the form of Eq. (2.65).

We subdivide quadrant C into two subregions just as in quadrant A. From Eqs. (2.60) and (2.65) we obtain for $t_2 > t_1$

$$\begin{aligned} \frac{b''_{C1}(t_1)}{b'_{C1}(t_1)} &= \frac{2b_{C2}^{*'}(t_1)}{b_{C1}^*(t_1) - b_{C2}^*(t_1)} - \frac{2b_{C2}^{*'}(t_1)}{b_{C1}^*(-\infty) - b_{C2}^*(t_1)} - \frac{2b'_{C1}(t_1)}{b_{C2}(t_1) - b_{C1}(t_1)} \\ &\quad - \frac{b_{C2}^{*'}(t_1)}{b_{C1}^*(t_1) - b_{C2}^*(t_1)} + \frac{b_{C2}^{*'}(t_1)}{b_{C1}^*(-\infty) - b_{C2}^*(t_1)} - \frac{b'_{C2}(t_1)}{b_{C1}(t_1) - b_{C2}(t_1)} + \frac{b'_{C2}(t_1)}{b_{C1}(-\infty) - b_{C2}(t_1)} \\ \frac{b''_{C2}(t_2)}{b'_{C2}(t_2)} &= -\frac{2b'_{C2}(t_2)}{b_{C1}(-\infty) - b_{C2}(t_2)} - \frac{b'_{C2}(t_2)}{b_{C1}(t_2) - b_{C2}(t_2)} + \frac{b'_{C2}(t_2)}{b_{C1}(-\infty) - b_{C2}(t_2)} \\ &\quad - \frac{b_{C2}^{*'}(t_2)}{b_{C1}^*(t_2) - b_{C2}^*(t_2)} + \frac{b_{C2}^{*'}(t_2)}{b_{C1}^*(-\infty) - b_{C2}^*(t_2)}. \end{aligned} \quad (\text{B.33})$$

Adding the equations in Eq. (B.33), we have

$$\frac{b''_{C1}(t)}{b'_{C1}(t)} + \frac{b''_{C2}(t)}{b'_{C2}(t)} = 2 \left(\frac{b'_{C1}(t) - b'_{C2}(t)}{b_{C1}(t) - b_{C2}(t)} \right) \quad (\text{B.34})$$

which integrates to the expected

$$b'_{C1}(t)b'_{C2}(t) = -\mathcal{J}^2(b_{C1}(t) - b_{C2}(t))^2 \quad (\text{B.35})$$

The final equations for the thermal equilibrium state are

$$\begin{aligned} \frac{b_{C1}''(t)}{b_{C1}'(t)} &= \frac{2b_{C1}'(t_1)}{b_{C1}(t) - b_{C2}(t)} - \frac{b_{C2}^{*'}(t)}{b_{C1}^*(-\infty) - b_{C2}^*(t)} + \frac{b_{C2}'(t)}{b_{C1}(-\infty) - b_{C2}(t)} \\ &\quad - \frac{b_{C2}'(t)}{b_{C1}(t) - b_{C2}(t)} + \frac{b_{C2}^{*'}(t)}{b_{C1}^*(t) - b_{C2}^*(t)} \end{aligned} \quad (\text{B.36})$$

$$b_{C1}'(t)b_{C2}'(t) = -\mathcal{J}^2(b_{C1}(t) - b_{C2}(t))^2. \quad (\text{B.37})$$

Unlike Eqs. (2.84, 2.85), Eqs. (B.36, B.37) have to be integrated from $t = -\infty$. One solution of Eqs. (B.36, B.37) is

$$b_{C1}(t) = b_{C1}(-\infty) + Ae^{i\theta}e^{\sigma t} \quad b_{C2}(t) = b_{C1}(-\infty) + Ae^{-i\theta}e^{\sigma t} \quad (\text{B.38})$$

with

$$\sigma = 2\mathcal{J} \sin(\theta). \quad (\text{B.39})$$

Note that the $g_C(t_1, t_2)$ obtained from this solution agrees with Eq. (B.27) at $\mathcal{J}_2 = 0$ and with Eq. (B.30) at $\mathcal{J}_{1/2} = 0$.



Appendix for Chapter 3

C.1 REAL TIME POLARIZATION BUBBLE

In this appendix we will describe the calculation of the components photon polarization tensor

We start with the expression for the spatial components in Euclidean time

$$\Pi_{ij} = T \sum_{p_n} \int_p \left[\frac{2\delta_{ij}}{P^2} + \frac{-K^2\delta_{ij} + 2k_i k_j - 4p_i p_j}{P^2(K-P)^2} \right] \quad (\text{C.1})$$

where K and P are Euclidean vectors. Note that the numerator does not involve the summed-over Matsubara frequencies. The Matsubara sums can be performed to give

$$T \sum_{p_n} \frac{1}{p_n^2 + E_{\mathbf{p}}^2} = \frac{1 - 2n_F(E_{\mathbf{p}})}{2E_{\mathbf{p}}} \quad (\text{C.2})$$

and

$$T \sum_{p_n} \frac{1}{p_n^2 + E_{\mathbf{p}}^2} \frac{1}{(k_n - p_n)^2 + E_{\mathbf{k}-\mathbf{p}}^2} = \frac{1}{4E_{\mathbf{p}}E_{\mathbf{k}-\mathbf{p}}} \sum_{s,s'} \frac{ss' [n_F(sE_{\mathbf{p}}) + n_F(s'E_{\mathbf{k}-\mathbf{p}}) - 1]}{ik_n - sE_{\mathbf{p}} - s'E_{\mathbf{k}-\mathbf{p}}}. \quad (\text{C.3})$$

The first term, after subtracting the zero temperature value, gives

$$2 \int_p \frac{1 - 2n_F(E_{\mathbf{p}})}{2E_{\mathbf{p}}} - 2 \int_p \frac{1}{2E_{\mathbf{p}}} = -\frac{T}{\pi} \int_0^\infty du \frac{1}{e^u + 1} = -\frac{T \log 2}{\pi}. \quad (\text{C.4})$$

where $\mathbf{k} = (|\mathbf{k}|, 0)$ and $\mathbf{p} = |\mathbf{p}|(\cos \theta, \sin \theta)$. The numerators of the second term are

$$L^{xx} = -(k_n^2 + |\mathbf{k}|^2) + 2|\mathbf{k}|^2 - 4|\mathbf{p}|^2 \cos^2 \theta \quad (\text{C.5})$$

and

$$L^{yy} = -(k_n^2 + |\mathbf{k}|^2) - 4|\mathbf{p}|^2 \sin^2 \theta. \quad (\text{C.6})$$

C.1.1 TENSOR STRUCTURE

The full real-time polarization can be expanded in two tensor structures. While the transverse component is spatially transverse, the longitudinal component is spacetime transverse. This tensor structure ensures that the full polarization bubble obeys the Ward identity required by current conservation.

At finite temperature there is a new vector available set by the rest frame of the medium. The

tensor structures are

$$P_L^{\mu\nu} = -g^{\mu\nu} + \frac{k^\mu k^\nu}{k^2} - P_T^{\mu\nu} \quad (\text{C.7})$$

and

$$P_T^{\mu\nu} = \delta^{\mu i} \delta^{\nu j} \left(\delta^{ij} - \frac{k^i k^j}{|\mathbf{k}|^2} \right). \quad (\text{C.8})$$

From these two structures we can define

$$\tilde{P}^{\mu\nu} = P_L^{\mu\nu} + P_T^{\mu\nu} \quad (\text{C.9})$$

These obey

$$k_\mu \tilde{P}^{\mu\nu} = 0, \quad (\text{C.10})$$

$$k_\mu P_T^{\mu\nu} = 0, \quad (\text{C.11})$$

and

$$k_i P_T^{ij} = 0. \quad (\text{C.12})$$

To give an invariant definition of $P_T^{\mu\nu}$, let u^μ denote the velocity of the medium, which in the medium's rest frame is $u = (1, 0, 0)$. Let the transverse u projector be

$$P_T^{\mu\nu}(u) = -g^{\mu\nu} + \frac{u^\mu u^\nu}{u^2}, \quad (\text{C.13})$$

and define

$$\bar{k}^\mu = P_T^{\mu\nu}(u) k_\nu = -k^\mu + (k \cdot u) u^\mu. \quad (\text{C.14})$$

In the medium's rest frame, $k = (k^0, \mathbf{k})$, then

$$\bar{k} = (0, -\mathbf{k}). \quad (\text{C.15})$$

The transverse projector to $\bar{k}$ is

$$P_T^{\mu\nu}(\bar{k}) = -g^{\mu\nu} + \frac{\bar{k}^\mu \bar{k}^\nu}{\bar{k}^2}, \quad (\text{C.16})$$

which in the medium's rest frame is

$$P_L^{00}(\bar{k}) = -1 = -u^0 u^0, \quad (\text{C.17})$$

$$P_L^{0i}(\bar{k}) = 0, \quad (\text{C.18})$$

and

$$P_T^{ij}(\bar{k}) = \delta^{ij} - \frac{k^i k^j}{|\mathbf{k}|^2} = P_T^{ij}. \quad (\text{C.19})$$

Hence we may identify $P_T^{\mu\nu}$ with $u^\mu u^\nu + P_T^{\mu\nu}(\bar{k})$ (with $\tilde{P}^{\mu\nu}$ with $\tilde{P}^{\mu\nu}(k)$). We write the full polarization as

$$\Pi_R^{\mu\nu} = \tilde{\Pi} \tilde{P}^{\mu\nu} + \Pi \tilde{P}_T^{\mu\nu}. \quad (\text{C.20})$$

The non-vanishing terms are

$$\Pi_R^{00} = \Pi \frac{|\mathbf{k}|^2}{(k^0)^2 - |\mathbf{k}|^2}, \quad (\text{C.21})$$

$$\Pi_R^{xx} = \Pi \left(1 + \frac{(k^x)^2}{(k^0)^2 - |\mathbf{k}|^2} \right) + \tilde{\Pi} \left(1 - \frac{(k^x)^2}{|\mathbf{k}|^2} \right), \quad (\text{C.22})$$

and

$$\Pi_R^{\gamma\gamma} = \Pi \left(1 + \frac{(k^0)^2}{(k^0)^2 - |\mathbf{k}|^2} \right) + \tilde{\Pi} \left(1 - \frac{(k^0)^2}{|\mathbf{k}|^2} \right). \quad (\text{C.23})$$

For kinematics where $\mathbf{k} = (|\mathbf{k}|, 0)$, these expressions simplify to

$$\Pi_R^{00} = \Pi \frac{|\mathbf{k}|^2}{(k^0)^2 - |\mathbf{k}|^2}, \quad (\text{C.24})$$

$$\Pi_R^{xx} = \Pi \frac{(k^0)^2}{(k^0)^2 - |\mathbf{k}|^2}, \quad (\text{C.25})$$

and

$$\Pi_R^{\gamma\gamma} = \Pi + \tilde{\Pi}. \quad (\text{C.26})$$

Hence Π_R^{xx} and $\Pi_R^{\gamma\gamma}$ are sufficient to determine the two functions Π and $\tilde{\Pi}$.

C.1.2 IMAGINARY PART, FULL EXPRESSIONS

The first contribution to both Π_R^{xx} and $\Pi_R^{\gamma\gamma}$ is the constant term,

$$2 \int_{\mathbf{p}} \frac{1 - 2n_F(E_{\mathbf{p}})}{2E_{\mathbf{p}}} = -\frac{T \log 2}{\pi} + \Pi_{T=0}. \quad (\text{C.27})$$

where $\Pi_{T=0}$ is the zero temperature contribution. Because this contribution is pure real, it drops out of the imaginary part.

Thus the full imaginary part of Π_R^{ij} can be obtained from

$$\text{Im } \Pi_R^{ij} = \text{Im} \sum_{s,s'} \int_{\mathbf{p}} \frac{L^{ij}}{4E_{\mathbf{p}}E_{\mathbf{k}-\mathbf{p}}} \frac{ss' [n_F(sE_{\mathbf{p}}) + n_F(s'E_{\mathbf{k}-\mathbf{p}}) - 1]}{\omega + i\varepsilon - sE_{\mathbf{p}} - s'E_{\mathbf{k}-\mathbf{p}}}, \quad (\text{C.28})$$

where the numerator is

$$L^{ij} = -(-(k^0)^2 + |\mathbf{k}|^2) \delta^{ij} + 2k^i k^j - 4p^i p^j. \quad (\text{C.29})$$

Using kinematics where $\mathbf{k} = (|\mathbf{k}|, 0)$ and $\mathbf{p} = (|\mathbf{p}| \cos \theta, |\mathbf{p}| \sin \theta)$, we find

$$L^{xx} = (k^0)^2 + |\mathbf{k}|^2 - 4|\mathbf{p}|^2 \cos^2 \theta \quad (\text{C.30})$$

and

$$L^{yy} = (k^0)^2 - |\mathbf{k}|^2 - 4|\mathbf{p}|^2 \sin^2 \theta. \quad (\text{C.31})$$

We can check that $\text{Im } \Pi_R^{ij}$ is an odd function of frequency. For positive k^0 , only the $(1, 1)$, $(1, -1)$, and $(-1, 1)$ terms contribute. Furthermore, the $(1, -1)$ and $(-1, 1)$ terms give the same result. The $(1, 1)$ contribution is

$$\text{Im } \Pi_R^{ij} \Big|_{s'=1} = - \int_p \frac{\pi L^{ij}}{4E_p E_{\mathbf{k}-\mathbf{p}}} [n_F(E_p) + n_F(E_{\mathbf{k}-\mathbf{p}}) - 1] \delta(k^0 - E_p - E_{\mathbf{k}-\mathbf{p}}). \quad (\text{C.32})$$

The $(1, -1)$ contribution (counted twice) is

$$\text{Im } \Pi_R^{ij} \Big|_{s'=-1} = 2 \int_p \frac{\pi L^{ij}}{4E_p E_{\mathbf{p}-\mathbf{k}}} [n_F(E_p) - n_F(E_{\mathbf{k}-\mathbf{p}})] \delta(k^0 - E_p + E_{\mathbf{k}-\mathbf{p}}). \quad (\text{C.33})$$

The delta function condition (with $s = 1$) is

$$k^0 - E_p = s' E_{\mathbf{k}-\mathbf{p}}. \quad (\text{C.34})$$

To solve it, let us square both sides,

$$(k^0 - |\mathbf{p}|)^2 = (|\mathbf{k}| - |\mathbf{p}| \cos \theta)^2 + |\mathbf{p}|^2 \sin^2 \theta. \quad (\text{C.35})$$

The result is

$$(k^0)^2 - 2k^0|\mathbf{p}| = |\mathbf{k}|^2 - 2|\mathbf{p}||\mathbf{k}|\cos\theta \quad (\text{C.36})$$

or

$$|\mathbf{p}| = |\mathbf{p}|_0 = \frac{(k^0)^2 - |\mathbf{k}|^2}{2(k^0 - |\mathbf{k}|\cos\theta)}. \quad (\text{C.37})$$

We need either $k^0 > |\mathbf{k}|$ or $k^0 < |\mathbf{k}|\cos\theta$ to have a positive solution for $|\mathbf{p}|$. The former is realized for $s' = 1$ while the latter is realized for $s' = -1$. To complete the evaluation of the delta function, we also need

$$v_{s'} = \left. \frac{\partial(E_{\mathbf{p}} + s'E_{\mathbf{k}-\mathbf{p}})}{\partial|\mathbf{p}|} \right|_{|\mathbf{p}|=|\mathbf{p}|_0} = 1 + s' \frac{|\mathbf{p}| - |\mathbf{k}|\cos\theta}{E_{\mathbf{k}-\mathbf{p}}}. \quad (\text{C.38})$$

The $s' = 1$ contribution is

$$\text{Im } \Pi_R^{ij} \Big|_{s'=1} = -\frac{1}{(2\pi)^2} \int_0^{2\pi} d\theta \int_0^\infty d|\mathbf{p}||\mathbf{p}| \frac{\delta(|\mathbf{p}| - |\mathbf{p}|_0)}{|v_1|} \frac{\pi L^{ij} (n_F(E_{\mathbf{p}}) + n_F(E_{\mathbf{k}-\mathbf{p}}) - 1)}{4E_{\mathbf{p}}E_{\mathbf{k}-\mathbf{p}}}. \quad (\text{C.39})$$

The $s' = -1$ contribution is

$$\text{Im } \Pi_R^{ij} \Big|_{s'=-1} = \frac{1}{(2\pi)^2} \int_{-\cos^{-1} \frac{k^0}{|\mathbf{k}|}}^{\cos^{-1} \frac{k^0}{|\mathbf{k}|}} d\theta \int_0^\infty d|\mathbf{p}||\mathbf{p}| \frac{\delta(|\mathbf{p}| - |\mathbf{p}|_0)}{|v_{-1}|} \frac{\pi L^{ij} (n_F(E_{\mathbf{p}}) - n_F(E_{\mathbf{k}-\mathbf{p}}))}{4E_{\mathbf{p}}E_{\mathbf{k}-\mathbf{p}}}. \quad (\text{C.40})$$

The real part is then obtained from a subtracted Kramers-Kronig relation. For a function $\chi(\omega) = \chi_1 + i\chi_2$ obeying the assumptions of Kramers-Kronig, the real part in terms of the imaginary part is

$$\chi_1(\omega) = \frac{2}{\pi} \int_0^\infty d\nu \frac{\omega \chi_2(\omega) - \nu \chi_2(\nu)}{\omega^2 - \nu^2}. \quad (\text{C.41})$$

C.1.3 LOW TEMPERATURE LIMIT

Suppose $\omega \gg T$ and $|\mathbf{k}| = 0$. Only the $ss' = 1$ term contributes to $\text{Im } \Pi_R^{ij}$. The contribution is

$$\text{Im } \Pi_R^{ij} = \frac{1}{(2\pi)^2} \int d|\mathbf{p}| |\mathbf{p}| \frac{\partial(|\mathbf{p}| - k^0/2)}{2} \frac{\pi L^{ij}}{4|\mathbf{p}|^2} \quad (\text{C.42})$$

with $L^{xx} \rightarrow (k^0)^2 - 4|\mathbf{p}|^2 \cos^2 \theta$ and $L^{yy} \rightarrow (k^0)^2 - 4|\mathbf{p}|^2 \sin^2 \theta$. Because of the simple dependence on θ , we conclude the $\text{Im } \Pi_R^{xx} = \text{Im } \Pi_R^{yy}$ in this limit. (Physically, since $|\mathbf{k}| = 0$, there is no difference between longitudinal and transverse.)

We find

$$\text{Im } \Pi_R^{yy} = \frac{1}{16\pi k^0} \int d\theta ((k^0)^2 - (k^0)^2 \sin^2 \theta) = \frac{k^0}{16}. \quad (\text{C.43})$$

C.1.4 HIGH TEMPERATURE LIMIT

In the high temperature limit, $k/T \ll 1$, we neglect $|\mathbf{k}|$ and k^0 relative to T and the loop momentum whenever the dependence on k is non-singular. The numerators limit to

$$L^{xx} = -4|\mathbf{p}|^2 \cos^2 \theta \quad (\text{C.44})$$

and

$$L^{yy} = -4|\mathbf{p}|^2 \sin^2 \theta. \quad (\text{C.45})$$

The xx polarization is

$$\Pi_R^{xx} = \left(-\frac{T \log 2}{\pi} + \sum_{s,s'} \int_{\mathbf{p}} \frac{L^{xx}}{4|\mathbf{p}|^2} \frac{ss' [n_F(sE_{\mathbf{p}}) + n_F(s'E_{\mathbf{k}-\mathbf{p}}) - 1]}{k^0 + i\varepsilon - sE_{\mathbf{p}} - s'E_{\mathbf{k}-\mathbf{p}}} \right).$$

oth $ss' = 1$ terms inside the parenthesis contribute the same value (in the small k/T limit, both are

real), as do both $ss' = -1$ terms. These are

$$ss' = 1 : 2 \int_{\mathbf{p}} \frac{-4 \cos^2 \theta}{4} \frac{2n_F(|\mathbf{p}|) - 1}{-2|\mathbf{p}|} = \frac{T \log 2}{2\pi} + (T = 0 \text{ value}) \quad (\text{C.47})$$

and

$$ss' = -1 : 2 \int_{\mathbf{p}} \frac{-4 \cos^2 \theta}{4} \frac{-\frac{\partial n_F}{\partial |\mathbf{p}|} |\mathbf{k}| \cos \theta}{k^0 + i\varepsilon - |\mathbf{k}| \cos \theta} = -\frac{T \log 2}{2\pi^2} \int d\theta \cos^2 \theta \frac{\cos \theta}{z - \cos \theta}, \quad (\text{C.48})$$

where $z = (k^0 + i\varepsilon)/|k|$. Further processing the $ss' = -1$ term gives

$$ss' = -1 : -\frac{T \log 2}{2\pi^2} \left(-\pi + \int d\theta \cos^2 \theta \frac{z}{z - \cos \theta} \right). \quad (\text{C.49})$$

Then we see that the constant term, the $ss' = 1$ terms, and the constant part of the $ss' = -1$ terms combine to give zero. Thus the xx polarization is

$$\Pi_R^{xx} = -\frac{T \log 2}{2\pi^2} \int d\theta \frac{z \cos^2 \theta}{z - \cos \theta}. \quad (\text{C.50})$$

The only difference between xx and yy is the replacement $\cos^2 \theta \rightarrow \sin^2 \theta$ when going from L^{xx} to L^{yy} . Hence it follows that

$$\Pi_R^{yy} = -\frac{T \log 2}{2\pi^2} \int d\theta \frac{z \sin^2 \theta}{z - \cos \theta}. \quad (\text{C.51})$$

The two integrals we must calculate are

$$I_1 = \int d\theta \frac{z \cos^2 \theta}{z - \cos \theta} \quad (\text{C.52})$$

and

$$I_2 = \int d\theta \frac{z \sin^2 \theta}{z - \cos \theta}. \quad (\text{C.53})$$

The sum of I_1 and I_2 is

$$I_1 + I_2 = - \int \frac{dw}{iw} \frac{2wz}{w^2 - 2wz + 1}, \quad (\text{C.54})$$

where we converted the expression to a contour integral around the unit circle. The integrand has poles at $w = w_{\pm} = z \pm \sqrt{z^2 - 1}$. The $w = w_-$ pole is inside the unit circle for $\text{Im } z > 0$. The integral is thus

$$I_1 + I_2 = -2\pi \frac{2z}{w_- - w_+} = \frac{2\pi z}{\sqrt{z^2 - 1}}. \quad (\text{C.55})$$

The difference of I_1 and I_2 is

$$I_1 - I_2 = - \int \frac{dw}{iw} \frac{2wz}{w^2 - 2wz + 1} \frac{w^4 + 1}{2w^2}, \quad (\text{C.56})$$

where again we converted to a contour integral. The integrand now has poles at $w = 0$ (double pole) and at $w = w_{\pm} = z \pm \sqrt{z^2 - 1}$. The integral is thus

$$I_1 - I_2 = -2\pi \left(2z^2 + \frac{2z}{w_- - w_+} \frac{w_-^4 + 1}{2w_-^2} \right). \quad (\text{C.57})$$

Simplifying gives

$$I_1 - I_2 = -2\pi \left(2z^2 - \frac{z}{\sqrt{z^2 - 1}} \left(\frac{(w_-^2 + 1)^2}{2w_-^2} - 1 \right) \right), \quad (\text{C.58})$$

or using $w_-^2 + 1 = 2zw_-$,

$$I_1 - I_2 = 2\pi \left(\frac{z}{\sqrt{z^2 - 1}} (2z^2 - 1) - 2z^2 \right). \quad (\text{C.59})$$

Hence

$$\Pi_R^{xx} = -\frac{T \log 2}{2\pi} \left[\frac{z}{\sqrt{z^2 - 1}} + \frac{z}{\sqrt{z^2 - 1}} (2z^2 - 1) - 2z^2 \right] \quad (\text{C.60})$$

or

$$\Pi_R^{xx} = -\frac{T \log 2}{2\pi} \left[\frac{2z^3}{\sqrt{z^2 - 1}} - 2z^2 \right]. \quad (\text{C.61})$$

Similarly,

$$\Pi_R^{yy} = -\frac{T \log 2}{2\pi} \left[\frac{z}{\sqrt{z^2 - 1}} - \frac{z}{\sqrt{z^2 - 1}} (2z^2 - 1) + 2z^2 \right] \quad (\text{C.62})$$

or

$$\Pi_R^{yy} = -\frac{T \log 2}{2\pi} \left[\frac{z(2 - 2z^2)}{\sqrt{z^2 - 1}} + 2z^2 \right]. \quad (\text{C.63})$$

C.2 FERMION DAMPING RATE

We can verify the expression for the fermion damping directly from the optical theorem using the finite temperature cutting rules²¹⁶. This relates the decay rate to the imaginary part of the tree level cross-section which in turn is related to the imaginary part of the self-energy. We can define $\mathcal{M}_L$ and $\mathcal{M}_T$ as the longitudinal and transverse matrix elements for the tree level scattering processes of an electron emitting a soft photon

$$\mathcal{M}_{L(T)} = \frac{1}{\sqrt{N_F}} \bar{u}(p) (-i\gamma_\mu) u(k) \varepsilon_{L(T)}^\mu(q^0, \mathbf{k} - \mathbf{p}) \quad (\text{C.64})$$

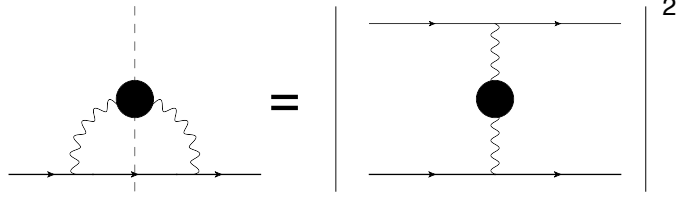


Figure C.1: The decay rate is related to the imaginary part of the fermion self energy via the optical theorem.

where $\bar{u}(p)$ and $u(k)$ are massless on shell spinors. We can write $\Gamma_{\mathbf{k}}$ in terms of these matrix elements as

$$\begin{aligned}
 \Gamma_{\mathbf{k}} &\equiv \frac{1}{2E_{\mathbf{k}}} \int_{\mathbf{p}} \int \frac{dp^0}{2\pi} \frac{dq^0}{2\pi} \sum_{\lambda=L,T} \mathcal{A}_{\lambda}(q^0, \mathbf{k}-\mathbf{p}) 2\pi \delta((p^0)^2 - E_{\mathbf{p}}^2) |\mathcal{M}_{\lambda}|^2 (1 - n_F(p^0) + n_B(q^0)) \\
 &\quad 2\pi \delta(k^0 - p^0 - q^0) \\
 &= \int_{\mathbf{p}} \int \frac{dq^0}{2\pi} \sum_{s=\pm} \sum_{\lambda=L,T} \frac{s}{8E_{\mathbf{k}}E_{\mathbf{p}}} \mathcal{A}_{\lambda}(q^0, \mathbf{k}-\mathbf{p}) |\mathcal{M}_{\lambda}|^2 (1 - n_F(sE_{\mathbf{p}}) + n_B(q^0)) 2\pi \delta(k^0 - sE_{\mathbf{p}} - q^0) \\
 &= \sum_{\lambda=L,T} \sum_{s=\pm} \int_{\mathbf{p}} \frac{s}{8E_{\mathbf{k}}E_{\mathbf{p}}} \mathcal{A}_{\lambda}(k-p_s) |\mathcal{M}_{\lambda}|^2 (1 - n_F(sE_{\mathbf{p}}) + n_B(k^0 - sE_{\mathbf{p}})) \quad (\text{C.65})
 \end{aligned}$$

where $k = (k^0, \mathbf{k})$ and $p_s = (sE_{\mathbf{p}}, \mathbf{p})$. Plugging $\mathcal{M}_{L(T)}$ into C.65 and using the relation $u(k)\bar{u}(k) = \not{k}$ we get

$$\begin{aligned}
 \Gamma_{\mathbf{k}} &= \sum_{\lambda=L,T} \sum_{s=\pm} \int_{\mathbf{p}} \frac{\text{Tr} \left[\bar{u}(k) \gamma_{\mu} \not{p}_s \gamma_{\nu} u(k) \varepsilon_{\lambda}^{\mu}(\mathbf{k}-\mathbf{p}) \varepsilon_{\lambda}^{\nu}(\mathbf{k}-\mathbf{p}) |\mathcal{M}_{\lambda}|^2 \right]}{8E_{\mathbf{k}}E_{\mathbf{p}}} \mathcal{A}_{\lambda}(k-p_s) \\
 &\quad (1 - n_F(sE_{\mathbf{p}}) + n_B(k^0 - sE_{\mathbf{p}})) \\
 &= -\frac{1}{2E_{\mathbf{k}}} \text{Tr} [\bar{u}(k) \text{Im} \Sigma_R(\not{k}) u(k)] \Big|_{k^0=E_{\mathbf{k}}} \\
 &= -\frac{1}{2E_{\mathbf{k}}} \text{Im} \text{Tr} [\not{k} \Sigma_R(\not{k})] \Big|_{k^0=E_{\mathbf{k}}} \quad (\text{C.66})
 \end{aligned}$$

we find the relation for the self energy and the decay rate given in Eqn. 3.39. We can now write the decay rate $\Gamma_{\mathbf{k}}$ in Coulomb gauge as

$$\begin{aligned} \Gamma_{\mathbf{k}} = & \frac{1}{N_f} \sum_{s=\pm} \int_{\mathbf{p}} \frac{s}{4E_{\mathbf{k}}E_{\mathbf{p}}} \left(\mathcal{A}_L(k-p_s)(2(k \cdot u)(p_s \cdot u) - k \cdot p_s u \cdot u) \right. \\ & + \mathcal{A}_T(k-p_s)(2(k \cdot \varepsilon_T(\mathbf{k}-\mathbf{p}))(p_s \cdot \varepsilon_T(\mathbf{k}-\mathbf{p})) - k \cdot p_s(\varepsilon_T(\mathbf{k}-\mathbf{p}) \cdot \varepsilon_T(\mathbf{k}-\mathbf{p}))) \\ & \left. (1 - n_F(sE_{\mathbf{p}}) + n_B(k^0 - sE_{\mathbf{p}})) \right) \Big|_{k^0=E_{\mathbf{k}}} \end{aligned} \quad (\text{C.67})$$

Defining the Lorentz vectors $k = (k^0, |\mathbf{k}|, 0)$, $p_s = (sE_{\mathbf{p}}, |\mathbf{p}| \cos \theta, |\mathbf{p}| \sin \theta)$, and the transverse polarization vector $\varepsilon_T^\mu(\mathbf{k}-\mathbf{p}) = \frac{1}{|\mathbf{k}-\mathbf{p}|} (0, |\mathbf{p}| \sin \theta, |\mathbf{k}| - \mathbf{p} \cos \theta)$, we get the final expression for $\Gamma_{\mathbf{k}}$ in Coulomb gauge

$$\begin{aligned} \Gamma_{\mathbf{k}} = & \frac{1}{N_f} \int_{\mathbf{p}} \sum_{s=\pm} \frac{s}{4E_{\mathbf{k}}E_{\mathbf{p}}} \left[\left(\mathcal{A}_L(k-\tilde{p}_s) (sE_{\mathbf{k}}E_{\mathbf{p}} + |\mathbf{k}||\mathbf{p}| \cos \theta) \right. \right. \\ & + \left. \left. \mathcal{A}_T(k-\tilde{p}_s) \left(sE_{\mathbf{k}}E_{\mathbf{p}} - |\mathbf{k}||\mathbf{p}| \cos \theta + \frac{2|\mathbf{k}|^2|\mathbf{p}|^2(1 - \cos^2 \theta)}{|\mathbf{k}|^2 + |\mathbf{p}|^2 - 2|\mathbf{k}||\mathbf{p}| \cos \theta} \right) \right) (1 - n_f(sE_{\mathbf{p}}) + n_b(E_{\mathbf{k}} - sE_{\mathbf{p}})) \right] \end{aligned}$$

C.3 FULL EXPRESSIONS FOR THE SCALAR KERNEL

Using the definition in Eq. 3.75, we can write an expression for the scalar kernel in the Bethe-Salpeter equation in terms of the rung matrix elements.

$$\begin{aligned} K_{ss'}(\mathbf{k}, \mathbf{k}') = & e^{2i(\theta_{\mathbf{k}'} - \theta_{\mathbf{k}})} \mathcal{R}_{11} - e^{i(\theta_{\mathbf{k}'} - \theta_{\mathbf{k}})} s_1 s_2 (\mathcal{R}_{22} + \mathcal{R}_{23} + \mathcal{R}_{32} + \mathcal{R}_{33}) + \mathcal{R}_{44} \\ & - e^{-2i\theta_{\mathbf{k}}} \mathcal{R}_{14} - e^{2i\theta_{\mathbf{k}'}} \mathcal{R}_{41} + i e^{i(2\theta_{\mathbf{k}'} - \theta_{\mathbf{k}})} s_1 (\mathcal{R}_{21} + \mathcal{R}_{31}) + i e^{i(\theta_{\mathbf{k}'} - 2\theta_{\mathbf{k}})} s_2 (\mathcal{R}_{12} + \mathcal{R}_{13}) \\ & - i e^{i\theta_{\mathbf{k}'}} s_2 (\mathcal{R}_{42} + \mathcal{R}_{43}) - i e^{-i\theta_{\mathbf{k}}} s_1 (\mathcal{R}_{24} + \mathcal{R}_{34}) \end{aligned} \quad (\text{C.69})$$

where $\theta_{\mathbf{k}}$ is the angle between $\mathbf{k}$ and the x axis and $\theta_{\mathbf{k}'}$ is the angle between $\mathbf{k}'$ and the x axis.

If we make a rotationally symmetric ansatz for the wavefunction, only terms depending on $\theta_{\mathbf{k}'} - \theta_{\mathbf{k}}$ remain, which leaves us

$$K_{ss'}(\mathbf{k}, \mathbf{k}') = e^{2i(\theta_{\mathbf{k}'} - \theta_{\mathbf{k}})} \mathcal{R}_{11} - e^{i(\theta_{\mathbf{k}'} - \theta_{\mathbf{k}})} s_1 s_2 (\mathcal{R}_{22} + \mathcal{R}_{23} + \mathcal{R}_{32} + \mathcal{R}_{33}) + \mathcal{R}_{44} \quad (\text{C.70})$$

Both rungs can be written in the form

$$\mathcal{R} = \mathcal{R}_{0,0} \gamma_0 \otimes \gamma_0 + \mathcal{R}_{i,j} \gamma_i \otimes \gamma_j + \mathcal{R}_{i,0} \gamma_i \otimes \gamma_0 + \mathcal{R}_{0,j} \gamma_0 \otimes \gamma_j \quad (\text{C.71})$$

For the first rung, we can write these coefficients as

$$\begin{aligned} \mathcal{R}_1(\bar{k})_{(0,0)} &= -\frac{\mathcal{Q}_b(\bar{k}^0)}{N_f} \mathcal{A}_L(\bar{k}) \\ \mathcal{R}_1(\bar{k})_{(x,x)} &= -\frac{\mathcal{Q}_b(\bar{k}^0)}{N_f} \mathcal{A}_T(\bar{k}) \left(1 - \frac{(|\mathbf{k}| - |\mathbf{k}'| \cos \theta_{\mathbf{k},\mathbf{k}'})^2}{|\bar{\mathbf{k}}|^2} \right) \\ \mathcal{R}_1(\bar{k})_{(y,y)} &= -\frac{\mathcal{Q}_b(\bar{k}^0)}{N_f} \mathcal{A}_T(\bar{k}) \left(1 - \frac{(|\mathbf{k}'| \sin \theta_{\mathbf{k},\mathbf{k}'})^2}{|\bar{\mathbf{k}}|^2} \right) \\ \mathcal{R}_1(\bar{k})_{(x,y)} &= -\frac{\mathcal{Q}_b(\bar{k}^0)}{N_f} \mathcal{A}_T(\bar{k}) \left(\frac{(|\mathbf{k}| - |\mathbf{k}'| \cos \theta_{\mathbf{k},\mathbf{k}'}) (|\mathbf{k}'| \sin \theta_{\mathbf{k},\mathbf{k}'})}{|\bar{\mathbf{k}}|^2} \right) \\ \mathcal{R}_1(\bar{k})_{(y,x)} &= -\frac{\mathcal{Q}_b(\bar{k}^0)}{N_f} (\mathcal{A}_T(\bar{k})) \left(\frac{(|\mathbf{k}| - |\mathbf{k}'| \cos \theta_{\mathbf{k},\mathbf{k}'}) (|\mathbf{k}'| \sin \theta_{\mathbf{k},\mathbf{k}'})}{|\bar{\mathbf{k}}|^2} \right) \end{aligned} \quad (\text{C.72})$$

where $\theta_{\mathbf{k},\mathbf{k}'}$ is the angle between $\mathbf{k}$ and $\mathbf{k}'$. For the second rung, we use the general expression given in 3.55. Defining $\theta_{\mathbf{p}}$ as the angle between $\mathbf{p}$ and $\bar{\mathbf{k}}$ and φ as the angle between $\bar{\mathbf{k}}$ and $\mathbf{K}$, the transverse polarization vector with respect to the $\bar{\mathbf{k}}$ axis can be written as

$$\varepsilon_T^\mu(\mathbf{p} + \mathbf{K}) = \frac{1}{|\mathbf{p} + \mathbf{K}|} (0, -(|\mathbf{p}| \sin \theta_{\mathbf{p}} + |\mathbf{K}| \sin \varphi), |\mathbf{p}| \cos \theta_{\mathbf{p}} + |\mathbf{K}| \cos \varphi) \quad (\text{C.73})$$

We use these to define the following Lorentz vectors which are contained in the numerators of the

longitudinal and transverse components of the second rung in

$$\mathcal{R}_{2,L}^{(1)}(\bar{k}, K, \mathbf{p})^\mu = (\bar{k}^0 - s'E_-, -\bar{\mathbf{k}}/2 - \mathbf{p}) \quad (\text{C.74a})$$

$$\mathcal{R}_{2,L}^{(2)}(\bar{k}, K, \mathbf{p})^\mu = (s'E_-, -\bar{\mathbf{k}}/2 + \mathbf{p}) \quad (\text{C.74b})$$

$$\mathcal{R}_{2,T}^{(1)}(\bar{k}, K, \mathbf{p})^\mu = (\bar{k}^0 - s'E_-, (\bar{\mathbf{k}} + 2\mathbf{p}) \cdot \varepsilon_T(\mathbf{p} + \mathbf{K}) \varepsilon_T(-\mathbf{p} - \mathbf{K}) + \bar{\mathbf{k}}/2 + \mathbf{p}) \quad (\text{C.74c})$$

$$\mathcal{R}_{2,T}^{(2)}(\bar{k}, K, \mathbf{p})^\mu = (s'E_-, (\bar{\mathbf{k}} - 2\mathbf{p}) \cdot \varepsilon_T(\mathbf{p} + \mathbf{K}) \varepsilon_T(-\mathbf{p} - \mathbf{K}) + \bar{\mathbf{k}}/2 - \mathbf{p}) \quad (\text{C.74d})$$

Plugging C.73 into C.74, we can write the spatial components of the Lorentz vectors for the longitudinal polarizations as

$$\mathcal{R}_{2,L}^{(1)}(\bar{k}, K, \mathbf{p})^x = -|\bar{\mathbf{k}}|/2 - |\mathbf{p}| \cos \theta_{\mathbf{p}} \quad (\text{C.75a})$$

$$\mathcal{R}_{2,L}^{(1)}(\bar{k}, K, \mathbf{p})^y = -|\mathbf{p}| \sin \theta_{\mathbf{p}} \quad (\text{C.75b})$$

$$\mathcal{R}_{2,L}^{(2)}(\bar{k}, K, \mathbf{p})^x = -|\bar{\mathbf{k}}|/2 + |\mathbf{p}| \cos \theta_{\mathbf{p}} \quad (\text{C.75c})$$

$$\mathcal{R}_{2,L}^{(2)}(\bar{k}, K, \mathbf{p})^y = |\mathbf{p}| \sin \theta_{\mathbf{p}} \quad (\text{C.75d})$$

and the transverse polarizations as

$$\mathcal{R}_{2,T}^{(1)}(\bar{k}, K, \mathbf{p})^x = \frac{(2|\mathbf{K}||\mathbf{p}|\sin(\theta_{\mathbf{p}} - \varphi) - |\mathbf{k}|(|\mathbf{p}|\sin\theta_{\mathbf{p}} + |\mathbf{K}|\sin\varphi))(|\mathbf{p}|\sin\theta_{\mathbf{p}} + |\mathbf{K}|\sin\varphi)}{|\mathbf{p} + \mathbf{K}|^2} + \frac{|\bar{k}| + 2|\mathbf{p}|\cos\theta_{\mathbf{p}}}{2} \quad (\text{C.76a})$$

$$\mathcal{R}_{2,T}^{(1)}(\bar{k}, K, \mathbf{p})^y = -\frac{(2|\mathbf{K}||\mathbf{p}|\sin(\theta_{\mathbf{p}} - \varphi) - |\mathbf{k}|(|\mathbf{p}|\sin\theta_{\mathbf{p}} + |\mathbf{K}|\sin\varphi))(|\mathbf{p}|\cos\theta_{\mathbf{p}} + |\mathbf{K}|\cos\varphi)}{|\mathbf{p} + \mathbf{K}|^2} + \mathbf{p}\sin\theta_{\mathbf{p}} \quad (\text{C.76b})$$

$$\mathcal{R}_{2,T}^{(2)}(\bar{k}, K, \mathbf{p})^x = \frac{(2|\mathbf{K}||\mathbf{p}|\sin(\theta_{\mathbf{p}} - \varphi) + |\mathbf{k}|(|\mathbf{p}|\sin\theta_{\mathbf{p}}|\mathbf{K}|\sin\varphi))(|\mathbf{p}|\sin\theta_{\mathbf{p}} + |\mathbf{K}|\sin\varphi)}{|\mathbf{p} + \mathbf{K}|^2} + \frac{|\bar{k}| - 2|\mathbf{p}|\cos\theta_{\mathbf{p}}}{2} \quad (\text{C.76c})$$

$$\mathcal{R}_{2,T}^{(2)}(\bar{k}, K, \mathbf{p})^y = -\frac{(2|\mathbf{K}||\mathbf{p}|\sin(\theta_{\mathbf{p}} - \varphi) + |\mathbf{k}|(|\mathbf{p}|\sin\theta_{\mathbf{p}}|\mathbf{K}|\sin\varphi))(|\mathbf{p}|\cos\theta_{\mathbf{p}} + |\mathbf{K}|\cos\varphi)}{|\mathbf{p} + \mathbf{K}|^2} - |\mathbf{p}|\sin\theta_{\mathbf{p}} \quad (\text{C.76d})$$

We can now now plug the expressions in C.76 into 3.55 and evaluate the momentum integrals to get complete expressions for the matrix coefficients of rung II.

C.4 FEYNMAN RULES FOR LADDER DIAGRAMS

We derive the Feynman rules for the ladder diagrams in the $\frac{1}{N_f}$ expansion. Ignoring thermal corrections, we can start by going into the interaction picture with respect to the the free Hamiltonian.

The time evolution operator is given by

$$U_I = \mathcal{T} \exp \left(-\frac{1}{\sqrt{N_f}} \sum_{a=1}^{N_f} \int_0^t ds \int_{\mathbf{x}} A_{\mu}(s, \mathbf{x}) f_a^{\mu}(s, \mathbf{x}) \right) \quad (\text{C.77})$$

here $\mathcal{A}$ is the free photon field and j_μ is the free fermion current $j^\mu(t) = \bar{\psi}(t)\gamma^\mu\psi(t)$. We can expand this (dropping spatial arguments) as

$$U_I = 1 + \frac{i}{2} \int_0^t ds_1 \mathcal{A}^\mu(s_1) j_\mu(s) + \left(\frac{i}{2}\right)^2 \int_0^t ds_1 \int_0^{s_1} ds_2 \mathcal{A}^\mu(s_1) j_\mu(s_1) \mathcal{A}^\mu(s_2) j_\mu(s_2) + \dots \quad (\text{C.78})$$

for $U_I^\dagger$ we have

$$U_I^\dagger = 1 - \frac{i}{2} \int_0^t ds_1 \mathcal{A}^\mu(s_1) j_\mu(s_1) + \left(\frac{-i}{2}\right)^2 \int_0^t ds_1 \int_0^{s_1} ds_2 \mathcal{A}^\mu(s_2) j_\mu(s_2) \mathcal{A}^\mu(s_1) j_\mu(s_1) + \dots \quad (\text{C.79})$$

We start by stating the usual anti-commutation relations for free fermions

$$\{\psi_a(t, \mathbf{x}), \psi_b(0)\} = 0 \quad (\text{C.80a})$$

$$\{\bar{\psi}_a(t, \mathbf{x}), \bar{\psi}_b(0)\} = 0 \quad (\text{C.80b})$$

$$\{\psi_a(t, \mathbf{x}), \bar{\psi}_b(0)\} = \gamma^0 \delta_{ab} \delta(t) \delta(\mathbf{x}). \quad (\text{C.80c})$$

For currents we have

$$[\psi_a(t, \mathbf{x}), j_b^\mu(0)] = \{\psi_a(t, \mathbf{x}), \bar{\psi}_b(0)\} \gamma^\mu \psi_b(0) \quad (\text{C.81a})$$

$$[\bar{\psi}_a(t, \mathbf{x}), j_b^\mu(0)] = -\bar{\psi}_b(0) \gamma^\mu \{\bar{\psi}_a(t, \mathbf{x}), \psi_b(0)\} \quad (\text{C.81b})$$

$$\begin{aligned} [j_a^\mu(t, \mathbf{x}), j_b^\nu(0)] &= \bar{\psi}_a(t, \mathbf{x}) \gamma^\mu \{\psi(t, \mathbf{x}), \bar{\psi}_b(0)\} \gamma^\nu \psi_b(0) \\ &\quad - \bar{\psi}_b(0) \gamma^\nu \{\bar{\psi}_a(t, \mathbf{x}), \psi_b(0)\} \gamma^\mu \psi_a(t, \mathbf{x}). \end{aligned} \quad (\text{C.81c})$$

The current-fermion and current-current commutators can be written in terms of the retarded propagator terms of the retarded propagator is now

$$[\psi_a(t, \mathbf{x}), j_b^\mu(0)] = G_{R,ab}(t, \mathbf{x}) \gamma^\mu \psi_b(0) \quad (\text{C.82a})$$

$$[\bar{\psi}_a(t, \mathbf{x}), j_b^\mu(0)] = -\bar{\psi}_b(0) \gamma^\mu G_{R,ab}^*(t, \mathbf{x}) \quad (\text{C.82b})$$

$$[j_a^\mu(t, \mathbf{x}), j_b^\nu(0)] = \bar{\psi}_a(t, \mathbf{x}) \gamma^\mu G_{R,ab}(t, \mathbf{x}) \gamma^\nu \psi_b(0) - \bar{\psi}_b(0) \gamma^\nu G_{R,ab}^*(t, \mathbf{x}) \gamma^\mu \psi_a(t, \mathbf{x}). \quad (\text{C.82c})$$

Lastly for free fields, both $j_a^\mu(t, \mathbf{x})$ and $\tilde{\psi}(t, \mathbf{x})$ commute with the photon operator so we have

$$[j_a^\mu(t, \mathbf{x}), A^\nu(0)] = [A_i^\mu(t, \mathbf{x}), \psi(0)] = [A_i^\mu(t, \mathbf{x}), \bar{\psi}(0)] = 0. \quad (\text{C.83})$$

C.4.1 FERMION LADDER

In the interaction picture, $\mathcal{F}(t, \mathbf{x})$ is given by

$$\mathcal{F}(t, \mathbf{x}) = \frac{1}{N_f^2} \sum_{i,j=1}^{N_f} \text{Tr} \left[\sqrt{\hat{\rho}} \{ U_I^\dagger \psi_i(t, \mathbf{x}) U_I, \bar{\psi}_j(0) \} \sqrt{\hat{\rho}} \{ U_I^\dagger \psi_i(t, \mathbf{x}) U_I, \bar{\psi}_j(0) \}^\dagger \right] \quad (\text{C.84})$$

where $U_I^\dagger \tilde{\psi}(t) U_I$ is given by the expansion

$$\begin{aligned} U_I^\dagger \psi(t) U_I &= \psi(t) + \frac{i}{2} \int_0^t ds_1 [\psi(t), iA^\rho(s_1) j_\rho(s_1)] \\ &+ \left(\frac{-i}{2} \right)^2 \int_0^t ds_1 \int_0^{s_1} ds_2 [[\psi(t), iA^\rho(s_1) j_\rho(s_1)], iA^\sigma(s_2) j_\sigma(s_2)] \\ &+ \dots \end{aligned} \quad (\text{C.85})$$

From Eqn. C.85 and using the definition of the retarded propagator, we can immediately see that the zeroth order term in $\mathcal{F}(t, \mathbf{x})$ is given by

$$\mathcal{F}_0(t, \mathbf{x}) = [iG_R(t, \mathbf{x})][iG_R^*(t, \mathbf{x})] \quad (\text{C.86})$$

The first nontrivial term is given by

$$\begin{aligned} \mathcal{F}_1(t, \mathbf{x}) \sim & \text{Tr} \left[\sqrt{\hat{\rho}} \left\{ \frac{i}{2} \int_0^t ds \int_{\mathbf{y}} [\psi(t, \mathbf{x}), iA^\mu(s, \mathbf{y}) j_\mu(s, \mathbf{y})], \bar{\psi}(0) \right\} \right. \\ & \left. \sqrt{\hat{\rho}} \left\{ \frac{i}{2} \int_0^t ds' \int_{\mathbf{y}'} [\psi(t, \mathbf{x}), iA^\nu(s', \mathbf{y}') j_\nu(s', \mathbf{y}')] \bar{\psi}(0) \right\}^\dagger \right] \quad (\text{C.87}) \end{aligned}$$

Expanding the anti-commutator for each time fold, we get

$$\begin{aligned} \{[\psi(t, \mathbf{x}), iA^\mu(s, \mathbf{y}) j_\mu(s, \mathbf{y})], \bar{\psi}(0)\} &= i\{A^\mu(s, \mathbf{y})[\psi(t, \mathbf{x}), j_\mu(s, \mathbf{y})], \bar{\psi}(0)\} \\ &= iA^\mu(s, \mathbf{y})\{\{\psi(t, \mathbf{x}), \bar{\psi}(s, \mathbf{y})\}\gamma_\mu \psi(s, \mathbf{y}), \bar{\psi}(0)\} \\ &= iA^\mu(s, \mathbf{y})(iG_R(t-s, \mathbf{x}-\mathbf{y}))\gamma_\mu(iG_R(s, \mathbf{y})) \quad (\text{C.88}) \end{aligned}$$

Putting everything together gives us

$$\begin{aligned} \mathcal{F}_1(t, \mathbf{x}) &= -\frac{1}{4N_f} \int_{ss'} \mathcal{D}_{W}^{\mu\nu}(s-s', \mathbf{y}-\mathbf{y}') G_R(s, \mathbf{y}) \gamma_\mu G_R(t-s, \mathbf{x}-\mathbf{y}) \\ &\times [G_R^*(s', \mathbf{y}')] \gamma_\nu [G_R^*(t-s', \mathbf{x}-\mathbf{y}')] \quad (\text{C.89}) \end{aligned}$$

which we identify as a ladder with one type one rung.

Now we can expand to second order on both time folds. We consider the expansion of the com-

mutator contained inside the anti-commutator on one time fold

$$\begin{aligned}
[[,],] &= [[\psi(t), iA^\mu(s_1)j_\mu(s_1)], iA^\nu(s_2)j_\nu(s_2)] \\
&= i^2\{\psi(t), \bar{\psi}(s_1)\}\gamma_\mu\psi(s_1)j_\nu(s_2)[A^\mu(s_1), A^\nu(s_2)] \\
&+ i^2\{\psi(t), \bar{\psi}(s_1)\}\gamma_\mu[\psi(s_1), j_\nu(s_2)]A^\mu(s_1)A^\nu(s_2)
\end{aligned} \tag{C.90}$$

From the first term we get

$$\begin{aligned}
\{[[,],], \}_1 &= i^2\{\{\psi(t), \bar{\psi}(s_1)\}\gamma_\mu\psi(s_1)j_\nu(s_2)[A^\mu(s_1), A^\nu(s_2)], \bar{\psi}(0)\} \\
&= i^2\{\psi(t), \bar{\psi}(s_1)\}\gamma_\mu\psi(s_1)\bar{\psi}(s_2)\gamma_\nu\{\bar{\psi}(0), \psi(s_2)\}[A^\mu(s_1), A^\nu(s_2)] \\
&= i^2(iG_R(t-s_1))\gamma_\mu\psi(s_1)\bar{\psi}(s_2)\gamma_\nu(iG_R^*(s_2)[A^\mu(s_1), A^\nu(s_2)])
\end{aligned} \tag{C.91}$$

which will give us the type II rung. This can be written as

$$\begin{aligned}
\mathcal{F}_2(t, \mathbf{x}) &= -\frac{1}{4N_f} \int_{s_1 s'_1} \int_{s_2 s'_2} \mathcal{D}^{\mu\nu}(s_1-s_2) \mathcal{D}^{\mu'\nu'}(s'_1-s'_2) \\
&G_R(t-s_1))\gamma_\mu G_W(s_1-s'_1)\gamma_{\mu'} G_W^*(s_2-s'_2)\gamma_\nu G_R^*(s_2)\gamma_{\nu'}
\end{aligned} \tag{C.92}$$

The second term gives

$$\begin{aligned}
\{[[,],], \}_2 &= i^2\{[[\psi(t), j_\mu(s_1)], j_\nu(s_2)]A^\mu(s_1)A^\nu(s_2), \bar{\psi}(0)\} \\
&= i^2\{\psi(t), \bar{\psi}(s_1)\}\gamma_\mu\{\psi(s_1), \bar{\psi}(s_2)\}\gamma_\nu\{\psi(s_2), \bar{\psi}(0)\}A^\mu(s_1)A^\nu(s_2) \\
&= i^2 G_R(t-s_1)\gamma_\mu G_R(s_1-s_2)\gamma_\nu G_R(s_2)A^\mu(s_1)A^\nu(s_2)
\end{aligned} \tag{C.93}$$

This will give two copies of the type I rung or a self energy term for ψ .

C.4.2 SQUARED CURRENT-CURRENT COMMUTATOR

In the interaction $\mathcal{C}(t, \mathbf{x})$ is given by

$$\mathcal{C}(t, \mathbf{x}) = \frac{1}{N_f^2} \sum_{i,j=1}^{N_f} \delta_{\mu\rho} \delta_{\nu\sigma} \text{Tr} \left[\sqrt{\hat{\rho}} \{ U_{I,i}^\dagger{}^\mu(t, \mathbf{x}) U_{I,j}^\nu(0) \} \sqrt{\hat{\rho}} \{ U_{I,i}^\rho(t, \mathbf{x}) U_{I,j}^\sigma(0) \}^\dagger \right] \quad (\text{C.94})$$

The zeroth order term can be deduced from the expression in C.82c

$$\begin{aligned} \mathcal{C}_0(t, \mathbf{x}) &= G_W^*(t, \mathbf{x}) \gamma^\mu i G_R(t, \mathbf{x}) \gamma^\nu G_W(0) \gamma^\nu G_R^*(t, \mathbf{x}) \gamma^\mu \\ &+ G_W^*(0) \gamma^\nu i G_R^*(t, \mathbf{x}) \gamma^\mu G_W(t, \mathbf{x}) \gamma^\mu i G_R(t, \mathbf{x}) \gamma^\nu \\ &- G_W^*(t, \mathbf{x}) \gamma^\mu i G_R(t, \mathbf{x}) \gamma^\nu G_W^*(t, \mathbf{x}) \gamma^\mu i G_R(t, \mathbf{x}) \gamma^\nu \\ &- G_W^*(t, \mathbf{x}) \gamma^\nu i G_R^*(t, \mathbf{x}) \gamma^\mu G_W(t, \mathbf{x}) \gamma^\nu i G_R^*(t, \mathbf{x}) \gamma^\mu \end{aligned} \quad (\text{C.95})$$

Diagrammatically the first terms each look like a box but have momentum flowing in the opposite directions. The second two terms also have opposite momentum flow but have Wightman functions which are crossed. However, ladders with crossed legs should not contribute to chaos given the pole structure⁴⁴.

The first nontrivial in the expansion for the current-current commutator is

$$\begin{aligned} \mathcal{C}_1(t, \mathbf{x}) &= \text{Tr} \left[\sqrt{\hat{\rho}} \left\{ \frac{i}{2} \int_0^t ds \int_{\mathbf{y}} [j^\mu(t, \mathbf{x}), iA^\nu(s, \mathbf{y}) j_\nu(s, \mathbf{y})], j^\rho(0) \right\} \right. \\ &\quad \left. \sqrt{\hat{\rho}} \left\{ \frac{i}{2} \int_0^t ds' \int_{\mathbf{y}'} [j^\mu(t, \mathbf{x}), iA^\nu(s', \mathbf{y}') j_\nu(s', \mathbf{y}')] \right\}^\dagger \right] \end{aligned} \quad (\text{C.96})$$

Expanding the commutator on one of the time folds (again supressing spatial arguments) gives us

$$\begin{aligned}
[[j_\mu(t), A^\nu(s_1)j_\nu(s_1)], j_\rho(0)] &= A^\nu(s_1)[[j_\mu(t), j_\nu(s_1)], j_\rho(0)] \\
&= A^\nu(s_1)\bar{\psi}(t)\gamma^\mu G_R(t-s_1)\gamma^\nu G_R(s_1)\gamma^\rho\psi(0) \\
&- A^\nu(s_1)\bar{\psi}(0)\gamma^\rho G_R^*(s_1)\gamma^\nu G_R^*(t-s_1)\gamma^\mu\psi(t) \\
&- A^\nu(s_1)\bar{\psi}(0)\gamma^\rho G_R^*(t)\gamma^\mu G_R(t-s_1)\gamma^\nu\psi(s_1) \\
&+ A^\nu(s_1)\bar{\psi}(s_1)\gamma^\nu G_R^*(t-s_1)\gamma^\mu G_R(t)\gamma^\rho\psi(0) \quad (C.97)
\end{aligned}$$

The whole first order term will have 16 terms. However, most contain a photon three point function and will vanish due to Furry's theorem. The nonvanishing terms will gives us

$$\begin{aligned}
C_1(t, \mathbf{x}) &= -\frac{1}{4N_f} \int_{s_i s'_i} \left[\mathcal{D}_W^{\nu\nu'}(s_1)(G_W^*(t)\gamma^\mu G_R(t-s_1)\gamma^\nu G_R(s_1)\gamma^\rho G_W(0) \right. \\
&\quad \gamma^{\mu'} G_R^*(t-s'_1)\gamma^\nu G_R^*(s'_1)\gamma^\rho \\
&+ G_W^*(0)\gamma^\rho G_R^*(s_1)\gamma^\nu G_R^*(t-s_1)\gamma^\mu G_W(t)\gamma^{\rho'} G_R^*(s'_1)\gamma^{\nu'} G_R^*(t'-s'_1)\gamma^{\mu'} \\
&- G_W^*(t)\gamma^\mu G_R(t-s_1)\gamma^\nu G_R(s_1)\gamma^\rho G_W^*(t)\gamma^{\rho'} G_R^*(s'_1)\gamma^{\nu'} G_R^*(t'-s'_1)\gamma^{\mu'} \\
&- \left. G_W(t)\gamma^\rho G_R^*(s_1)\gamma^\nu G_R^*(t-s_1)\gamma^\mu G_W(t)\gamma^{\mu'} G_R^*(t-s'_1)\gamma^\nu G_R^*(s'_1)\gamma^\rho) + \dots \right] \quad (C.98)
\end{aligned}$$

where the primes will technically be contracted with the unprimed terms. These correspond to boxes with one rung with momenta flowing in opposite directions. The parts in between the Wightman functions are the same as the first order correction to the fermion ladder. These are pictured in

Fig. 3.7. To second order

$$\begin{aligned}
[[[j_\mu(t), A^\nu j_\nu(s_1)], A^\rho j_\rho(s_2)], j_\sigma(0)] &= A^\nu(s_1) A^\rho(s_2) [[j_\mu(t), j_\nu(s_1)], j_\rho(s_2)], j_\sigma(0)] \\
&+ [A^\nu(s_1), A^\rho(s_2)] [j_\mu(t), j_\nu(s_1)] [j_\rho(s_2), j_\sigma(0)] \\
&+ [A^\nu(s_1), A^\rho(s_2)] [[j_\mu(t), \tilde{j}_\nu(s_1)], j_\sigma(0)] j_\rho(s_2) \quad (\text{C.99})
\end{aligned}$$

By inspection, the first term will give us terms with two photon rails or a self energy correction.

The second term will give the second type of rung appearing in the fermion diagram. This give four terms where the momentum flow in the boxes at either end can go in the opposite direction. The third term can give us a ladder with self energy corrections on the rails.

C.5 DETAILS OF NUMERICAL COMPUTATIONS

Here we discuss the details of our numerical computation of Γ_k and λ_L . Our methods are similar to those used in ⁴⁴, with some adjustments made to address various issues. The most challenging aspect of the numerics is managing the light-cone singularities that appear throughout the computation, which require additional care. To compute the components of the polarization bubble, we dealt with this issue by a slight off-shift in the grid. From our analysis in App. C.1, we know that all singularities that appear in the decay rate and the kernel should be integrable, so we can manually remove them during each integration.

The first step in our computation is to compute the analytic continuation of the xx and yy components of the polarization bubble. The imaginary parts are computed by numerical integration of Eqn. C.28 using a standard MATLAB routine. The integral is computed separately for $k^0 < |\mathbf{k}|$ and $k^0 > |\mathbf{k}|$ because of the non-analyticity at $k^0 = |\mathbf{k}|$. The real parts are then obtained by a stable version of Kramers-Kronig given in the Eqn. C.41. These are then used to precompute the values

of the spectral function, retarded propagator, and Wightman propagator on 200×200 grids for $k^0/T \in [0, 12.1]$, $|\mathbf{k}|/T \in [0, 12.21]$. These grids are then used to create two dimensional cubic interpolations for all propagators. $\Gamma_{\mathbf{k}}$ is calculated from Eqn. C.68 on a 60 point grid for $|\mathbf{k}|/T \in [0, 12]$ where 60 grid points are used for the angular integral. To calculate the exponent on varying grid sizes, we use a one dimensional spline interpolation for the decay rate.

The kernel is computed using Eqn. C.70 on a 40×40 grid for $|\mathbf{k}|/T, |\mathbf{k}'|/T \in [0, 12]$ where 40 grid points are used for the angular integral. We find that most eigenvalues are complex with negative real parts, but the eigenvalue with the largest real part is completely real. This eigenvalue is identified as the Lyapunov exponent. There are also two eigenvalues which are exactly zero. We also compute the kernel using an interpolation on a grid that is slightly offset and find $\lambda_L \approx 4.06$.

D

Appendix for Chapter 4

D.1 DIFFUSION IN AN INHOMOGENEOUS FLUID

There is a rather controversial history of Einstein relations in inhomogeneous media (in many cases this involves situations with inhomogeneous temperature²⁰⁸). In this appendix, we directly confirm that the Einstein relation employed in the main text is correct in the hydrodynamic limit, suitable for our holographic computation.

As in the main text, we consider a background charge neutral fluid, and study the linearized propagation of charge fluctuations around this background. The equations of hydrodynamics simplify to “resistor network” equations (in the continuum)¹³⁷:

$$\partial_t n = \partial_x \left(\tilde{\sigma}(x) \partial_x \left(\frac{n}{\tilde{\chi}(x)} \right) \right), \quad (\text{D.1})$$

where $\tilde{\sigma}$ is the local conductivity of the fluid (a dissipative coefficient within hydrodynamics) and $\tilde{\chi}$ is the local charge susceptibility, which are related to the global σ and χ through (4.29) and (4.36). Our goal is now to show that on long time and length scales, in some suitable sense,

$$n(x, t) \sim e^{iqx - Dq^2 t} \quad (\text{D.2})$$

is a “solution” to (D.1), with D given by (4.21). More precisely, we show that there is an asymptotic solution to (D.1) of the form

$$n(x, t) = e^{iqx - Dq^2 t} \left[b_0(x) + iq b_1(x) - q^2 b_2(x) + \dots \right]. \quad (\text{D.3})$$

By comparing (D.3) into (D.1) at $q = 0$, we start with a simple solution

$$b_0 = \tilde{\chi}. \quad (\text{D.4})$$

At $\mathcal{O}(q)$, we find that (D.1) reads

$$0 = \partial_x \left(\tilde{\sigma} \partial_x \left(\frac{b_1}{\tilde{\chi}} \right) + \tilde{\sigma} \right). \quad (\text{D.5})$$

This equation is readily integrated:

$$\partial_x \left(\frac{b_1}{\tilde{\chi}} \right) = -1 + \frac{C}{\tilde{\sigma}}, \quad (\text{D.6})$$

with C an integration constant. We fix C by demanding that b_1 not diverge with x :

$$\frac{1}{C} = \mathbb{E} \left[\frac{1}{\tilde{\sigma}} \right]. \quad (\text{D.7})$$

At $\mathcal{O}(q^2)$ we find

$$D\tilde{\chi} = \tilde{\sigma} \left(1 + \partial_x \left(\frac{b_1}{\tilde{\chi}} \right) \right) + \partial_x \left(\tilde{\sigma} \partial_x \left(\frac{b_2}{\tilde{\chi}} \right) + \frac{\tilde{\sigma} b_1}{\tilde{\chi}} \right). \quad (\text{D.8})$$

Again assuming $b_{1,2}$ do not diverge, we may spatially average both sides of (D.8). Using (D.6) and (D.7) along with (4.29) and (4.36) we recover the Einstein relation (4.21).

We conclude with a technical comment. The true eigenstates of an inhomogeneous diffusion equation in one spatial dimension are spatially localized, with a frequency dependent localization length that diverges as $Dq^2 \rightarrow 0$ ²³². Subject to mild assumptions about the distribution of $\tilde{\sigma}$ ²³², the localization length diverges fast enough that the diffusion constant and conductivity are finite. While the ansatz (D.3) appears to describe the time evolution of a delocalized eigenstate of (D.1), in principle an expansion of (D.3) to all orders in $b_{0,1,2,\dots}$ can be consistent with localization; above we only computed $b_{0,1}$ explicitly. The assumptions which we made that $b_{1,2}$ did not diverge with x amount to the assumption that the localization length grows quickly enough at low frequencies, are sensible in ordinary models²³². An alternative way to think about this problem is to assume $q \in e^{i/4}\mathbb{R}$, so that the driving is periodic in time and leads to a spatial decay of $n(x)$. This spatial decay will be dominated at low driving frequencies not by the localization of eigenstates, but by dissipative diffusion¹³⁸.

sectionFluid-Gravity Expansion in Kruskal Coordinates In this appendix, we carry out the fluid-

gravity expansion to second order in Kruskal coordinates, for our static striped black holes. For simplicity, we work in $d = 2$ and assume that the background is governed by the Einstein-dilaton system (4.59). The zeroth order metric is given by (4.42) and to this order, after making the change of variables

$$A = e^{\widehat{a}}, \quad B = e^{\widehat{b}}, \quad (\text{D.9})$$

we have the following equations of motion (denoting $UV \equiv \rho$, and ∂_ρ with primes):

$$e^{-\widehat{b}}(\rho e^{\widehat{b}} \varphi')' = \frac{1}{2} e^{\widehat{a}} \frac{\partial \mathcal{V}(\varphi)}{\partial \varphi}, \quad (\text{D.10a})$$

$$\widehat{a}' \widehat{b}' - \frac{1}{2} (\widehat{b}')^2 - \frac{1}{2} (\varphi')^2 - \widehat{b}'' = 0, \quad (\text{D.10b})$$

$$\frac{1}{2} e^{\widehat{a}} \mathcal{V}(\varphi) + \rho \widehat{b}'' + \widehat{b}' + \rho \widehat{b}'' = 0, \quad (\text{D.10c})$$

$$\frac{1}{2} e^{\widehat{a}} \mathcal{V}(\varphi) + \frac{1}{2} \rho \varphi'^2 + \frac{1}{2} \rho \widehat{b}''^2 + \widehat{a}' + \widehat{b}' + \rho \widehat{a}'' + \rho \widehat{b}'' = 0. \quad (\text{D.10d})$$

As we have explained in the main text, subleading corrections to these equations in ξ^{-1} are regular everywhere, and hence this forms the basis for a well-behaved perturbative expansion. As we go through this appendix, we will see explicitly what these regular corrections are.

D.1.1 FIRST ORDER CORRECTION

By parity symmetry under a local change $x \rightarrow -x$, and due to the fact that there is a Killing vector $V\partial_U - U\partial_V$, the only corrections to the metric which could arise at $\mathcal{O}(\xi^{-1})$ are

$$ds_1^2 = \zeta(UV)(VdUdx + UdVdx) \quad (\text{D.11})$$

These are simply coordinate artifacts. To see this, we note that at first order,

$$\begin{aligned}
0 &= \frac{1}{V} \left(R_{Ux} - \frac{R}{2} g_{Ux} - 8G_N T_{Ux} \right) \\
&= \zeta e^{-\widehat{a}} \left[\frac{e^{\widehat{a}} \mathcal{V}}{2} + \widehat{a}' + \widehat{b}' + \rho (\widehat{a}'' + \widehat{b}'') + \frac{\rho}{2} (\widehat{b}'^2 + \varphi'^2) \right] + \frac{\widehat{b}' \partial_x \widehat{a} - \varphi' \partial_x \varphi - \partial_x (\widehat{a}' + \widehat{b}')}{2}. \quad (\text{D.12})
\end{aligned}$$

Using (D.10d) the coefficient of ζ identically vanishes. Indeed, this is a consequence of the fact that ζ can be removed by a change in coordinates, so for simplicity we set it to vanish. There is another contribution which is ζ -independent. Let us now consider the following particular radial derivative:

$$\begin{aligned}
e^{-\widehat{b}} \left[\rho e^{\widehat{b}} (\widehat{b}' \partial_x \widehat{a} - \varphi' \partial_x \varphi - \rho \partial_x (\widehat{a}' + \widehat{b}')) \right]' &= -\partial_x \varphi [\varphi'' + (1 + \rho \widehat{b}') \varphi'] - \partial_x \left(\rho \frac{\varphi'^2}{2} + \rho \widehat{a}'' + \rho \widehat{b}'' + \widehat{a}' + \widehat{b}' + \frac{\rho}{2} \widehat{b}'^2 \right) \\
&+ \partial_x \widehat{a} (\widehat{b}' + \rho \widehat{b}'^2 + \rho \widehat{b}'') = -\frac{e^{\widehat{a}}}{2} \frac{\partial \mathcal{V}}{\partial \varphi} \partial_x \varphi + \partial_x \left(\frac{e^{\widehat{a}}}{2} \mathcal{V} \right) - \frac{e^{\widehat{a}}}{2} \mathcal{V} \partial_x a = 0. \quad (\text{D.13})
\end{aligned}$$

In the last step above we have used the zeroth order equations of motion (D.10a), (D.10d) and (D.10c) respectively. We see that (up to a factor $\rho e^{\widehat{b}}$) the non-vanishing contribution to (D.12) is constant in ρ . We may evaluate it at the horizon, $\rho = 0$ – since all fields are regular at the horizon on the fluid-gravity ansatz, we conclude that the entire contribution in square brackets above vanishes. Hence, away from the horizon, where $\rho e^{\widehat{b}}$ is strictly finite, we conclude that the remaining contribution to (D.12) vanishes. Thus, there is no first order correction to our geometry.

D.1.2 SECOND ORDER CORRECTION

The first nonvanishing corrections to the metric and dilaton fields occur at second order and are of the following form

$$\begin{aligned} ds_2^2 = & V^2 \eta(UV) dU^2 + U^2 \eta(UV) dV^2 + \alpha(UV) dU dV + [\beta(UV) + \gamma(UV)] dx^2 \\ & + [\beta(UV) - \gamma(UV)] dy^2 + \mathcal{O}\left(\frac{1}{\xi^3}\right) \end{aligned} \quad (\text{D.14a})$$

$$\Phi_2 = \varphi_2(UV) \quad (\text{D.14b})$$

After using the background equations of motion to simplify the results somewhat, we obtain the following equations of motion for the perturbations:

$$\gamma \left[\frac{B'}{B} - \frac{\rho B'^2}{B^2} + \frac{\rho B''}{B} \right] + \gamma' \left(\frac{\rho B'}{B} - 1 \right) - \gamma'' = -\frac{(\partial_x A)^2}{4A} - \frac{\partial_x A \partial_x B}{2B} - \frac{A}{4} (\partial_x \varphi)^2 - \frac{\partial_x^2 A}{2} \quad (\text{D.15a})$$

$$\begin{aligned} & \frac{2B}{A^2} \eta \left(\frac{\rho A'}{A} - 1 \right) + \left(\frac{\rho^2 B A'}{A^3} - \frac{4\rho B}{A^2} - \frac{\rho^2 B'}{A^2} \right) \eta' - \frac{\rho B''}{A^2} \eta - \beta \left[\frac{\rho B'' + B'}{AB} - \frac{\rho B'^2}{AB^2} \right] \\ & + \frac{\beta'}{A} \left(1 - \frac{\rho B'}{B} \right) + \frac{\rho}{A} \beta'' - \alpha \left[\frac{\rho B''}{A^2} + \frac{\rho B'^2}{2A^2 B} + \frac{B'}{A^2} - \frac{\rho B A'^2}{A^4} \right] + \frac{B \alpha'}{A^2} \left(1 - \frac{2\rho A'}{A} \right) + \frac{\rho B}{A^2} \alpha'' \\ & + \frac{B}{2} \frac{\partial \mathcal{V}}{\partial \varphi} \varphi_2 + \frac{\rho B \varphi' \varphi_2'}{A} = -\frac{\partial_x^2 A}{2A}. \end{aligned} \quad (\text{D.15b})$$

$$\beta \left[\frac{B''}{B^2} - \frac{A' B'}{AB^2} - \frac{B'^2}{B^3} \right] + \left(\frac{A'}{AB} + \frac{B'}{B^2} \right) \beta' - \frac{\beta''}{B} - \alpha \frac{A' B'}{A^2 B} + \frac{B' \alpha'}{AB} - \varphi' \varphi_2' = 0 \quad (\text{D.15c})$$

$$\begin{aligned} & \frac{B \mathcal{V} \alpha}{2} - \frac{\beta}{B} (B' + \rho B'') + \beta' + \rho \beta'' - \left(\frac{\rho^2 \eta B'}{A} \right)' + \frac{B A}{2} \frac{\partial \mathcal{V}}{\partial \varphi} \varphi_2' = \frac{(\partial_x A)^2}{4A} + \frac{A (\partial_x B)^2}{2B^2} - \frac{A (\partial_x \varphi)^2}{4} \\ & - \frac{\partial_x^2 A}{2} - \frac{A \partial_x^2 B}{2B} \end{aligned} \quad (\text{D.15d})$$

$$\begin{aligned} & \frac{2\rho^2 A' \varphi'}{A^3} \eta - \frac{2\rho^2 \varphi' \eta'}{A^2} - \frac{2\rho B' \varphi'}{AB^2} \beta + \frac{2\rho \varphi'}{AB} \beta' + \frac{2\rho}{A} \varphi_2'' + \frac{2\rho B' \varphi_2'}{AB} + \frac{2\varphi_2'}{A} - \frac{\partial^2 \mathcal{V}}{\partial \varphi^2} \varphi_2 = -\frac{\partial_x^2 \varphi}{B} - \frac{\partial_x A \partial_x \varphi}{AB} \end{aligned} \quad (\text{D.15e})$$

As in the standard fluid-gravity correspondence, we see that these perturbations obey ordinary differential equations depending on ρ alone, pointwise at each x .

Although we are not able to solve these equations analytically, we do note that they are completely regular as $\rho \rightarrow 0$ (near the black hole horizon), supporting our claim that this expansion of Einstein's equations is well-behaved. We also note that there is some gauge redundancy in the above equations of motion. One useful gauge will be to set $\eta = 0$ – this can be done through a coordinate change of the form $U \rightarrow U[1 + \Xi(UV, H(x))]$, $V \rightarrow V[1 + \Xi(UV, H(x))]$ for a small $\Xi(UV, H(x)) \sim \xi^{-2}$.

D.2 HIGHER ORDER CORRECTIONS TO THE BUTTERFLY VELOCITY

In this appendix, we will assume that we have found the regular solution to the fluid-gravity expansion in Kruskal coordinates up to second order. We choose the gauge $\eta = 0$, which simplifies the computation of the shock wave geometry.

We can now use (4.44) to calculate the butterfly velocity to next order for $d = 2$. The only additional metric correction at $\mathcal{O}(\xi^{-2})$ comes in the UU component of Einstein's equation. After employing (4.49), only the UU component of Einstein's equation is altered by the shockwave. Looking at this component, and demanding that the coefficient of (U) vanishes, we find the differential equation

$$\left(1 + \frac{\alpha}{A} - \frac{\beta + \gamma}{B}\right) \partial_x^2 b + \frac{\partial_x A \partial_x b}{A} - \left(\frac{B'}{A} + \frac{B}{A} \left(\frac{\beta}{B}\right)'\right) b = 8G_N \frac{B(0,0)}{A(0,0)} \mathcal{E} \epsilon^{2Tt}(x) \quad (\text{D.16})$$

after employing (D.15), at $\rho = 0$, to simplify the result. Following the derivation in the main text, this leads to an effective butterfly velocity

$$\frac{1}{v_B} = \frac{1}{2T} \mathbb{E}[\tilde{m}], \quad (\text{D.17})$$

where

$$\widetilde{m}^2 = \frac{B'}{A} \left[1 - \frac{\alpha}{A} + \frac{\beta + \gamma}{B} \right] + \frac{B}{A} \left(\frac{\beta}{B} \right)' + \frac{1}{\sqrt{A}} \partial_x^2 \sqrt{A} + \mathcal{O} \left(\frac{1}{\xi^3} \right). \quad (\text{D.18})$$

To get this result, we have divided through (D.16) by the coefficient of $\partial_x^2 b$, and then used that (for $x \neq 0$)

$$\begin{aligned} & \partial_x^2 b + \frac{\partial_x A \partial_x b}{A} - \left[\frac{B'}{A} \left[1 - \frac{\alpha}{A} + \frac{\beta + \gamma}{B} \right] + \frac{B}{A} \left(\frac{\beta}{B} \right)' \right] b \\ &= \partial_x^2 (b \sqrt{A}) - b \sqrt{A} \left\{ \left[\frac{B'}{A} \left[1 - \frac{\alpha}{A} + \frac{\beta + \gamma}{B} \right] + \frac{B}{A} \left(\frac{\beta}{B} \right)' \right] + \frac{1}{\sqrt{A}} \partial_x^2 \sqrt{A} \right\} = \mathcal{O} \left(\frac{1}{\xi^3} \right). \end{aligned} \quad (\text{D.19})$$

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